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RESEARCH METHODOLOGY

Unit 1

RESEARCH PROBLEM FORMULATION

MEANING OF RESEARCH

According to a typical English Dictionary, the meaning of 'research' is "a careful investigation or inquiry specifically through search for new facts in any branch of knowledge."

One can also define research "as a scientific and systematic search for pertinent information on a specific topic."

It is a voyage of discovery propelled by the instinct of inquisitiveness of human mind. It is an academic activity and as such the term should be used in technical sense. According to Clifford Woody, research comprises defining and redefining problems, formulating suitable solutions or hypothesis; collecting, organizing and evaluating data; making deduction and reaching conclusions; and at last carefully testing the conclusions to decide whether they satisfy the hypothesis formulated.

Research is an original contribution to the existing stock of knowledge making for its advancement. We try to establish the truth with the help of study, observation, comparison and experiment. In short, the search for knowledge through objective and systematic method of finding solution to a problem is research.

OBJECTIVES AND MOTIVATION

The primary objective of researches is to find answers to questions through the application of scientific procedures. The truth is hidden and has not been found out so far. Every research has its own specific purpose, but one can broadly group the research objectives as,

- To gain familiarity with the phenomenon or to get new insights into it. These studies are termed as exploratory or formative research studies.
- To determine accurately the characteristics of a particular individual, system, situation or a group. These studies are known as descriptive research studies.

- To decide the frequency with which a phenomenon occur or with which it is associated with something else. These studies are known as diagnostic research studies.
- To establish the causal relationship between variables by forming suitable hypothesis. They are termed as hypothesis testing research studies.

There are several motives for somebody undertaking a research work. The possible ones include,

- Desire to get a research degree along with its consequential benefits.
- Desire to face challenges in solving the unsolved problems.
- To get intellectual joy of doing some creative work.
- To do service to the society and get respectability.
- Direction by the employer, Govt. etc.

TECHNOLOGICAL INNOVATION

Engineering is concerned with problems whose solution is needed and / or desired by the society. The capacity to innovate, manage information, and nourish knowledge as a resource will dominate the economic domain as natural resources, capital and labour ones did. This places high premium on development of technology and delivery systems.

The advancement of technology has three phases.

Invention : The creative act whereby an idea is conceived

Innovation :The process by which an invention or idea is translated into successful practice and is utilized by the economy.

Diffusion : The successive and widespread initiation of successful innovation.

Science based innovation has made significant contributions to the development of industries like aircraft, computers, plastics and television in developed countries like USA. Traditionally, engineers play a major role in technological innovation. A strong basic research is needed to maintain the storehouse of new knowledge and ideas. Innovation in response to a need of the society has greater probability of success than innovation in response to technological research opportunity. The scope for doing applied research is high, which involves innovative solution to the problem of the society.

TYPES OF RESEARCH

The basic types of research are as follows:

i) Descriptive Vs Analytical

The major purpose of descriptive research is description of the state of affairs as it exists at present. This method includes surveys and fact finding enquiries of different kinds. The main characteristic of this method is that the researcher has no control over the variables, he can report only what is happening or what has happened (eg: frequency of arrival of vehicles for repairs in a workshop). In the analytical, researcher has to use facts or information already available from secondary sources and analyze them to make critical evaluation of the material.

ii) Applied Vs Fundamental

Applied research is also known as 'action research'. It aims at finding a solution for an immediate problem facing the society or on Industrial / business organization. On the other hand, the fundamental research is concerned with generalization and with the formulation of a theory. Research concerning some natural phenomenon or relating to pure mathematics are examples or fundamental research. The central aim of applied research is to discover a solution to some pressing practical problems, whereas basic research is directed towards finding information that has a broad base of applications and thus, adds to the already existing organized body of scientific knowledge.

iii) Quantitative Vs Qualitative

Quantitative research is based on the measurement of quantity or amount. This is applicable to phenomena that can be expressed in terms of quantity, amount, size or weight etc. Qualitative research on the other hand is concerned about the qualitative phenomena, that is, something relating to or involving quality. For example, research designed to find out how, people feel or what they think about a particular subject or institution is qualitative research.

iv) Conceptual Vs Empirical

A concept is an abstract idea, a visionary expression of a proposed or planned action that leads to achievement of an objective. Therefore, a conceptual research is related to some abstract idea or theory. It is generally used by thinkers to develop new concepts or reinterpret existing ones. On the other hand, empirical research relies on experience or observation alone, often without due regard for system and theory. It is data based research ending up with conclusions, which are capable of being verified by observation or experiment.

Many engineering oriented research work are of empirical type involving physical experiments or simulation. In such a research, the researcher tries to establish a working hypothesis (probable) and then try to get enough facts (data) to prove or disprove his / her hypothesis. He then sets up experimental design to manipulate the material concerned so as to bring forth the desired information. In this type of research, the experimenter has control over the variables under study and he can deliberately manipulate any one of them to study its effects. Evidence gathered through experiments or empirical studies is considered to be the most powerful support for a given hypothesis.

v) Other types of research

Other types of researchers are there, which are variations of one or more of the above stated approaches they include,

- ✓ **One time or longitudinal** : One time is confined to a single period whereas the longitudinal is carried over several time periods.
- ✓ **Field setting or laboratory or simulation**: This is based on the environment in which the research work is carried out.
- ✓ **Conclusion oriented** and decision oriented: In conclusion oriented, a researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes. Decision oriented research is always to meet the needs of decision maker and the researcher is not free to embark upon research according to his own inclination. Operations research is an example of decision oriented research.

From the above description of the types of research, it is obvious that there are two basic approaches to research viz: Quantitative approach and Qualitative approach.

As already stated the quantitative approach involves generation of data in quantitative form (ordinal numbers), which are subjected to rigorous quantitative analysis in a formal and rigid fashion. This approach can be further sub-divided into (i) Inferential (ii) experimental (iii) simulation approaches. In the inferential approach, the characteristics of the population is inferred from the database. The variables are manipulated by the researcher in the experimental approach to observe the effert of a variable on others. Greater control is over the experiment. The simulation approach involves building a numerical model and manipulating the model with the help of a computer. This permits observation of the dynamic behaviour of the system or sub-system under controlled conditions.

Qualitative approach to research is concerned with subjective assessment of attitudes, opinions and behaviour. This approach generally generates results in non-qualitative form or in a form, which are not subjected to rigorous quantitative analysis.

Research Vs Scientific method

These two are closely related. The research can be termed as "inquiry into the nature of reasons and the consequences of any particular set of circumstances, whether these circumstances are experimentally controlled or recorded just as they occur. The researcher is interested in more than one particular results; he is interested in the repeatability of the results and in their extension to more complicated general situations."

The term 'scientific method' covers the philosophy common to all research methods and techniques, although they may very considerably from one science to another. The scientific method is one and the same in the branches of a particular science. This method encourages a rigorous, impersonal made of procedure dictated by the demands of logic and objective procedure.

HOW RESEARCH METHODOLOGY DIFFERS FROM RESEARCH METHODS?

The Research methods refer to the methods the researcher use in performing research operations, For instance, in applied research, the objective is to arrive at a solution for a given problem, the available data and the unknown aspects of the problem have to be related to each other to make a solution possible. For this, the research methods employed can be grouped into i) methods used to collect data ii) statistical techniques used for establishing relationships between data and unknowns, and iii) methods used to evaluate the accuracy of the results.

The scope of Research methodology is wider than that of Research methods. Here, we not only talk of research methods, but also consider the logic behind the methods we use in the context of our research study and also explain why we are using a particular method or technique and why we are not using others. This is necessary to ensure that research results are capable of being evaluated either by the researcher himself or by others.

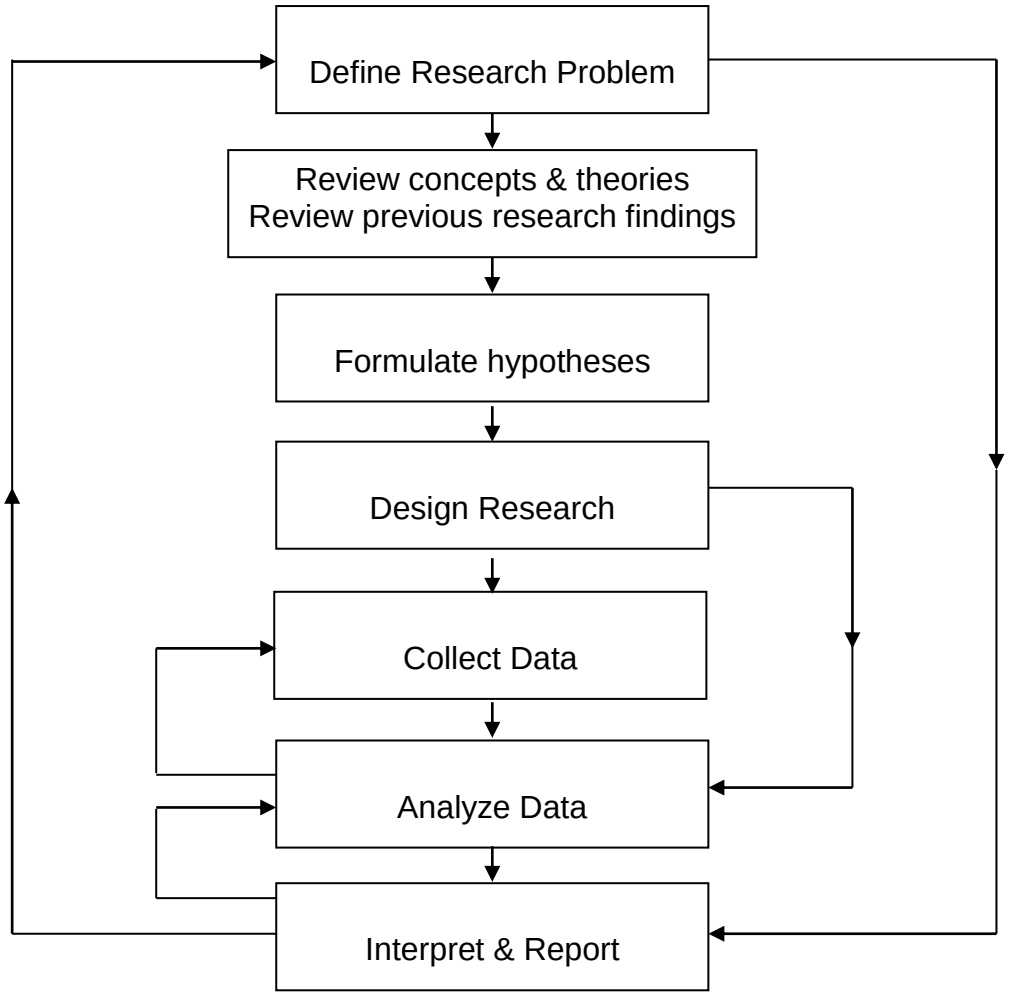
The researcher need to know which method or technique is relevant and which are not, and what would they mean and indicate and why. The researcher also needs to understand the assumptions underlying various techniques and they need to know the criteria by which they can decide that certain techniques and procedures will be applicable to certain problems and others will not. All this mean, the researcher has to design his methodology for his problems as the same may differ from problem to problem.

RESEARCH PROCESS

The Research process consists of series of actions or steps necessary to effectively carryout research and the desired sequencing of steps. The figure illustrates the research process. It may be seen that the research process consists of several closely related activities, which overlap considerably rather than strictly following a prescribed sequence. Many a times, the first step determines the nature of the last step to be undertaken. It is better that subsequent procedures are taken into account in the early stages, to avoid serious difficulties later. The various steps involved in the research process are not mutually exclusive, nor are they separate and distinct. They do not

necessarily follow each other in any specific order and the researcher has to constantly anticipate at each step the requirements of subsequent steps.

The following sequence concerning various steps provides a useful procedural guideline regarding research process.



i) Formulating the research problem

ii) Extensive literature survey

iii) Developing the hypothesis

iv) Preparing the research design

v) Determining sample design

vi) Collection of data

vii) Project execution

viii) Data analysis

ix) Testing the hypothesis

x) Interpretation and generalization

xi) Report preparation

Formulating the research problem

At the outset the researcher must decide the general area of interest or aspect of subject matter that he would like to inquire into. The formulation of a general topic into a specific research problem constitutes the first step in a scientific enquiry. Essentially two steps are involved in formulating the research problem, viz; understanding the problem thoroughly, and rephrasing the same into meaningful terms from an analytical point of view.

Literature survey

Literature survey connected with the problem is very essential for a Ph.D. research work. One of the major reason is that the researcher should get himself acquainted with the selected problem both conceptually, ie. concepts and theories and empirically by reviewing studies made which are similar to the one proposed. This gives confidence in defining the problem and also to identify and narrow the topic of his research.

Development of working hypothesis

A working hypothesis is a tentative assumption made in order to draw out and test its logical or empirical consequences. In most of the research work, it plays an important role. The role of hypothesis is to guide the researchers by delimiting the area of research work.

Research Design

The function of research design is to plan for the collection of relevant data / evidence with minimum expenditure of effort, time and money. But, this depends mainly on the research purpose. Research purposes may be grouped into i) Exploration, ii) Description, iii) Diagnosis, and iv) Experimentation.

Sample Design

The field of inquiry cannot constitute the entire 'universe' or population as it involves great deal of time, money and energy and many a times impossible. Hence, quite often we select a few items from the universe ie. sample for our study purpose. The way a researcher decides selecting a sample is popularly

known as 'sample design'. There are several ways of selecting sample such as random sampling, systematic, stratified, sequential, multistage, cluster etc.

Data Collection

Primary data required for the research study can be collected either through experiment or through survey. It the researcher conducts an experiment, he observes some quantitative measurements or the data, with the help of which he examines the truth contained in the hypothesis. The secondary data are gathered from published reports, handbooks and literature.

Project execution

This is an important step in the research process. The execution of the entire project should have planned properly. If the execution proceed on correct lines, the data collected would be adequate and dependable. Many a times, it will be useful if a pilot experiment or preliminary survey conducted before proceeding to actual production run.

Data analysis

The analysis of data requires a number of closely related operations such as establishment of categories, the application of these categories to raw data through coding, tabulation and then drawing statistical inferences. If the data is voluminous, use of computers and statistical packages can be resorted to.

Testing of hypothesis

In this step, the hypothesis formed already is tested using various statistical tests such as Chi-square, T-test, F-test etc. The hypothesis testing will result in either accepting the hypothesis or rejecting it.

Interpretation and generalization

If the researcher had no hypothesis to start with, he might explain his findings on the basis of some theory. It is known as interpretation. Quite often the process of interpretation may lead to new questions, which in turn may lead to further researches. If a hypothesis is tested and upheld several times, it may be

possible for the researchers to arrive at generalization that is to build a theory. The real value of research lies in its ability to arrive at certain generalization

Reporting or thesis writing

Reporting or thesis writing is the culmination of the research work done. Generally it is a complete, stand-alone document aimed at persons having good background of knowledge especially in the case of a Ph.D. thesis. Enough time should be devoted for writing thesis and it should be in proper format. Manuals are available to guide the researcher. It is always better that a researcher should go through some of the well written thesis to grasp the nuances.

PROBLEM SOLVING IN ENGINEERING

While the research approach presented in the previous section provides a general framework, there are two fundamental approaches to solving engineering problems that arise in the discovery of knowledge and its application to society's needs.

i) Theoretical (Physical / Mathematical) modeling

ii) Experimental measurement.

In engineering, this is true regardless of the discipline such as mechanical, electrical etc. or the engineering function such as design, development, research, manufacturing etc. While some problems are adequately treated by using only theory or only experiment, most require a judiciously chosen mix of these techniques. Even though each specific application has its own peculiarities, there are some important general characteristics of the theoretical and experimental methods which will be helpful in deciding the proper blend when a choice is necessary. This will help us in organizing our thinking about the whole process.

COMPARISON BETWEEN THEORETICAL AND EXPERIMENTAL PROCESS

i) All physical principles and their mathematical expressions (mathematical models) are approximations of the real world behavior. Though by making the model more complex, the quality of approximations can be improved, perfection is not always possible. The practical engineering always labors under constraints-of time, money,

facility etc. But, however depending upon the overall project objective, a researcher can choose an experimental method, which is simpler and less accurate and may be judged as and quote for the purpose.

Thus, this comparison centre's on the fact that theories are always approximations involving simplifying assumptions; whereas experiments when properly designed and executed or run on the actual system reveal true behaviour. This comparison favours experiment over theory.

ii) The second comparison is based on the generality of results. The main reason we prefer theoretical approaches (when their accuracy is adequate) is that theoretical results are usually applicable to a whole class of problems whereas experimental results are peculiar to the specific apparatus on which the measurements were made. The ability to solve entire classes of problems with a single analysis gives theoretical methods an efficiency which is their major advantage. A significant price paid for the generality of theory is the universal need for simplifying assumptions.

But, however when models are made more closer to real world, leads to mathematical formulations of deeper complexity and mathematical intractability. Today, of course, this problem has been greatly alleviated by the development of computerized numerical methods (eg: FEA), these methods sacrifice generality; they do not provide formula solutions, ie they give only numerical answers for individual specific cases, analogeus to physical experiment.

Just as pursuit of accuracy in mathematical modeling leads to undesired complexity, the experimental approaches also generally become more intricate when we require higher accuracy in the measured result. Thus, this comparison shows no obvious or uniform advantage for each approach.

iii) The third comparison can be based on facilities needed to undertake theoretical or experimental studies. Theoretical studies may be initiated with less commitment of financial resources, while experimentation requires investment in equipment, laboratory space

etc. This apparent advantage partially disappears when we realize that most current theoretical studies require good computing facility supported by suitable software's. Thus, neither approach has clear superiority.

iv) The next comparison is based on the time required to achieve results. Any technical endeavour has an appropriate time scale associated with its success. Deciding the time frame is difficult at the beginning of the project. Theory seems to offer advantages, since one can start immediately, whereas experimentation requires the building of experimental set up, debugging, pilot run, calibration before the first data available. But, however, if the chosen theoretical approach turns out fruitle, we may have to reformulate, which again requires more time.

It should be clear that sometimes, a theoretical approach is quickest and at other times experimentation gives the most rapid solution. In general, simple problems are more likely to yield quickly to theory, whereas puzzling phenomena may require a strong component of experiment.

IDENTIFICATION OF RESEARCH TOPIC

Selecting the problem

In the research process, the first and foremost step happens to be that of selecting and properly defining a research problem. A researcher must find the problem and formulate it so that it becomes susceptible to research. The research problem undertaken for study should be carefully selected. The task is difficult, though it may not appear to be so. The guide may be of some help to the researcher in this. Nevertheless, every researcher must find out his own salvation, as research problems cannot be borrowed. A problem must spring from the researcher's mind. For instance while prescribing glasses for eyes, the optician alone cannot decide, we have to cooperate with him to prescribe the right type of glass. Thus, a research guide can be at the most only help a researcher choose a subject.

The following pointes may be taken into account a researcher while selecting a research problem or a subject for his research.

- Subject which is overdone should not be normally chosen, as it will be difficult to throw any new light in such a case.
- Controversial subject should not become the choice of an average researcher
- Avoid too narrow or too vague problem
- Choose a familiar and feasible subject so that the related research material or sources of research are within one's reach. An expert or supervisor already engaged in similar research work or journal publications will be of use in this.
- Importance of the subject, the background of the researcher, costs and time factor involved are a few other aspects which should be considered.
- Selection of a problem must be preceded by a preliminary study if needed.

What constitutes research?

In a research work leading to the award of Ph.D degree, the researcher always looks for some original contribution. The originality under normal circumstances may be confined to,

- New interpretation or application of known facts.
- Bringing new evidences to support a known fact
- Empirical work of one's choice and not a repetition of what someone has already done.
- Using a known technique in an untested area
- Furthering knowledge by continuing the work suggested by another scholar at the end of his research work.
- Propounding a new concept or a formulae in the chosen discipline or subject
- Evolving a new methodology to solve an existing problem.

A suggested approach to identify research problem

It appears there is no universal approach to identify the research problem especially as how to start defining the research topic and zooming in on specific

research problem. The following are general guidelines, which may be followed by research students of M.S. and Ph.D. programmes.

i) First choose a direction or area of interest based on the background knowledge of the researcher obtained through courses undertaken, readings, conferences attended and discussions with professors etc. There could be a list of topics of interest to the researcher, but it could be limited to not more than three.

ii) Compile a set of 'key words' to start searching for higher quality readings for each of the previously selected topics. May be one can refer to library on-line or search on the web. But, one should check the publication details for quality.

iii) Out of the search hits, select 15 - 20 papers that are most related to what the researcher had in mind and are of highest quality. One read not need all these papers. Check the title, abstract, names of authors, their affiliations and most importantly the conference or journal. Try to refine the set of key words and perform multiple searches to most related quality work. To check for quality one can use references and citations. Usually the most cited work by high quality papers is also of high quality. The list of references of a specific paper will provide a good direction to follow.

iv) For the selected 15 - 20 papers, read only the abstract, introduction and conclusion in detail. If necessary skim the rest of the paper to get general idea. Identify the emphasis of each paper.

- Which problem it addresses
- What solution it proposes
- How the solution differs from previous solutions, and
- What are the major contributions and conclusions

After going through these 15 – 20 papers and understanding select a list of 4 – 6 papers that are of highest quality and addresses the researcher's interest and challenges in the field most appropriately.

v) Read these 4 – 6 papers from beginning to end and identify (1) major approaches (2) methods of analysis (a) metrics (b) evaluation

tools (c) analysis and interpretation of resulting measured data, and (3) conclusions. At the same time try to keep a list of what you think that the authors may have missed in the paper / study, gaps or limitations that could be improved upon and any ideas on how to accomplish these improvements. One can ask these questions.

- Did all / some papers use similar approaches?
- Have they used the same evaluation criteria or method of analysis?
- If not, than what are the strengths / weaknesses of each method?

Also, keep a list of ideas that you want to explore, further, or background material you want to brush upon. This will create another list of readings for you in later stages.

vi) Write a two page proposal defining as clearly as possible the following items.

- Motivation
- Research challenges
- Overview of existing work
- Limitations of existing work
- Potential directions and ideas for improvement
- Expected results and impact on the field.

The two page report can be reviewed by knowledgeable friends and with the research supervisor. Based on their comments the proposal may be rewritten. If some major items missed, then go back to the 15 – 20 list and select another 3 – 5 good papers, read in detail and rewrite parts of your proposal.

Other sources to look for research problems:

Many industries and government organizations indicate the research areas of interest to them. By going through website of the Government organizations, the thrust areas or specific topics of interest can be identified. Similarly, company's annual reports do provide information about their R & D activities, from which some useful hints can be picked up and explored.

One can find many excellent research topics by considering specific products or processes already in existence and looking into ways of better understanding their behaviour and improve their performance. When properly defined this is not 'routine testing' inappropriate for academic research study but rather the essence of most experimental study in engineering. Many industrial undertakings may be willing to contribute by way of sample products and company engineers willing to share their expertise.

In many standard books on the broad area of research interest, the authors would have indicated the current status and what further exploration needs to be made. It is better to go through the latest edition of such books to identify possible specific area, which can be explored.

It is a normal practice in the Ph.D theses, the researcher indicate scope for further research and the limitations of his study due to certain constraints. By going through these theses, one can identify many research problems for investigation.

Discussion with experts, who have been working in the proposed area for several years and attending of seminars, conferences and workshops related to the areas of interest to the researcher will provide ample opportunities to choose a topic of research.

PROBLEM DEFINITION

The selection of a topic for research obviously includes a certain level of definition of the problem selected. It is most important to develop a thoughtful and detailed problem description early. Quite often one hears "A problem well defined is a problem half solved". The problem to be investigated must be defined unambiguously for that will help to discriminate relevant data from the irrelevant ones and what sort approach we have to adopt. It helps to answer questions like,

- What data are to be collected?
- What characteristics of data are relevant and need to be studied.
- What relations one to be explored?
- What techniques, experiments etc are to be used?

In an academic environment, choice of problem topics can be wider since no economic pay off is needed. But, however, the major constraints are time frame available facilities and preparation for studies. The faculty, who is a guide to the scholar, is likely to exert influence on the choice of topic.

Defining a problem involves the task of laying down boundaries within which a researcher shall study the problem with a predetermined objective in view. Though, this is a little difficult task, it has to be tackled intelligently to avoid the perplexity encountered in the research operation. The usual approach of researcher posing a question himself and set up techniques and procedures for throwing light on the question concerned for defining the research problem may not produce definite results. Defining a research problem properly and clearly is a crucial part of research study and must in no case be accomplished hurriedly.

The research problem should be defined in a systematic manner giving weight age to all related points. A suggested technique for the purpose involves the following steps generally one after the other.

i) State the problem in a general way
ii) Understand the nature of the problem
iii) Survey the available literature
iv) Develop ideas through discussions
v) Rephrase the research problem into a working proposition

General statement of the problem

Keeping in view either some practical concern or some scientific or intellectual interest or taking the clue from other published sources, state the problem in a general way broadly. The researcher should thoroughly immerse first in the subject matter, which concern with the aspect of the problem he wants to pose. Sometimes, the guide gives a broad hint, and it is then up to the researcher to narrow it down and phrase it in operational terms. There may some ambiguities in the problem stated, which must be resolved by cool thinking Also, the feasibility of a particular solution has to be considered and the some should be kept in view while stating the problem. If necessary a pilot survey or cursory simulation could be attempted.

Understanding the problem

The next step in defining the problem is to understand its origin and nature clearly. The best way of understanding the problem thoroughly is by going through bench mark or base papers, connected books and other literatures. The researcher can discuss with other researchers who worked in similar problems or with a persons who have a good knowledge of the problem concerned. The researcher should also keep in view the environment and facilities within which the problem is to be studied.

Surveying the available literature

Study of available literature concerning the problem broadly considered will be very useful for identifying the type of difficulties that may be encountered in the proposed approach as also the possible analytical shortcomings. At times such studies may also suggest useful and even new lines of approach to the problem being considered.

Develop ideas through discussions

Discussion concerning a problem often produce useful information, As already stated, the researcher should discuss his problem with his colleagues or others who have enough experience in the same area or working on similar problems. Such discussions should not be confined only to formulation of specific problem at hand, but should also be concerned with the general approach to the given problem, techniques and tools that might be used, possible solutions etc.,

Rephrasing the problem

The researcher must rephrase the research problem into a working proposition. Once, one goes through the steps stated above, rephrasing the problem into analytical or operational terms is not a difficult task. By rephrasing, the researcher puts the problem in specific terms as possible so that it may become operationally viable and may lead to suggested solutions of research problem, which may or may not lead to the real solutions. These propositions

should be amenable to testing within a reasonable time and should be consistent with most of the known facts.

Apart from the above, the following points should also be taken into account while defining a research problem.

- Technical terms, words which have special meanings used in the statement of the problem be defined.
- Basic assumptions relating to the research problem should be clearly stated.
- The criteria for the selection of the problem should be stated.
- The source of data, such as survey or experiment etc. and the time period should also be considered in defining the problem
- Scope and limitations of the problem should be mentioned explicitly.

LITERATURE SURVEY AND REVIEW

Literature survey:

Once we have agreed upon a clear problem definition, the logical next step will be survey of background literature. This is essential whether the proposed problem deals only with theoretical or with experimental aspects or both. The survey should be extensive which means that all available literature concerning the problem at hand must necessarily be surveyed and examined before the definition of the problem made. The researcher must be well conversant with relevant theories in the field, reports and records and all other relevant literature. The researcher must spent sufficient time in reviewing of literature of research already undertaken on related problems.

The literature review is mainly to find out what material and other data are available for operational purposes. If somebody has done experimental study or simulation, he might have mentions the problem and difficulties faced in his study, which may be useful for the current study. Knowing what data are available often serves to narrow the problem itself as well as the technique that might be using. This would also help a researcher to know if there are certain gaps in the theories, or whether the existing theories applicable to the problem under study

are inconsistent with each other, or whether the findings of the different studies do not follow a pattern consistent with the theoretical expectations and so on. All this will enable a researcher to take new strides in the field of furtherance of knowledge i.e., he can move up starting from the existing premise. Studies on related problems are useful for indicating the difficulties that he may encounter in the present study as also the possible analytical shortcomings. At times such studies may also suggest useful and even new lines of approach to the present problem.

Sources of information

The sources of information can be classified as indicated below.

a) Public sources

i. Govt. Departments (Defence, Energy, Set etc) ii) State and local Government (Highway, PCB, etc.) iii) Libraries iv) Universities v) Internet

b) Private sources

i. Non profit organizations and services (professional societies, trade associations, membership organizations)

ii. Profit oriented organizations (manufacturers, vendors catalogs, samples, test data etc., consultants)

iii. Individuals (Direct conversation or correspondence, personal friends, faculty)

Depending upon the nature of the problem undertaken for research study, the researcher has to seek the needed information form different sources. But, however major sources of information nowadays is library and internet. When we look for information in the library, there are hierarchy of information sources as given below.

- Technical dictionaries
- Encyclopedias
- Handbooks
- Bibiliographies
- Indexing and abstract services
- Technical and professional journals

- Translations
- Technical reports
- Books
- Patents
- Catalogs and manufacturer's brochures.

When we deal with a project in a new technical area, there may be a need to have a broad overview of the subject. Technical dictionaries and Handbooks provide useful technical data. Many handbooks also provide ample technical description of theory and its application, so they are good refreshers of material once studied in greater detail.

Some libraries and research institutes provide indexing and abstracting services. These provide current information on periodical literature and they also provide a way to retrieve published literature. An 'indexing service' cites the article by title, author, and bibliographic data. An 'abstracting service' also provides a summary of the content of the article. Some of the indexes and abstracts include,

Ceramic abstracts,
Chemical abstracts,
Corrosion abstracts,
Fuels and energy abstracts,
Highway research abstracts,
Applied science and Technology Index,
Engineering Index

Another useful source of much detailed information is 'Dissertation abstract', which gives abstract of most doctoral dissertations compiled in the United States and Canada. A copy of the dissertation can be ordered at a nominal cost. In India, the UGC started putting abstract of research thesis in their website. Now, they are making it mandatory all Universities to send them a soft copy of the thesis to UGC.

Conducting a search of the published literature is like putting together a complex puzzle. One has to select a starting place, but some starts are better than others. A good strategy is to start with the most recent subject indexes and abstracts and try to find a current review article or general purpose technical

paper. The references cited in it will be helpful in searching back along the 'ancestor references', to find the search that led to the current state of knowledge. However, this search path will miss any references that were over looked or ignored by the original researchers.

The next step should therefore involve citation sending to find the descendent references using "Science Citation Index". Once you have a reference of interest, you can use citation index to find all other references published in a given year that cited the key reference. These two search strategies will uncover as many references as possible about the topic.

The next step is to identify the key documents. One way to do this is to identify the references which has the greatest number citations, or those that other experts in the field cite as particularly important. One should keep in mind that take 6 to 12 months for a reference to be included in a index or abstract service, so current research will not be picked up using this strategy. Current awareness can be achieved by searching current context on regular basis using key words, subject headings, journal titles and authors already identified from literature search.

Information through Internet

The fastest growing communication medium is the Internet. The internet is a computer network inter-connecting numerous computers or local computer networks linked by common technical protocols. The most recent and most rapidly growing component of the internet is the world wide webs (www), an enormous, far-flung collection of colourful on screen documents that are linked to each other by high lighted words called 'hypertext'. The popularity of www comes from the fact that it makes distributing and accessing digital information simple and inexpensive. Since a large collection of documents are involved, using the web requires a 'search engine'. Nowadays a number of user friendly highly graphical search tools – often called 'web
browsers' are extensively used. The most commonly used include Netscape communicator, Microsoft Internet explorer.

Locations on the internet are identified by Universal Resource Locators (URL). For example, a URL that gives a brief history of the Internet is http://

www.isoc.org internet – history. The prefix http://www indicates we are typing to access a HTTP server on the World Wide Web at a computer with the domain name "isoc.org". The document is stored in that computer in a file called 'internet – history'.

To search the World Wide Web, we need a search engine. Each search engine works differently in combing through the web. Some scan the information in the title or header of the document, while others look at the bold headings on the page. In addition the way information is sorted, indexed and categorized differs between search engines. Therefore all search engines will not produce the some result for a specific enquiry, The most commonly used general purpose search engines include,

Alta Vista	http:// www.altavista.digital.com
Hot Bot	http://www.hotbot.com
Yohoo	http://www.yahoo.com

It is important to understand how search engines obtain and maintain their databases and also the search methods. Another important aspect is the degree to which a search engine covers all the materials out of the web. A recent study estimated that even the best search engine covered about one third of the web pages. Therefore, the best strategy for maximizing information retrieval from internet is to pick two or three search engines.

The researcher should also have knowledge of many URLs devoted to engineering topics and the need of the user. Some examples of URLs are,

NASA Technical Information service http://tech reports.lors.nasa.gov / egi-bin / NTRs

This covers all NASA centres and provides abstracts only. Reports must be ordered.

National Technical Information service http://ntis.gov

Large number of reports are available on file

The purpose of a literature search are several. A main purpose is ofcourse, to find whether the study you are about to undertake has already been done. We do not want to waste time and energy repeating someone work, unless we question their results for some reasons. Because of the vastness of open literatures and the

inaccessibility of many proprietary files, we can never be sure that we have uncovered all works that has been done. Therefore, a sincere effort in doing a thorough literature survey is essential.

Literature Review

It will be every useful while preparing the thesis if a proper review of literature gathered is made by the researcher. The review should encompass genesis of the problem in the researcher's mind, the extent of knowledge already available and a fairly extensive list of significant papers or other works in the area along with summary of the relevant findings. This can be done easily if the researcher keeps a proper record of these. A suggested form of literature survey / review is given below

Sl.No.	Author (s)	Paper title	Publication details	points	Conclusion& Future work
1	Hung-Yu Chin	Secure Access Control schemes for RFID system with anonymity	Mobile Data Management, 2006.MDM2006.7th International Conference10-12 May 2006 pp 90-96	- Forward Vs Backward channel - Hash based access control - Randomized access control	- NewHash scheme preposed
2					

Preparations of a review paper will help to infuse confidence in the mind of the researcher about the problem he proposes to explore. If necessary, he can publish such a review paper in journals or present in conferences.

Unit - II

RESEARCH DESIGN

RESEARCH DESIGN: WHAT IT IS?

A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure. It constitutes the blue print for the collection, measurement and analysis of data. The overall research design can be split into the following parts.

i) Evolving a suitable method for selecting items to be observed for the study -**sampling design**
ii) Identifying the conditions under which the observations are to be made - **observational design**
iii) Methods dealing with as how many items are to be observed and how to gather information / data and how to analysis them – **statistical design**
iv) Procedures specified (i) to (iii) can be carried out by using which techniques - **Operational design**.

WHY WE NEED RESEARCH DESIGN?

Research design is needed because, it facilitates the smooth sailing of the various research operations, thereby making research as efficient as possible yielding maximum information with minimum expenditure of effort, time and money. It also stands for advanced planning of methods to be adopted for collecting the relevant data and techniques to be used in their analysis, keeping in view the objective of the research In fact preparation of the research design should be done with great care, as any error in it may upset the entire project. Not only, this improves the reliability of the results arrived at, but also constitute a firm foundation of the entire edifice of the research work.

A good design is often characterized by adjectives like flexible, appropriate, efficient, economical and so on. It is related to the purpose or objective of the research problem and also with the nature of the problem studied.

A design, which is quite suitable in one case, may be inadequate in the context of some other problem. In fact, a single design cannot serve the purpose of all types of problems.

In an exploratory research study, where the major emphasis is on discovery of ideas and insights, the research design must be flexible enough to permit consideration of many different aspects of a phenomenon. But in a study dealing with accurate description of a situation or an association between variables, accuracy becomes a major consideration. Studies involving the testing of hypothesis of causal relationship between variables require a design which will permit inferences about causality, ensuring minimum bias and maximum reliability.

We may not be able to put a particular study in a particular group in practice, as a given study may have in it elements of two or more of the functions of different studies. Only on the basis of primary function, a study can be categorized as exploratory or descriptive or hypothesis testing and accordingly the choice of a research design may be made in the case of a particular study.

SOME BASIC CONCEPTS

We use various terminology and concepts in Research design. Some of the important terminology with short explanation for each of them is given to ensure better understanding of the subjects

Variable: A concept which can take on different quantitative values is called a variable. For example weight, stress, strain are variables. Qualitative phenomena (or attributes) are also quantified on the basis of the presence or absence of the concerned attribute. Certain variable take continues values and they are called " continuous variables " and variables which takes only fixed, integer values are called as " discrete variables ". If one variable depends upon or is a consequence of another variable, it is termed as "dependent variable"; and the variable that is antecedent to the dependent variable is called as "independent variable".

Extraneous Variable: Independent variables that are not related to the purpose of study, but may affect the dependent variable are termed as 'extraneous variables'. Any effect noticed on the dependent variable as a consequence of extraneous variable is technically called as "experimental error".

Control: One important characteristic of a good research design is to minimize the influence or effect of extraneous variable(s). The technical term "control" is used when we design the study minimizing the effect of extraneous variables.

Confounded relationship: When the dependent variable is not free from the influence of extraneous variables, the relationship between dependent and independent variable is said to be confounded by an extraneous variable.

Research hypothesis: The research hypothesis is a prediction statement that relates an independent variable to the dependent variable.

Experimental and non-experimental hypothesis testing: When the purpose of research is to test a research hypothesis, it's termed as hypothesis - testing research. Research in which the independent variable is manipulated is termed as "experimental hypothesis testing research". On the other hand, a research in which an independent variable is not manipulated is called "non-experimental hypothesis testing research".

Experimental and Control groups: In an experimental hypothesis testing research, when a group or sample is exposed to usual conditions, it is termed as control groups, but when the group is exposed to some novel or special conditions, it is termed as experimental group. It is possible that certain studies involve only experimental groups or involve both experimental and control groups.

Treatments: The different conditions under which experimental and control groups usually exposed are referred to as "treatments".

Experiment: The process of examining the truth of a statistical hypothesis, relating to some research problem is known as experiment. The

experiments can be of two types viz, absolute experiment and comparative experiment.

Experimental units: The predetermined plots or the blocks, where different treatments are used are known as experimental units. Such experimental units must be selected (defined) very carefully.

DIFFERENT RESEARCH DESIGNS

The research designs can be categorized as i) Designs involving exploratory research studies, ii) Descriptive and diagnostic type research design studies, and iii) Research studies involving hypothesis – testing.

Design involving exploratory research

This type of study is also termed as formulative research studies. The major emphasis of such studies is the discovery of concepts (ideas) and better insights. Since, this is of exploratory nature; study design should be flexible enough to provide opportunity for considering different aspects of the problem under study. In this, the research problem is initially defined broadly, and may get transformed into one with more precise meaning. This may require changes in the research procedure for gathering data, hence inbuilt flexibility in research design is needed.

In studies relating to engineering, the experimental work may be tentative and somewhat unfocused nature initially or in the early stages of familiarization with a physical phenomena. Hence, one wants to keep an open mind, in the hope of observing some unusual or potentially useful behaviors, which might become the subject matter later for more focussed investigation. In routine work we usually look for reasons to throw out data points that don't repeat themselves (outliers); in exploratory work the "wild" point may be the basis of an important new product or process, and hence we should not be too quick to dismiss it.

For exploratory research work, the following three methods of research designs are contemplated i) survey of concerned literature, ii) experience survey, and iii) analysis of " insight – stimulating " examples.

Survey of concerned literature

The survey of literature concerning the issue is the most simple and fruitful method of formulating the research problem or developing the necessary premise. The premise or hypothesis stated by earlier researchers may be reviewed and their usefulness be evaluated as a basis for further research. Thus, the researcher should review and build upon the work already done by others, but in cases where no premise has been formulated, the task of the researcher will be to review the available material for deriving the premise from it.

Apart from the biographical survey of studies made in one's area of interest, the researcher should cross breed concepts and theories developed in different research areas to the area in which the researcher is currently working. For Example many of the concepts in biology have been successfully adopted in engineering designs, optimization etc.

Experience survey

This implies survey of people or experts who have practical experience with the problem to be studied. The object of the survey is to obtain insight into the relationships between variables and new ideas relating to the research problem. This survey may enable the researcher to define the problem more concisely and help in the formulation of research hypothesis.

Analysis of insight – stimulating examples

This is particularly suitable in issues where there is little experience to serve as guide. In this approach an intensive study of selected instances of phenomenon in which the researcher is interested will be useful.

Thus, in an exploratory research study, which leads to insights or premises, whatever method of research design outlined above is adopted, the only thing essential is that it must continue to remain flexible, so that many different facets of the problem be considered as and then they arise and come to the notice of the researcher.

Descriptive and diagnostic research studies

Descriptive research studies are those studies, which are concerned with describing the characteristics of a particular material, process, individual or a group. In diagnostic research studies, we are concerned with the frequency with which something occurs or its association with something else. Studies concerning whether certain variables are associated are examples of diagnostic research studies. Investigation which leads to characterization of materials in chemistry, polymer etc. comes under descriptive study. Most of the social research comes under this group.

From the point of view of research design, the descriptive as well as diagnostic studies share common requirements. The researcher in both the cases should define clearly what he wants to measure and must decide suitable methods for measuring it along with clear cut definition of 'population', he wants to study. The research design must make enough provision for protection against bias and must maximize reliability, with due concern for the economic completion of the research study.

In both the cases, the design must be rigid and inflexible and should focus attention on the following.

- Formulating the objective of the study (to indicate what is the study about and why it is being made)
- Designing the methods of data collection (what techniques to gather data will be adopted)
- What should be the sample size?
- Manner of collecting data (periodicity etc.)
- How to process the data collected?
- Reporting of findings.

All data collection equipments should be properly calibrated and pretested to minimize bias. The data collected should be examined for completeness, consistency, comprehensiveness and reliability. Since, sample design is involved; it should be tackled in such a fashion that samples yield accurate data / information with a minimum amount of research effort. Usually one or more forms of probability sampling like random sampling may be used. The

appropriate statistical operations, along with appropriate tests of significant should be carried out for drawing conclusions.

The difference between research design in respect of the above two types of research studies can be summarised as given below:

Design	Exploratory	Descriptive / diagnostic
Overall aspects	Flexible design to consider different aspects of the problem	Rigid design to protect against bias and should maximize reliability
Sampling	Non-probability sampling design	Probability sampling design
Statistical aspect	No pre-planned design for analysis	Pre-planned design for analysis
Observation	Unstructured instruments for data collection	Structured and well thought out instrument of data collection
Operational	No fixed decisions about the operational procedures	Prior decision about operational procedures

Hypothesis – testing research studies

Hypothesis – testing research studies, generally known as experimental studies, are these where the researcher test the hypothesis of causal relationship between variables. Such studies require procedures that will not only reduce bias and increase reliability, but should permit drawing inferences about causality. Usually experiments meet this requirement, when we talk of research design in such studies; we often mean the ‘design of experiments’.

EXPERIMENTAL DESIGNS

The study of experimental design has its origin in Agricultural research and Professor R.A. Fisher's name is associated with experimental designs. Prof. Fisher found that by dividing agricultural fields into different blocks and then by conducting experiments in each of these blocks, the information gathered and inferences drawn happens to be more reliable. Today the experimental designs are being used in researches relating to phenomena of several disciplines.

Basic principles

Professor **Fisher** has enumerated the following three principles of experimental designs.

i) The principle of replication
ii) The principle of randomization
iii) The principle of local control

The principle of Replication implies that the experiment should be repeated more than once. The experiment (treatment) is applied to many experimental units instead of one. By doing so, the statistical accuracy of the experiment is increased. Conceptually replication does not present any difficulty, but computationally it does. For example if an experiment requiring two way analysis of variance is replicated, it will then require three way analysis of variance since replication itself may be a source of variation in the data. Nevertheless, replication is introduced in order to increase the precision of a study ie, to increase the accuracy with which the main effects and interactions can be estimated.

The principle of randomization provides protection, when we conduct an experiment against the effects of extraneous factors. This principle indicates that we should design the experiment in such a way that the variations caused by extraneous factors can all be combined under the general heading 'chance'. In an experimental program involving a large number of tests, it is important to randomize the order in which the specimens are selected for testing. By this, any one of the many specimens involved in the experiment have an equal chance of

being selected for the test. The bias due to uncontrolled second order factors is reduced.

The principle of local control is another important principle of experimental design. In this, the extraneous factor, the known source of variability, is made to vary deliberately over a wide a range as necessary and this needs to be done in such a way that the variability it causes can be measured and can be eliminated from experimental error. This means that we should plan the experiment in a manner that we can perform a two way analysis of variance, in which the total variability is divided into three components attributed to treatments, the extraneous factor and experimental error. Here, the experiment is divided into homogeneous blocks, each block subdivided into parts equal to the number of treatments and treatments are randomly assigned to these parts of a block. In general blocks are the levels at which we held an extraneous factor fixed, so that we can measure its contribution to the total variability of the data by way of two-way ANOVA. Thus, through the principle of local control, we can eliminate the variability due to extraneous factor(s) from experimental error.

Important Experimental Designs

The most important benefit from statistically designed experiments is that more information per experiment will be obtained than with unplanned experiments. Second benefit is that statistical design results in an organized approach to the collection and analysis of information. Another advantage is the credibility that is given to the conclusions of an experimental programme when the variability and sources of experimental error are identified. Finally an important benefit of proper design of experiment is the ability to discover interaction between experimental variables. A major impetus to the experimental design has been given by the strong emphasis being shown in the design for quality (DFQ).

The experimental designs can be broadly devided into two categories, viz, i) Informal experimental designs and ii) Formal experimental designs. Informal experimental designs are those designs that normally use less sophisticated form of analysis based on difference in magnitudes. Formal designs or statistically

designed experiments offer relatively better control and use precise statistical procedures for analysis.

i) Informal experimental designs

a) Before – and – after without control design
b) After only with control design
c) Before – and – after with control design.

ii) Formal experimental designs.

a) Blocking designs
   - Completely randomized block (CRD)
   - Randomized block design (RBD)
   - Latin square Design (LSD)
   - Graeco – Latin square design
b) Factorial designs
   - Fractional factorial
   - Full factorial
   - Use of orthogonal arrays
c) Response surface designs.

Informal experimental designs will be covered in this unit, while formed experimental designs will be covered in the next unit after introducing some statistical concepts.

Before – and - after without control design

In this design a single test group is selected and the dependent variable is measured before the introduction of the treatment. The treatment ‘is then introduced and the dependent variable is measured again after the treatment has been introduced. The effect of the treatment is equal to the difference between after exposure to treatment and prior to treatment. The main difficulty of such a design is that with the passage of time considerable extraneous variations may be there in the treatment effect.

After-only with control design

In this design two groups (test group and control group) are selected and treatment is given to the test group only. The impact of the treatment is assessed by subtracting the value of the dependent variable in the control group from its value in the test group. The basic assumption is that the two groups are identical with respect to the behaviour of interest. If this assumption is not true, there is the possibility of extraneous variation entering into the treatment effect. This design is superior to the previous one.

Before-and-after with control design

In this design two groups are selected and the dependent variable is measured in both the groups for an identical period of time-period before the treatment. The treatment is introduced in the test group only, and the dependent variable is measured in both the groups for an identical period of time. The treatment effect is determined by finding the difference in the change in the dependent variable in the control group from the change in the dependent variable in the test group.

	Time period – I		**Time period II**
Test group	Level of dependant variable before treatment (x)	treatment →	Level of dependant variable after treatment (y)
Control group	Level of dependant variable without treatment (a)		Level of dependant variable without treatment (b)

$\therefore$ Treatment effect = [(y – x) – (b – a)]

This design is superior to both the previous cases for the simple reason that it avoids extraneous variation resulting from both the passage of time and from non-comparability of test and control groups.

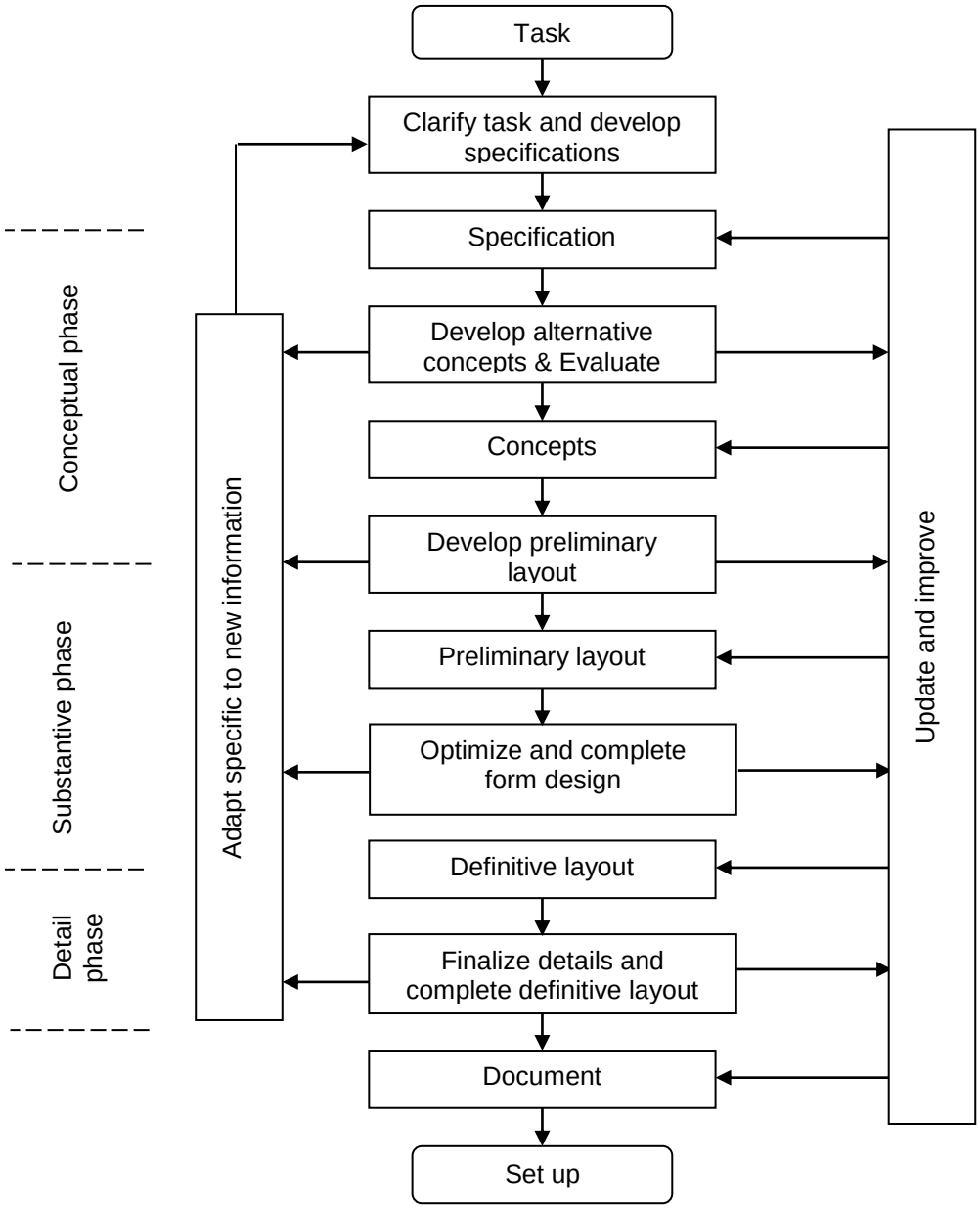
DESIGN OF EXPERIMENTAL SETUPS

Many research studies may involve development of experimental setup to meet the specific needs of the researcher. Using the experimental set up the researcher finds out the effect of independent variable(s) on the dependent variable or response variable. Of course, the experimental set up may vary depending upon the objective of the experiment. Arranging an experiment shares many common features with product design. Design of an experimental set up or developing a new product is a one-of-a-kind rather than a mass produced, since it is built for a specific purpose.

For developing an experimental set up, we can follow the approaches recommended for a new product design. The flow - chart for product design process is given in the figure. The whole process is divided into three major phases viz: conceptual, substantive and detail design.

During the conceptual phase, we think of all the different ways we could accomplish the basic task. Evaluate each approach and select those which are feasible ones and move to the next phase. In the substantive phase, we need to embody our concept in an actual physical device which can be analyzed for function and cost. Each of the approach must be carefully analyzed, improved and adopted to our specific needs and then evaluated critically against performance and cost specifications. When we reach this stage, we would have learned about the problems, so that we may return to the conceptual stage and add some new concepts to our initial list.

After completing the substantive design phase one of the alternative designs stands out as clearly superior and we may reject others. If several alternative concepts have survived to this stage, an intelligent choice among them can be made. For experimental set up more weightage be given to technical functioning. It is generally better to start with a complete list of considerations to ensure that nothing important to specific set up has been overlooked.



During conceptual phase one should avoid thinking about physical hardware and should concentrate on functional requirements. The steps that may help moving from the specification to one or more concepts include:

- abstract to identify the essential functions
- establish function structures (overall and sub functions)
- search for solution principles to fulfill sub functions
- combine solution principles to fulfill overall function
- selection of suitable combinations
- firm up concept variants

The approaches that may help to search for solution principles to fulfill sub functions are: (i) literature search (ii) known technical systems (iii) analogies (iv) previous experimental studies (v) design catalogs (vi) discussion with colleagues and experts etc.

Use of standards and codes

There are many standard practices produced and disseminated by various engineering societies and committees. These documents are a vast store house of accumulated engineering practices, which have been critically sifted and evaluated by knowledgeable experts. Annual book of standards of American Society for Testing Materials (ASTM) is a source. It deals with several subject areas such as iron and steel, non ferrous metals, petroleum products, plastics, rubber, water and environment, instrumentation, chemicals etc. There are many international standard books which the researcher can explore.

The significance and use of standards describe its practical importance and the various ways it might be applied. Standards cover instrument accuracy test apparatus and environment. They also specify number of sample specimens, detailed discussion are included on statistic tools which are useful to the researcher in sampling.

Use of readymade apparatus

An important aspect is the need for the researcher to be familiar with the existing devices, which might provide all or part of the requirements for an experimental set up needed by the researcher. This familiarity can lead to purchase of such parts or can serve as a source of idea for a part that will enable us to make ourselves. We can also combine and adopt features from existing devices in an optimum fashion for our specific needs.

The useful sources of information include standardized testing procedures, vendors of testing equipments, buyer's guides etc.

MATHEMATICAL MODELLING

Models

A model is an idealization of the part of real world that aids in the analysis of a problem. In engineering we come across several conceptual models like free body diagram, electric circuit diagram, crystal lactic etc.

A model may be either descriptive or predictive. A descriptive model enables us to understand the real-world system or phenomenon (eg: sectional model of an aircraft gas turbine). A predictive model on the other hand is used primarily in engineering design because, it helps us to both understand and predict the performance of the system.

Models can also be classified as follows: (i) Dynamic or static (ii) Deterministic or probabilistic and (iii) Iconic-analog-symbolic.

Static or Dynamic: A static model is one whose properties do not change with time; a model in which time-varying effects are considered is dynamic.

Deterministic or Probabilistic: A deterministic model describes the behaviour of a system in which the outcome of an event occurs with certainty. In many real-world situations, the outcome of an event is not known with certainty and these must be treated with probabilistic models.

Iconic model: It is a model that looks like the real thing (eg: model of an aircraft for wind tunnel test or enlarged model of a polymer molecule). These models are used to describe the static characteristics of a system. The experimental model is a functioning model, which embody the ideas of the design concept. It is as nearly like the proposed design as possible, but it may be incomplete in appearance. This model is subjected to extensive testing and modification. A prototype model is a full scale working model of the design, which is technically and usually complete.

Analog models: They behave like real systems. They are used to compare something that is unfamiliar with something that is very familiar. Unlike iconic model, analog model may look nothing like a real system.

Symbolic models: They are abstraction of the important quantifiable components of a physical system. A mathematical equation expressing the

dependence of the system output parameter on the input parameter is a common symbolic model. A symbol is a shorthand label for a class of objects, a specific object, a state of nature, or simply a number. Symbols are useful because, they are convenient, add to simplicity of explanation, and increase the generality of the situation. A symbolic model probably is the most important class of models, because it provides the greatest generality in attacking a problem.

Symbolic models are further distinguished between models that are "theoretical models", which are based on established and universally accepted laws of nature, and empirical models, which are the best approximation of mathematical representations based on existing experimental data.

MATHEMATICAL MODELS

A mathematical model, as already stated, uses mathematical language to describe a system. Mathematical models are not only used in natural sciences and engineering disciplines but also in social sciences. Physicists, engineers, computer scientist and economists use mathematical models extensively. The process of developing a mathematical model is termed as "mathematical modeling".

Mathematical models can take many forms, including but not limited to dynamical system statistical models, differential equations, or game theory models.

For example, a frequently used model in engineering is Linear-multiple regression equation.

$$y = a_0 + a_1x_1 + a_2x_2 + \ldots\ldots a_mx_m$$

In case y represents the yield strength of hot rolled ferritic- pearlitic steel based on chemical composition volume, fraction of pearlite and the grain size, than the model is

$$y = 13.29 + 5.30\ \text{Mn} + 10.21\ \text{Si} + 0.22\ (\%\ \text{of pearlite}) + 0.47\sqrt{d}\ .$$

The term‘d’ is the mean linear ferrite intercept grain size inches. The correlation co-efficient is 0.89 and 95% confidence limit for prediction of yield strength is $\pm$3800 psi.

Another example is a model which shows the steady – state relationship between the temperature and heat flow in a material is given by Fourier’s Law of Heat conduction.

$$\frac{Qh}{A} = -k_1 \frac{dT}{dx}$$

The heat transfer Qh is related to the temperature gradient in the direction of heat flow $\frac{dT}{dx}$. The negative sign indicates that there is a temperature drop in the direction of flow.

Background to modeling

Often, when engineers analyze a system to be controlled or optimized, they use a mathematical model. In analysis, engineers can build a descriptive model of the system as a hypothesis of how the system could work or try to estimate how an unforeseeable event could affect the system. Similarly, in control of a system engineers can try out different control approaches using simulation.

A mathematical model usually describes a system by a set of variables and a set of equations that establish relationships between variables. The values of variables can be practically anything; real numbers, integer numbers, boolean values or strings. The variables represent some properties of the system, for example measured system outputs often in the form of signals, timing data, counters and even occurrences (yes / no). The actual model is a set of functions that describe the relations between the different variables.

Building blocks

There are six basic groups of variables: decision variables, input variables, state variables, exogenous variables, random variables and output variables.

Since there can be too many variables of each type, the variables are generally represented by vectors.

Decision variables are also known as independent variables. Exogenous variables are sometime known as parameters or constants. The variables are not independent of each other as the state variables are dependent on the decision, input, random and exogenous variables. Further more, the output variables are dependent on the state of system (represented by state variables).

Objectives and constraints of the system and its users can be represented as functions of the output variables or state variables. The objective function will depend on the perspective of the model's user. Although, there is no limit to the number of objective functions and constraints a model can have, using or optimizing the model becomes more computationally involved.

Model classifications:

The mathematical models can be classified in the following ways.

Linear Vs non-linear: Mathematical models are composed of variables, which are abstractions of quantities of interest in the described systems and operators that act on those variables, which can be algebraic operators, functions, differential operators, etc. If all the operators in a mathematical model present linearity, the resulting mathematical model is defined as linear. A model is considered to be non-linear otherwise. Examples of linear models are,

$L = ax + by$; where 'a' and 'b' are constants and 'x, y' are variables

$L = a\frac{d^2x}{dt} + b.\frac{dx}{dt} + cx$; where a, b, c are constants

The question of linearity and non-linearity dependent on context, and linear models may have non-linear expressions in them. For example a differential equation is said to be linear if it can be written with linear differential operators, but still it can have non-linear expressions in it. A common approach to non-linear problems is linearization, but this can be problematic if one is trying to study aspects such as irreversibility, which are strongly tied to non-linearity.

Deterministic Vs probabilistic: A deterministic model is one in which every set of variable states is uniquely decided by parameters in the model and by sets of previous states of these variables. Therefore deterministic models perform the same way for a given set of initial conditions. Conversely, in a probabilistic or stochastic model, randomness is present, and variable states are not described by unique valves, but rather by probabilistic distributions.

Static Vs dynamic: A static model does not account for the element of time, while dynamic model does. Dynamic models typically represented with different equations or differential equations.

Lumped Vs Distributed parameters: If the model is homogeneous (consistent state throughout the entire system), the parameters are distributed. If the model is heterogeneous (varying state within the system), then the parameters are lumped. Distributed parameters are typically represented with partial differential equations.

A priori / subjective information: Mathematical modeling problems are often classified into 'black box or white box' models, according to how much a priori information is available of the system. A black-box model is a system of which there is no priori information available. A white-box model (glass box) is a system where all necessary information is available. Practically all systems are somewhere between the black-box and white-box models.

It is always preferable to use as much a priori information as possible to make the model more accurate. Therefore, white-box models usually considered easier, because if you have used the information correctly than the model will behave correctly.

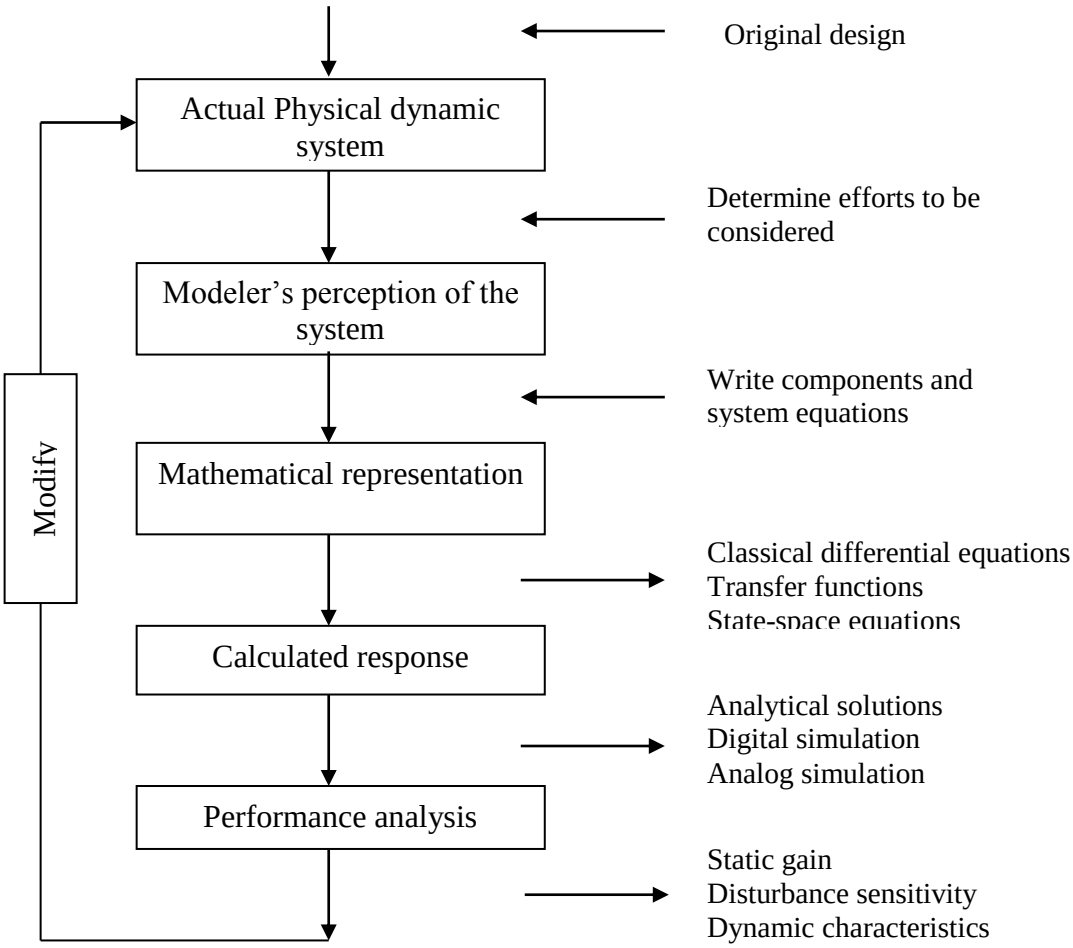
In black-box models one tries to estimate both the functional form of relations between variables and the numerical parameters in those functions. An often used approach for black-box models are neural network which usually do not make assumptions about incoming data.

Sometimes it is useful to incorporate subjective information into a mathematical model. This can be done based on intuition, experience, or expert opinion, or based on convenience of mathematical form. Bayesian statistics provides a theoretical framework for incorporating such subjectivity into a rigorous

analysis: One specifies a prior probability distribution (which can be subjective) and then update this distribution based on empirical data.

Steps in Modeling:

The figure illustrates the several stages involved in modeling of dynamic systems



The actual dynamic system of interest has all the dynamic characteristics that correspond to the exact linear or non-linear behaviour of the system. The actual system of course possesses the true response we wish to determine.

Second is the engineer's perception of the system and its dynamic characteristics. The modeler may neglect some non-linearities or higher order dynamic characteristics for simplicity; however, the actual system does not

include all of these effects ad characteristics. Hence, the engineer; perception might not truly represent the actual dynamic system.

The third is the mathematical model of the system represented by the differential equations derived from the conservation and property laws of appropriate disciplines. If the actual system is linear, the development of a suitable mathematical model is quite straight forward. However, if the system is non-linear, the mathematical model may include some approximation to simplify analysis; therefore the equations may not represent exactly the engineer's perception of real system.

In the fourth step, the system response expressed by analytic solutions to differential equations is an exact solution of the equations. Some error might exist between the solution computed by numerical and the actual solution of differential equations.

The fifth step involves the analysis of performance of the dynamic system, as expressed by specific response measures. If after the performance has been evaluated, the system does not display the desired response, the system and its model should be modified or the components of the existing model adjusted to obtain the desired output.

Model Complexity:

The model complexity involves a trade-off between simplicity and accuracy of the model. The essential idea is that among the models with roughly equal predictive power, the simplest one is the most desirable. The added complexity improves the fit of the model; it can make the model difficult to understand and work with and can also pose computational problems, including numerical instability.

It is therefore usually appropriate to make some approximations to reduce the model to a sensible size. Engineers generally accept some approximations in order to get a more robust and simple model. For example Newton's classical mechanics is an approximated model of the real world. Still, Newton's model is

quite sufficient for most ordinary life situations that is as long as particles speed are well below the speed of light, and we study macro-particles only.

Model evaluation

A crucial part of modeling process is the evaluation whether or not a given mathematical model describes a system accurately. Answering this question involves several different types of evaluation.

Any model which is not pure white-box contains some parameters that can be used to fit the model to the system it describes. If the modeling is done by neural network, the optimization of parameters is called training. In more conventional modeling through explicitly given mathematical functions, parameters are determined by curve fitting.

An easier way of model evaluation is to check whether a model fits experimental measurements or other empirical data. In models with parameters, a common approach is to split the data in to two sets; one set for training and the other set for verification. The training data are used to estimate the model parameters. Any accurate model will closely match the verification data even this set of data was not used to estimate the parameters. This practice is known as cross-validation in statistics.

Defining a metric to measure the deviation between observed and projected data is a useful tool of assessing a model fit.

It is difficult to test the validity of the general mathematical form of a model. Mathematical tools are available to test the fit of statistical models than models involving differential equations. Tools from non – parametric Statistics can sometimes be used to evaluate how well data fits known distribution.

Assessing the scope of the model, that is determining what situations the model is applicable to, can be less straight forward. If the model was constructed based on a set of data, one must determine for what system or situations the data is a typical set of data from, the model should also describe well the properties of the system between data points, which is called ‘interpolation’. If it is capable of describing events or data points outside the observed data, it is called

‘extrapolation’. As the purpose of modeling is to increase our understanding of the situation / phenomena of real work, the validity of the made rests not only on its fit to empirical observations, but also on its ability to extrapolate situations or data beyond those originally described in the model.

SIMULATION

Importance of Simulation:

Simulation refers to the application of computational models to the study and prediction of physical evens or the behaviour of engineered systems. The development of computer simulation has drawn from a deep pool of scientific, mathematical, computational and emerging knowledge and methodologies with the depth of its intellectual development and its wide range of applications, computer simulation has emerged as a powerful tool, one that promises to revolutionize the way engineering and science are conducted in the 21st century.

Computer simulation represents on extents of theoretical science in that it is based on mathematical models. Such models attempt to characterize the physical predictions or consequences of scientific theories. Simulation can be much more however. For example, it can be used to explore new theories and to design new experiments to test these theories. Simulation also provides a powerful alternative to the techniques of experimental science and observation when phenomena are not observable or when measurements are impracticable or too expensive.

Although the use of computer simulation in engineering science began over a half a century ago, only in the past decade or so have simulation theory and technology made a dramatic impact across the engineering fields. That remarkable change has come about mainly because of developments in the computational and computer sciences and the rapid advancement in computing equipment and systems.

Simulation – Based Engineering Science (SBES) is considered as a discipline that provides the scientific and mathematical basis for the simulation of engineered system. Such systems range from micro electronic devices to

automobiles, aircraft, and even infrastructures of oil fields and cities. SBES fuse the knowledge and techniques of the traditional engineering fields – electrical, civil, mechanical, chemical, aerospace, biomedical, nuclear and material science – with the knowledge and techniques of fields like computer science, mathematics and the physical and social sciences. As a result, engineers are better able to predict and optimize systems affecting almost all aspects of our lives and work, including our environment, security and safety and the products we use.

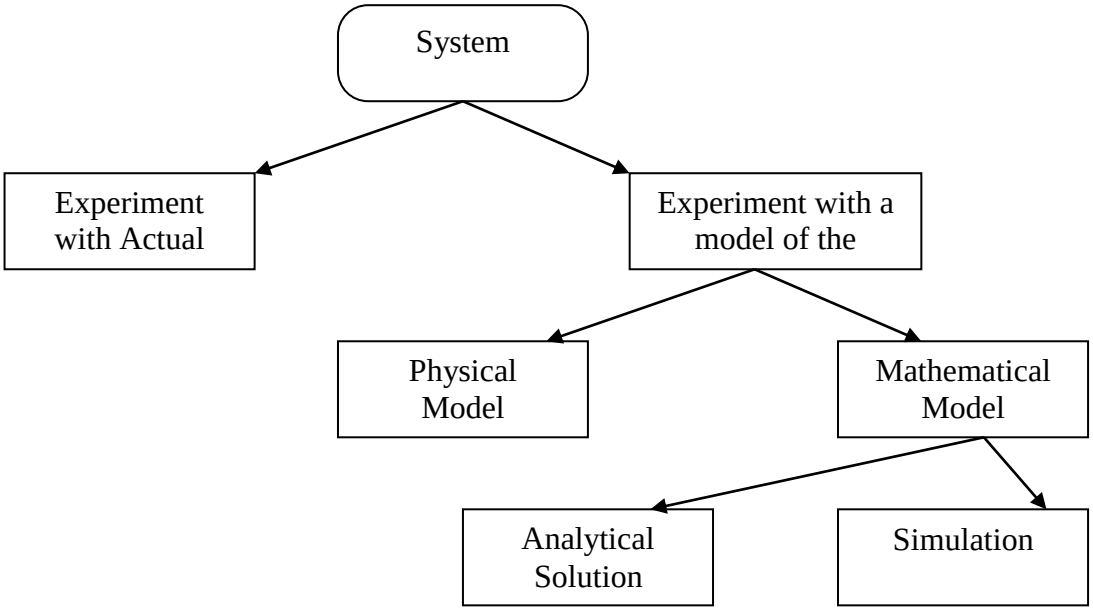
A panel appointed by USA Government in 2004, observed that SBES us an indispensable discipline to the nations continued leadership in science and Engineering. They have further added that it is central to advances in biomedicine, manufacturing, security of the homeland, micro electronics, energy and environmental sciences, advanced materials and product development. This new discipline could significantly impact every aspect of human.

System simulation:

System simulation comprises of a mathematics model of a real world situation called 'the system', in such a way that the model behaves like the actual system to events and inputs that takes place over time. It is the process of designing a model of a real system and conducting experiments with this model for the purpose of understanding the behaviour of the system or evaluating various strategies for the operation of the system.

By collecting data on system's response under various conditions, it is possible to learn how the real system may perform, without having to attempt costly experiments with the actual system. The main value of simulation is that it can be conveniently manipulated until the designed results are obtained, simulation can be performed on existing systems to identify options for problem solving or system improvements. It is also an excellent forward planning tool are acquired and installed.

Figure given below maps out different ways in which a system might be studies and the position of simulation.



Classification of simulation models:

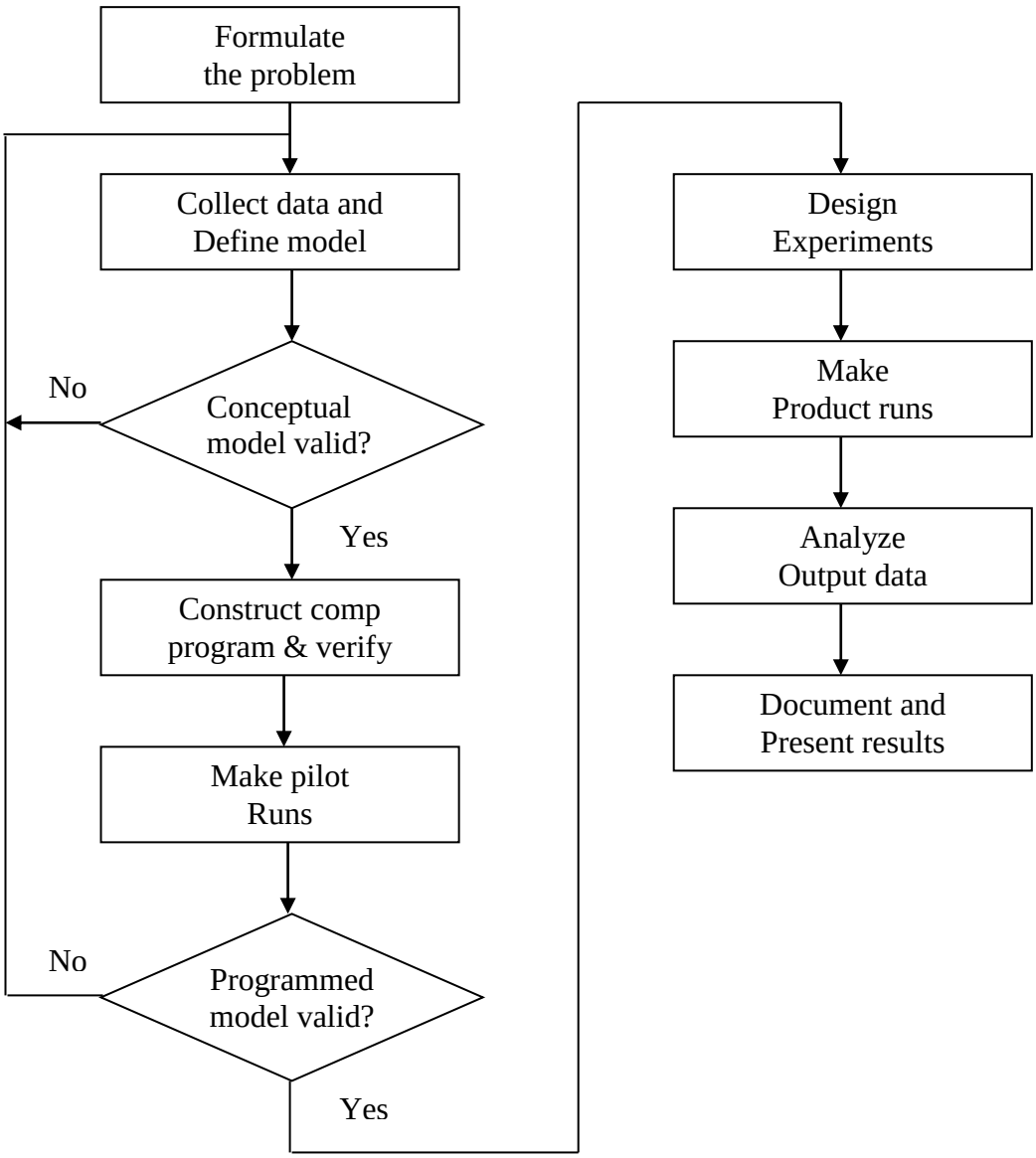
We have a mathematical model to be studied by means of simulation and then we must look for a particular tool to do this. It is useful for this purpose to classify simulation models along three different dimensions.

- **Static vs Dynamic simulation models:** A static simulation model is a representation of a system at a particular point in time. A dynamic simulation model represents a system as it evolves over time such as a conveyor system in a factory.
- **Deterministic vs Stochastic:** If a simulation study does not contain any probabilistic (ie. random) components, it is called deterministic (e.g.: a system of differential equation describing a chemical reaction) If a system modeled has at least one random component (input), this gives rise to stochastic simulation.
- **Continuous vs Discrete simulation:** Discrete event simulation concerns the modeling of a system as it evolves over time by a representation in which the state variables change instantaneously at separate points in time. Continuous simulation concerns the modeling over time of a system by a representation in which the state variable changes continuously with respect to time.

Steps in a simulation study:

There are various steps involved in a discrete event simulation study. The figure shows the steps that will compose a typical sound simulation study. It should be noted that a simulation study is not a simple sequential process. As one proceeding with the study, it may be necessary to go back to a previous step.

Steps in Simulation Study



Even though the figure self explanatory, some of the steps may need some explanation. For instance, to check conceptual model validity, we have to perform a structured Walk-though of the conceptual model using the assumptions to ensure that the model's assumptions are correct and complete. The model may be programmed in any programme language (eg: c or FORTRAN) or in simulation software (eg: Arev Automod, Extend MATLAB, etc.). The use of

simulation software not only reduces the programming time, but also the cost of the project.

To check the validity of programmed model, if the is an existing system, compare model and system performance the model results should be reviewed for correctness us sensitivity analysis to determine what model factors have significant impact on performance measures and thus, have to be modeled carefully. In Design of experiments, the length of each run, length of warm up period, number of independent simulation runs using different random number has to be taken into account.

Simulation softwares:

The simulation packages are classified as two major types viz. simulation languages and application oriented simulators. Simulation languages are general in nature, and the model development is done by writing code by the user. Simulation languages, in general provided a great deal of modeling flexibility, but are often difficult to use as it consumer time. On the other hand simulators are oriented towards a particular application, and a model can be developed by using graphics, dialog boxes and pull down manus. Simulators are easier to learn, but may not be flexible enough for some problems. In recent years vendors of simulation languages are attempting to make their software easier to use by employing a graphical model building approach.

Similarly, the vendors of simulators make their software more flexible by allowing programming in certain model locations using an internal pseudo – language thus, a general purpose simulation package can be used for any applications, but might have special features for certain ones (eg. for manufacturing, communications). On the other hand application oriented simulation package is designed to be used for a certain class of applications. Examples of some application oriented simulation package, include; manufacturing – Automod, Extend, Pro Model, QUEST, Arena: communication Networks – COMNET, OPTNET; Process Engineering – Process Model, SIMROCESS; Health care – Med Model;

SOME MAJOR APPLICATIONS

Infrastructure:

- Simulation based studies can help detail and measure the presence of biological or chemical contaminants in the air, and given the weather data, identify the likely release location and magnitude of the release. This will help to evolve an Optional response plan.
- Optimize the designs of buildings and other infrastructure elements such designs could be cite specific, interact well with natural and manuals surrounding and blend with other urban systems of which they are a part.
- Can predict the effects of effluents from existing and proposed facilities in urban and natural environments. This will increase the reliability and usefulness of environmental impact studies.
- Can be used to assist in the design and placement of air and water contaminant disposal and flood abetment.

Energy and environment:

- The energy related industrial rely on modern simulation methods to monitor the production of oil reservoirs, plan pollution remedial measures, and devise control strategies.

Material science:

- Multiscale modeling and simulation are transforming the science and technology of new material development and improvement of existing materials. The new methods enable unprecedented ability to manipulate metallic, ceramic, semiconductor and polymeric materials.
- The principle of materials design is rooted in the correlation of molecular structure with physical properties. From these correlations, models can be formulated that predict micro structural evolutions. Such models all on the researches to investigate the mechanisms underlying the critical behaviors of materials and to systematically arrive at improved designs.

Industry:

- Plays a significant role in the design of materials, manufacturing processes and products. Increasingly, simulation is replacing physical tests to ensure product reliability and quality.
- In automotive industry crash worthiness studies are made using simulation instead of using real vehicles and dummies.
- Simulation is used in perto chemical plants to develop study – state process simulation models to perform detailed optimization. As a result chemical plants are more emerge and environmentally efficient.
- Over the past two decades the integrated circuit industry has been a major player in simulation based engineering. Highly integrated easy – to – use software such as SPICE used for circuit analysis. With the clock rates move into gigahertz range, circuit theory may not be of much use. Future generation transistors such as single electron transistors, low threshold transistors and quantum computing devices will be based on new physics that links quantum mechanics and electro magnetics.

DIMENSIONAL ANALYSIS AND SIMILITUDE

Dimensional analysis

Every physical quality is a combination of mass, length, time electric charge and temperature quantity, denoted by M, L, T, Q, θ respectively. For example velocity is measured by metre / sec, which has a dimension L/T. Dimensional analysis is a tool to understand the properties of physical quantities independent of the units used to measure them. Dimensional is routinely used to check the plausibility of derived equations and computation. It is also used to form reasonable hypothesis about complex physical situations that can be tested by experiment or by more developed theories of the phenomena. It can be used to categories the types of physical quantities and units basis on their relations to or dependence on other units, or their “dimensions”’ or their lock thereof.

Dimensional analysis was developed by the 19th century French mathematician Joseph Fourier based on the idea that physical lens like **F = m.a** should be independent of units employed for measure physical variables.

Dimensional analysis is a form of symbolic modelling. It is an important aid in modelling and in experimentation an important reason for working with dimensionless variable is that it greatly reduces the number of courses that needs to be plotted and referenced in design calculations. For example if we have an experiment with a independent (design) variables, each at 6 different fixes valves, it would require 36 curves plotted on 6 graphics. If these 4 variables are expressed by two dimensionless units, then all the data can be presented in 2 graphics.

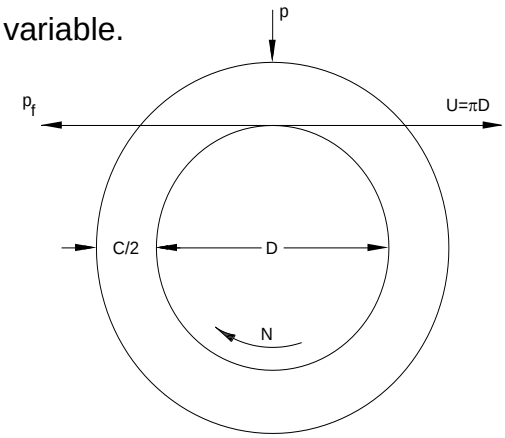
The Buckingham π theorem is usually used for arriving at dimensionless groups of variables this theorem describes how every physically meaningful equation involving "n" variables can be equivalently rewritten as an equation of (n-m) dimensionless parameters, where 'm' is the number of fundamental discussions used most importantly, it provides a method for computing there dimensionless parameters from the given variables. The use of this concept is explained by a case example.

Example:

A lubricated bearing runs at 'N' rpm with a supporting pressure 'p'. The important variables of interest are as follows:

Variable	**Unit**
Diameter (D)	L
Clearance on diameter (C)	L
Speed (N)	T^{-1}
Load supported per projected area (p)	$\frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$
Friction force per projected area (p_f)	$ML^{-1}\ T^{-2}$
Fluid velocity (μ)	$ML^{-1}\ T^{-1}$

The basic group of variables selected is D, N and p based on our feel for the problem. The remaining variables are p_f, c, and μ.We expect to find 6-3=3 dimensionless terms. We take each of the remaining variables p_f, c and μ in term and equate them to a π term multiplied by the basic variables, raised to an exponent. We then substitute the units for each variable.



$$p_f = \pi_1 (D)^a (N)^b (p)^c \quad - \quad (1)$$

$$\therefore \pi_1 = \frac{p_f}{(D)^a (N)^b (p)^c}$$

In dimensional terms

$$\pi_1 = \frac{ML^{-1}T^{-2}}{(L)^a (T^{-1})^b (ML^{-1}T^{-2})^c}$$

To find the values of exponents that makes π_1 dimensionless take each variable in now and then add the exponents.

M: 1 = 0 + 0 + c ∴ c = 1

L: -1 = a + 0 – c, a = 0

T: -2 = 0 – b – 2c, b = 0

$$\therefore p_f = \pi_1 (D)^0 (N)^0 (p)^1 = \pi_1 p$$

$$\therefore \pi_1 = {}^{p_f}/_{p} \text{ Coeff of friction}$$

$$C = \pi_2 (D)^a (N)^b (p)^c \quad - \quad (2)$$

In dimensional terms,

$$\pi_2 = \frac{L}{(L)^a (T^{-1})^b (ML^{-1}T^{-2})^c}$$

M: 0 = 0 + 0 + c ∴ c = 0

L: 1 = a + 0 – c, a = 1

T: 0 = 0 – b – 2c, b = 0

$$C = \pi_2 D \ , \quad \pi_2 = {}^{C}/_{D} \text{ clearance ratio}$$

$$\mu = \pi_3 (D)^a (N)^b (p)^c \quad - \quad (3)$$

In dimensional terms

$$\pi_3 = \frac{ML^{-1}T^{-1}}{(L)^a(T^{-1})^b(ML^{-1}T^{-2})^c}$$

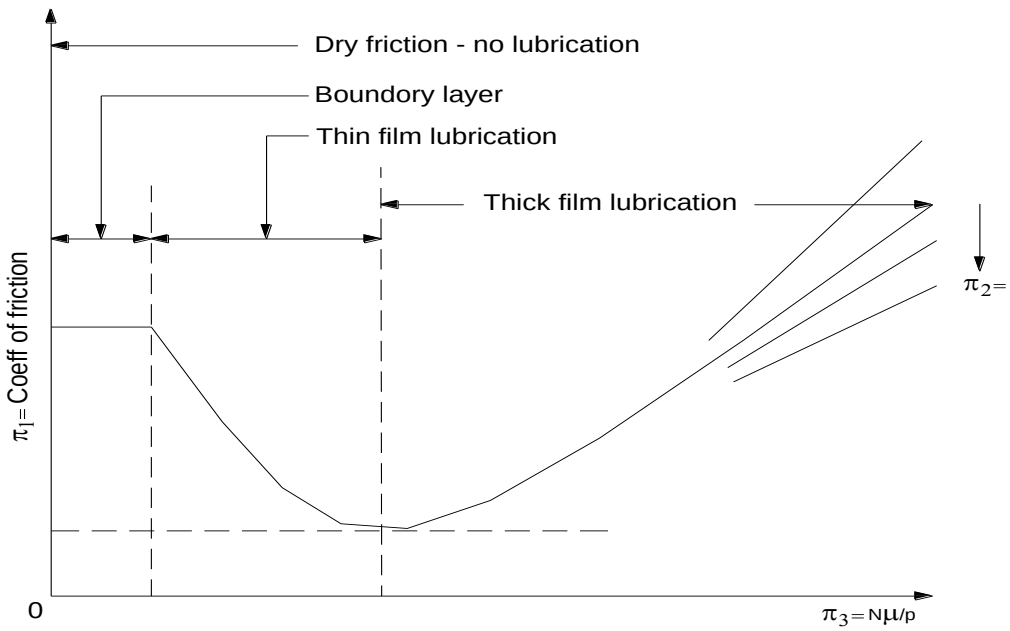
M: 1 = 0 + 0 + c $\quad\therefore\quad$ c = 1

L: -1 = a + 0 – c, $\quad$ a = 0

T: -1 = 0 – b – 2c, $\quad$ b = -1

$\mu = \pi_3 N^{-1} p \qquad \therefore \pi_3 = \frac{\mu N}{p}$

The Buckingham TT theorem has identified three dimensionless quantities. Two are simple ratio and third is a parameter that wraps a number of variables into a single term. We can call these dimensionless groups. To get a comprehensive picture of bearing performance we can plot is as groups as shown in the figure. Dimensionless variables can also be generated by matrix methods.



Dimensional analysis is very useful in applications where our goal is to reduce the number of independent variables that need to be studies some device or process. Actually the reduction is usually beneficial to both theoretical and experimental studies, which is quite common in engineering analysis. Particularly in optimization studies much benefit achieved by reducing the variables. If we

are trying to fine a combination of parameter values, which optimizes some performance criterion the optimum is found much more quickly and easily if the number of parameters is small.

Similitude:

Similitude is a concept used in the testing of engineering models. A model is said to have similitude with the real application if the two share geometric similarity, kinematic similarity and dynamic similarity, similarity and similitude are interchangeable in this control the term dynamic similitude's often used to catch – all because it implies that geometric and kinematics' similitude have already been met. Similitude's main application in hydraulic and aerospace engineering to test fluid flew conditions with scale models.

Engineering models are used to study complete fluid dynamics problems where calculations and computer simulations are not reliable. Models are usually smaller than the final design, but not always. Scale models allow testing of a design prior to building, and in many cases are a critical step in the development process. Construction of a scale model, however, must be accompanied by an analysis to determine under what conditions it is tested. While geometry may be simply scaled, other parameters such as pressure, temp, velocities and fluid type may need to be altered. Similitude is achieved when testing conditions are created such that the test results are applicable to real design.

Geometric similarity is the first type of similitude. The conditions are identity of shape, equality of corresponding analysis or area, and a constant proportionality or scale factor corresponding to linear dimensions.

Model dimension = scale factor x proto type dimension consider the example of a cylindrical bar of length 'L' that is classically deformed an amount 'δ' by an axial force 'p'. s = scale factor = $\delta_m / \delta_p = L_m / L_p$ where the subscript on refers to model and p to proto type.

The critical variables are δ, p and cross sectional area A, elastic modulus E and length L. These variables can be represented by dimensionless ratios from the relation

$$\delta = \frac{PL}{AE}$$

In case, the functional relation is not known, the dimensional ratio are developed using the procedure of dimensional analysis.

$$\pi_1 = \frac{\delta}{L} = \frac{\delta_p}{L_p} = \frac{\delta_m}{L_m}$$

$$\pi_2 = \frac{P}{EL^2} = \frac{P_p}{E_p L_p^2} = \frac{P_m}{E_m L_m^2}$$

$$\pi_3 = \frac{A}{L^2} = \frac{A_p}{L_p^2} = \frac{A_m}{L_m^2}$$

From the second dimensionless ratio, we determine how the load must be adjusted to maintain similarity.

$$\pi_2 = \frac{P_m}{E_m L_m^2} = \frac{P_p}{E_p L_p^2} \; and \; P_m = P_p \frac{E_m L_m^2}{E_p L_p^2} = P_p \frac{E_m}{E_p} \; , s^2$$

In addition to geometric similarity, we have static similarity. This implies that, under constant stress, the relative deformation will be in the some proportion as the characteristic geometric dimension.

Kinematics similarity is based on the ratio of time proportionality between corresponding events in the model and the proto type. Often time scale ratio is combined with the length – scale, ratio to express the ratio of velocities at equivalent positions in similar systems. Typical kinematics similarity ratios are,

$$Acceleration: \; a_r = \frac{a_m}{a_p} = \frac{L_m T_m^{-2}}{L_p T_p^{-2}} = L_r T_r^{-2}$$

$$Velocity: \; v_r = \frac{v_m}{v_p} = \frac{L_m T_m^{-1}}{L_p T_p^{-1}} = L_r T_r^{-1}$$

In dynamic similarity, the forces acting on the corresponding locations in the model and the proto type are in a fixed ratio. In fluid flew situation, the forces arise from inertia, viscosity, gravity, pressure, vibration, centrifugal force, surface tension etc. some dynamic similarity ratios are,

Inertial force F_i = mass x acceleration

$$= PL^3 \frac{L}{T_2} = PL^2 v^2$$

$$Viscons\ force\ F_\mu = \tau L^2 = \mu \frac{du}{dy} L^2 = \mu \frac{v}{L^2} = \mu L v$$

There are other similarities such as thermal similarity, chemical similarity, chemical similarity requires that the rate of a chemical reaction at any location in the model much be proportional to the rate of the some reaction at the corresponding time and location in the proto type thermal similarity requires that the temperature profiles in geometrically similar at corresponding times.

Dimensional analysis and similitude are necessary requirements for physical modeling and experimentation. The use of dimensional analysis provides a way of reducing the number of variable needed to describe a system. It also allows data from scale models to be properly interpreted with respect to actual system. Similitude can be used to predict the performance of a new design based on the data from an existing design. Another use of similitude and models is in validation of computer simulations with the ultimate goal of eliminating the need for physical models altogether.

Unit III

USE OF STATISTICAL TOOLS IN RESEARCH

IMPORTANCE OF STATISTICS TO RESEARCHERS

Statistics in Engineering

Engineers are finding increasing practical usefulness for some of the basic concepts of the branch of mathematics called probability and statistics. While this unit is devoted to engineering experimentation involved in research, the statistical concepts and methods discussed are also applicable to other areas. In order to give some practically useful statistical tools, the approach adopted here is to concentrate on general concepts and specific calculation method for commonly occurring problems without emphasis on proofs.

Suppose the researcher buy a carbon steel rod, the manufacturer says that modulus of elasticity (E) of the steel is 206.8 GPa, while actual testing of specimen of several rods, the E value is found varying between 205.7 GPa to 207.9 GPa. Hence, the statistics may be helpful in deciding the probability of the E value falling below a chosen value. Similarly when a specimen is prepared using machining with a value of 25mm, diameter, in actual practice when several specimens are prepared, the variation between specimens could be from 24.5mm to 25.5mm. This uncertainty will produce an associated uncertainty in results.

In the engineering experimental analysis, one can find that statistical methods have much important application such as specifying the uncertainty of measured data, planning of experiments to get the maximum amount of significant data with the least effort and expense. Moreover the hypothesis could be tested rationally to decide the probability of their being true or false.

Every measurement we make have some inaccuracy and this can be broken down into so called systematic and random or chance errors. Systematic error repeat themselves if the measurement is repeated and could be removed by calibrating the instrument and the experimental set up. Random errors do not behave in this way rather they exhibit scatter. The best that we can do is to statistically estimate what their largest values might reasonably be. We can then quote the measurement as, say 25.20 + .06 to indicate that we are, say 95% sure that the value is somewhere between 25.14 and 25.26.

Many of the statistical methods were originally developed for use in biological and social sciences, where the interrelationship between variables are not clear-cut as they often are in engineering. In many circumstances the engineers and physicists have to deal with poorly defined situations or forced to use data that have low precision. Proper application of statistical analysis can greatly help them in such situations.

Scope of Statistical Analysis

Often physicists and engineers run experiments but their variables are subject to much more precise control than those in biological and social sciences. Statistics have been found very useful, indeed necessary, in sorting out the subtle effects of various interacting variables in situations of this sort. As engineering is becoming more involved with biological and social system, the need for engineers for some statistical background increases. Even in pure technical problems, statistics has been found very helpful in giving a rational basis for reaching judgments when the interaction of variables is subtle.

There are several uses of statistical tools. Some of the important application of statistical tools in engineering include,

- At the most basic level, we are interested in the analysis of experimental data so that the results can be unequivocally described by appropriate statistical parameters.
- In statistical inference, we use statistics for making reliable decisions utilizing tests of hypothesis and confidence limits. Hypothesis tests are used to determine whether there is a significant difference between the characteristics of an observed set of data and a proposed mathematical model of the data.
- Confidence limits allow us to determine the range in which the true characteristic of a population is likely to lie.
- Analysis of variance is a test for the equality of means and / or variances of group of observations, such as the means resulting from different experimental procedures

- The statistical design of experiments helps the researcher to collect data in a more efficient and economic way that gives value of information per experiment optimally.
- Regression analysis is another tool used in experimentation to determine whether there is a relation between two or more variables and this can also be used for prediction purposes.

BASIC PROBABILITY CONCEPTS

Probability Definition

Probability is a number between 0 and 1 related to a given event. If the event is absolutely certain, its probability is 1 and if the event is impossible, its probability is zero. Three different definitions of probability are in common use, each one is appropriate for certain types of applications.

Classical definition

If an event can occur in 'N' equally likely and mutually exclusive ways, and if 'n' of these ways have an attribute 'A', than the probability P(A) of the occurrence of 'A' is defined to be n/N. This definition is applicable to study of games of chance. For example, if there are 4 aces in a deck of 52 cards, the probability of drawing on ace in one try is 4/52 = 0.777.

Frequency definition

If an experiment is carried out 'N' times and an event 'A' occurs 'n' times, then if we let N approach infinity, the limit of n/N is defined to be the probability P(A) of an event 'A'. This is the most popular definition and allows treatment of many practical problems in which classical definition could not be applied.

For example, it is neither desirable nor economical to test all the finished components in a continuous production system. Theoretically, it is not possible to study an infinite population; hence a sample size large enough to be reliable but small enough to be economical is chosen to make decision about the quality of components produced.

Subjective definition

In this the probability P (A) of a proposition “A” is a measure of the “degree of belief” one holds in the proposition. This is perhaps the broadest definition and a necessary one. Suppose we are trying to choose between two alternative strategies, each of which is likely to produce certain results, and we cannot experimentally try out each case, then one has to rely on ‘expert judgment’ to assign numerical probability in such cases. (eg: betting in horse race, dropping a bomb or not dropping on enemy target). Here again, the classical and frequency interpretation are useless and a subjective judgment is necessary.

Irrespective of the interpretation one puts on probability, once a number has been chosen, any further calculations are independent of the definition.

Probability laws

A special and precise system of language and notation is used in probability theory. Two events A and B are said to be ‘independent’ if the occurrence of one event (say A) has no effect on the probability of occurrence of the other (say B). The two events A and B are called ‘mutually exclusive’ if one of them happens, the other cannot, i.e. they have no elements in common.

Multiplicative low:

1. If A, B, C …….. are independent events then the probability that all events occur, is called joint probability, is the product of their respective probabilities.

   P (ABC…….) = P(A) P(B) P(C) ……..
2. If A and B are mutually exclusive than P (AB) = 0. (or) P (ABC…..) = 0.
3. If two events A and B are not independent, then

   [P(A and B) = P (AB) = P(A). P (B/A) = P (B) P (A/B). Where,]

   P (B/A): conditional probability of B with respect to A. that is, probability of B occurring, assuming that A has occurred.

   P (A/B: Conditional probability of A with respect B. that is, probability of A occurring, assuming that B has occurred.

Whether events are independent or not is not always obvious or clearcut in practical applications. By critically examining the situation, one has to decide

whether the events are dependent or independent. When the correct interpretation is not obvious after brief consideration, more theoretical and/or experimental studies may be helpful in reaching a decision.

Additive law:

The probability that event 'A' or 'B' occur is given by the relation,

P (A and / or B) = P (A) + P (B) – P(AB)

The events A and B need not be independent, so long as we know their joint probability

P (AB) i.e., not mutually exclusive events. If A and B are mutually exclusive (means that if one of them happens, the other cannot), then P (AB) = 0. In that case,

P (A+B) = P (A and/or B) = P (A) + P (B) (or) P (A+B) = P (A) +P (B)

This can be generalized to any number of events by a process of continued replication,

P (A and/or B and/or C) = P (A+B+C) = P(A)+P(B)+P(C) - P(AC)-P(BC)-P(ABC).

Baye's rule:

Baye's method allow us to modify a probability estimate as additional information becomes available. If B is an event, which may be possible by k mutually exclusive and exhaustive ways of Ai, i = 1,2…..k, then

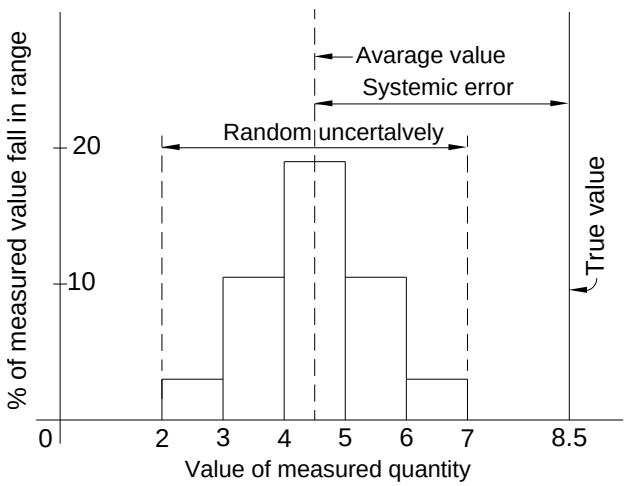
$$P\left({}^{Ai}/_{B}\right) = \frac{p(Ai)p\left({}^{B}/_{Ai}\right)}{\sum p(Ai)p\left({}^{B}/_{Ai}\right)}$$

This is known as posterior probability

PROBABILITY DISTRIBUTIONS

Errors and Samples

The act of making any type of experimental observations involve two types of errors: systematic errors (which exert a non random bias) and experimental or random errors. Systematic errors arise because of faulty control of the



environment. For instance in the case of an experimental study by properly calibrating the measuring instruments and improving the experimental setup, repetition of this type of error could be minimized.

Random errors are due to many unassignable causes and exhibit scatter as already stated. They have to be estimated. Infect the main objective of statistical analysis is to deal quantitatively with random or experimental error.

The data collected from an experiment represents a sample of the population from which it was drawn. The population in this case is collection of all possible specimens. As it is impossible to experiment with all specimens of the population, one of the main purpose of statistical analysis is to determine the best estimate of the population parameter from a randomly selected sample. While the population parameters are fixed and invariant, the parameter calculated from a sample contain random errors. Therefore, the sample provides only an estimate of the population parameter.

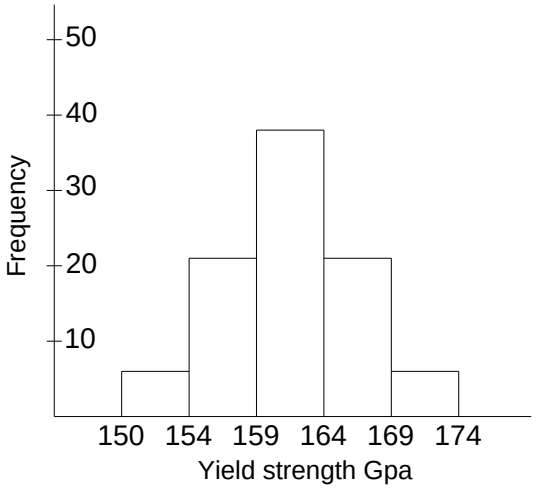
The statistical methods, therefore leads to conclusions having a given probability of being correct.

Frequency distributions

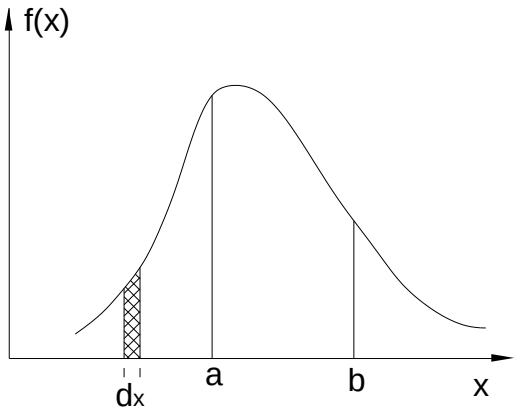
When large number of observations are made from a random sample, a method is needed to characterize the data. The most common method is to arrange the observations into a number of equal class intervals and then determine the frequency of the observations falling within each class interval. The table below shows the yield strength of a particular material after testing 100 specimens in GPa. This set of 100 data are arranged in the form of frequency tabulation. For instance out of 100 specimens, 5 specimens have value of yield strength in between 150 to 154 GPa.

Class Interval of Y. strength	Class mid points (xi)	Frequency (fi)	(xi) (fi)	Relative frequency	Cum Relative frequency	% of cum Relative frequency
150-154	152	5	760	0.05	5	5
155-159	157	18	2826	0.18	23	23
160-164	162	42	6804	0.42	65	65
165-169	167	27	4509	0.27	92	92
170-174	172	8	1376	0.08	100	100

An estimate of the frequency distribution of observations can be obtained by plotting the frequency of observations against class-interval of yields strength as a bar chart, shown in the figure. This type of bar chart is known as “histogram”. As the number of observations increases, the size of class interval can be reduced until we obtain a limiting curve which represents the frequency distribution of the sample. This smooth curve is a function of the random variable yield strength (X). This f(X) is called the probability density function. The sub range width now becomes the infinitesimal dx, but the bar area f(x) d x still has the meaning the probability of finding x in d x. Thus $P(a < x < b) = \int_a^b f(x)\,dx$. If we know the probability density f (x) for some random variable, many useful calculations can be made.

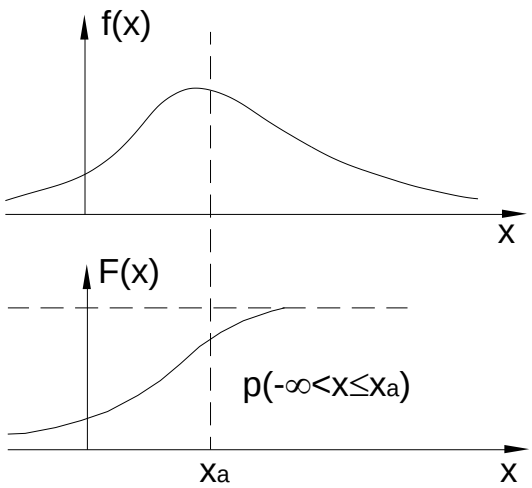


If the frequency of observations in each class interval is expressed as a percentage of total number of observations, the area under the curve, which is plotted on this basis, is equal to unity. If the curve range is from -∞ to +∞, then,



$P(-\infty < x < \infty) = \int_{-\infty}^{\infty} f(x)\,dx = 1.0$

Before inferring probabilities from the frequency distribution, we need to fit the experimental results to standard statistical distributions such as normal exponential lognormal etc.,



Another way to present the experimental data is to arrange in a cumulative manner. The presentation of

data as a cumulative distribution is preferred sometimes, because it is much less sensitive than the frequency distribution to the choice of class intervals. Thus for each probability density function f (x), there is an associated function F(X), called the cumulative distribution function denoted by.

$$f(x) = \int_{-\infty}^{x} f(x)dx$$

This is just the probability that x is less than some chosen value x_a

$$P(-\infty < x \leq x_a) = \int_{-\infty}^{x_a} f(x)\, dx$$

Measures of central tendency and dispersion

Before applying any more sophisticated statistical techniques, one always compute two simple parameters viz mean and variance.

Mean:

The arithmetic mean (AM) or average is the most common and important measure of the central value of any array of data. It is the best indicator of central tendency. If the observations are devoted by x_1, x_2….x_n, than the mean $\bar{x}$ is given by,

$$\bar{x} = \frac{\sum_{i=1}^{n} xi}{n}$$ where n is the number of observation

Variance: The most important measure of dispension of sample data is given by variance (s^2). It is calculated as

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$ where (x_i - $\bar{x}$) is the deviation of each observation from the mean $\bar{x}$ of n observations, It is a measure of the scatter. The quantity (n-1) is called the number of degrees of freedom and is equal to the number of observations minus the number of linear relations between the observations. Since the mean $\bar{x}$ represents one such relation used, the number of degrees of freedom for the variance about the mean is (n-1). The other formulas for calculating variance are

$$s^2 = \frac{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}{n(n-1)}$$ (For ungrouped data)

$$s^2 = \frac{\sum_{j=1}^{r} x_j^2 f_j - \frac{\left(\sum_{j=1}^{r} f_j x_j\right)^2}{n}}{n(n-1)}$$ (For grouped data)

x_j = midpoint of group

f_j = frequency of group

n = total no. of observations

For our example of analysing yield strength of 100 specimens, the average or mean value is,

$$\bar{x} = \frac{\sum_{j=1}^{n} x_i f_i}{\sum_{j=1}^{n} f_j} = \frac{16275}{100} = 162.75 GP_a$$

$$s^2 = \frac{\sum_{j=1}^{5} x_j^2 f_j - \frac{\left(\sum_{j=1}^{5} f_j x_j\right)^2}{100}}{100-1} = 48.67$$

It is usual practice to work with standard deviation (s) which is defined as the positive square root of variance. (absolute dispersion)

$$s = \left[\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}\right]^{1/2}$$

Sometimes, it is desirable to describe the variability relative to the average. If the absolute dispersion is the standard deviation (s) and the mean is (x), the relative dispersion is called the "coefficient of variation" of dispersion C and is given by $C_v = \left[\frac{s}{\bar{x}}\right]$ Suppose if there are two sets of data each having s = 100, but one has x = 200 and the other the x = 2000, the values for cvi = 100/200 = .5 and cv2 = 100/2000 = .05 obviously, the second data set clustered much more closely around the mean than the first one.

Range: The range is another measure of dispersion. It is simply the difference between the largest and the smallest observation $R = (X_u - X_l)$

There are two other common measures of central tendency other than average viz mode and median.

Mode: The mode is that value of observation which occur most frequently (M_o)

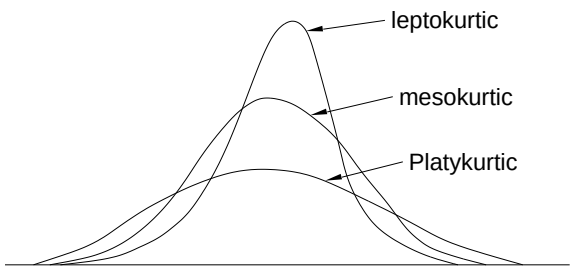
Median: The median is the middle value of a group of observations. For a frequency distribution, the median is the value that divides the area under the curve into two equal parts (M_d).

There are certain other measures depending upon the shape of the distribution which may be useful for analysis.

Skewness: Skewness indicates lack of symmetry. A distribution is said to be skewed if mean (M), median (M_d) and mode (M_o) fell at different points i.e. $(M) \neq (M_o) \neq (M_d)$.

Karl Pearson's coefficient (Sk) is one measure used to for skewness $Sk=3(M-M_o)/\sigma$ where σ = standard deviation of the distribution.

Kurtosis: Kurtosis enable us to have an idea about the flatness or peakedness of the curve. In is also known as "convexity of curve". It is measured by coefficient β_2 or its derivative δ_2. The shapes are defined as mesokurtic (neither flat nor peaked), platy kurtic (flatter than normal) and leptokurtic (more peaked)



POPULAR PROBABILITY DISTRIBUTIONS

We have defined the probability density functions in terms of a histogram based on real world data. In order to design a meaningful experiment or proper analysis of a set of data, it is necessary to make a realistic choice regarding some standard distributions to be used to fit the data gathered. The reason for using them include the generalization they provide plus the added speed and convenience of routine calculations. There are several standard distributions. Basically they are grouped under i) continuous probability distributions and ii) discrete probability distributions and iii) sampling distributions.

It is not possible to discuss all distributions and their scope of applications. To give an exposure, under the category of continuous distributions i) Normal or Gaussian ii) Weibull and iii) Exponential will be covered. Under the discrete

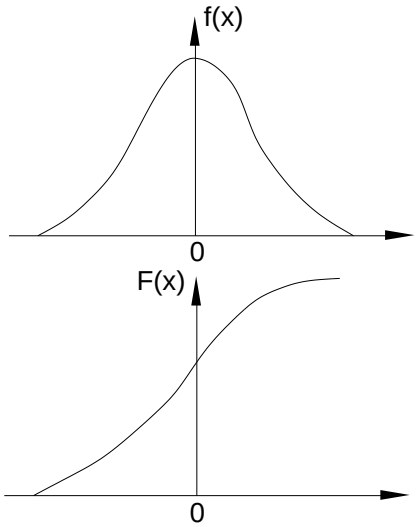
distributions, i) Binomial, ii) Poisson will be covered. In sampling, t-distribution, chi-square and f-distribution will be covered.

Gaussian or Normal distribution:

Gaussian or normal distribution is the most important and commonly used distribution to fit real world data. In addition to its goodness of fit for practical data, there are some theoretical reasons for its importance. Repeated measurements of lengths, diameter etc., distribution of yield strength, tensile strength etc. have been found to follow this curve. The shape of the curve is bell shaped and symmetrical. By definition, the probability density for the Gaussian distribution is,

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$; $-\infty<x<\infty$ where σ : standard deviation and μ : population average value (mean).

We can choose any positive value for σ and real (+, -, o) value for μ, giving sufficient adjustability to fit many sets of practical data. For any allowed values of σ and μ the curve is symmetrical about μ and has a total area of precisely 1.0. The value of σ decide the spread of the curve while μ .locates the centre. The cumulative distribution F (x) must be found by numerical integration since it cannot be integrated analytically.



In order to place all normal distributions on a common basis, the Gaussian distribution is frequently expressed in terms of the standard normal variable 'z' given by

$$z = \frac{x=\mu}{\sigma} \; giving \; f(x) = \frac{1}{\sqrt{2\pi}} exp\left(\frac{-z^2}{2}\right)$$

For this standard normal curve, μ-o and σ =1. Again the total area under the standardized curve is 1.0. Cumulative distributions $F(x) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} exp\left(-\frac{1}{2}z^2\right)$ values are available in the form of table for various values of 'Z' falling between -∞ and a specified value of 'Z'.

To use normal table for any calculations, we need to have numerical values μ and σ, the true mean and standard deviation of the "total population" (infinite size sample)

In practice, these are never available, and we must instead use their estimates $\bar{x}$ and s, calculated from the available sample. These estimates are more reliable for larger samples and we can quantify the reliability using confidence intervals.

For any variable that follows a Gaussian distribution it may be noticed that,

68.3% of valves fall within ± 1 σ of μ

95.4% of valves fall within ± 2 σ of μ

99.7% of valves fall within ± 3 σ of μ

Ex : It is established that the average response of a particular phenomena is 1.250 with standard deviation as 0.002. One experimenter conducted several repeated experiments and found the average valve varies between 1.245 and 1.255. Find what % of results non confirm.

We may assume the population mean (μ) = 1.250

and standard deviation of population (σ) = .002

using standard Gaussian variable, z, $z = \frac{x-\mu}{\sigma}$

$$z_u = \frac{1.255\text{-}1.250}{.002} = 2.5$$

$$z_L = \frac{1.245\text{-}1.250}{.002} = \text{-}2.5$$

P(1.245≤x≤1.250) = P(-2.5 ≤ z ≤ 2.5)

= 2(1.0000 – 0.9938)

= 2 x .0062 = .0124

= 1.24%

Weibull distribution

The normal or Gaussian distribution is an unbounded symmetrical distribution with long tails extending from −∞ to +∞. Many engineering random variables follow a bounded non – symmetrical distribution. Weibull distribution is one such distribution. It is used in many engineering problems because of its versatility. Originally used to describe the fatigue life of components, it is now widely used to describe the life of parts / components like ball bearings, gears

and electronic components. One of the reason for the popularity of this distribution is that the data can be conveniently plotted on the Weibull paper and the conformance of the data to Weibull distribution can be evaluated by the linearity of the cumulative distribution function in the same way as the normal distribution.

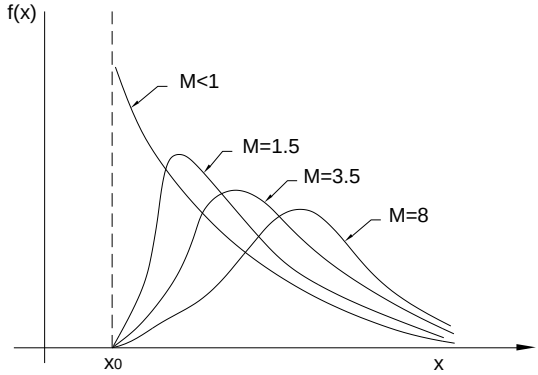
The Weibull distribution density function is given by

$$f(x) = \frac{m}{\theta}\left[\frac{x-x_o}{\theta}\right]^{m-1} exp\left[-\left(\frac{x-x_o}{\theta}\right)^{m}\right] \quad \text{where}$$

θ : characteristic or scale parameter f(x)

m : shape or slope parameter

x_0 :expected minimum value of x and is often referred as the threshold parameter



The cumulative distribution function

$$f(x) = 1 - exp\left[-\left(\frac{x-x_0}{\theta}\right)^{m}\right]$$

In the case of two parameter, $x_0 = 0$, then

$$f(x) = \frac{m}{\theta}\left(\frac{x}{\theta}\right)^{m-1} \exp\left[-\left(\frac{x}{\theta}\right)^{m}\right] \qquad f(x) = 1 - exp\left[-\left(\frac{x}{\theta}\right)^{m}\right]$$

The shape of the curve varies with change in shape parameter valves, i.e. m valves as shown in the figure.

Exponential distribution

The exponential distribution is a special case of Weibull distribution when m = 1 and $x_0 = 0$.

For Weibull distribution, the probability density function is

$$f(x) = \frac{m}{\theta}\left[\frac{x-x_0}{\theta}\right]^{m-1} exp\left[-\left(\frac{x-x_0}{\theta}\right)^{m}\right]$$

Putting m = 1 and $x_0 = 0$

$$f(x) = \frac{1}{\theta} exp\left(-\frac{x}{\theta}\right) = \frac{1}{2} e^{-x/\theta}$$

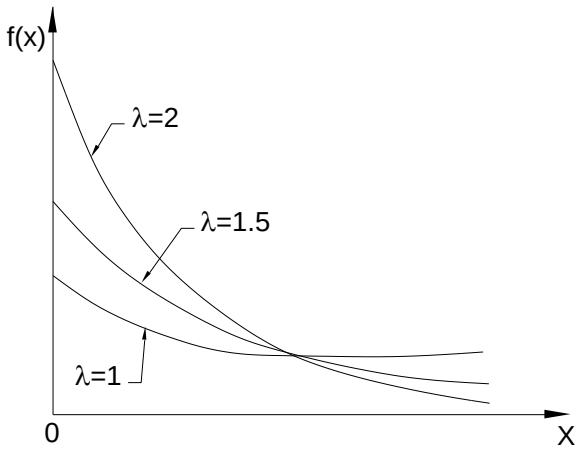
Putting $\lambda = \frac{1}{\theta}$ the exponential distribution becomes, $f(x) = \lambda\, e^{-\lambda x}$; for x>0 and $\lambda > 0$

The mean of exponential distribution $(\bar{x}) = \frac{1}{\lambda}$ and variance $(s^2) = \left(\frac{1}{\lambda}\right)^2 \therefore s = \frac{1}{\lambda}$. In this distribution both mean and standard deviation are equal i.e., X = s

The cumulative distribution function of exponential distribution is,

$$F(x) = 1 - e^{-\lambda x}; x > 0$$

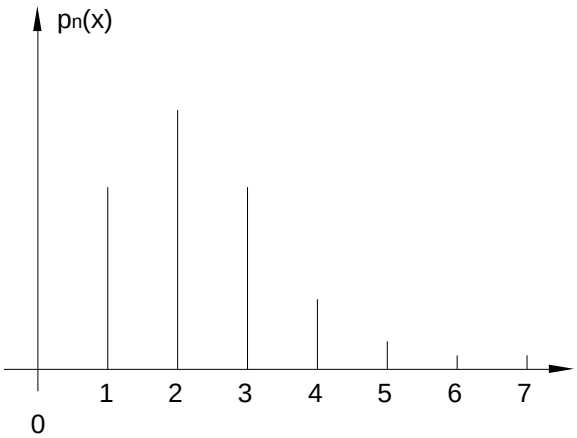
The exponential distribution plays a key role in the theory of queues and in reliability analysis. This is used to represent activity where most of the events take place in a relatively short time, while there are a few which take very long time. The service times in queuing systems, inter-arrival time of vehicles in a highway, the life of some electronic components are some of the examples where exponential distribution could be used. To model failure of complete system this distribution is better suited.



Binomial distribution

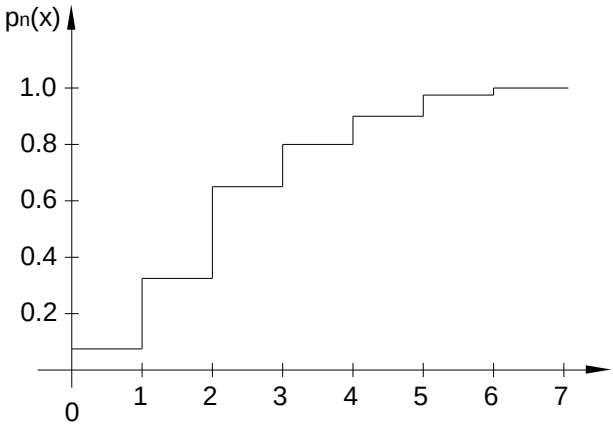
In continuous distribution like Gaussian the probability of the random variable taking on any specific value is zero. For those applications where such probabilities are non-zero, we use discrete distributions. Binomial and Poisson are most commonly used discrete distributions.

The binomial distribution applies to situations where events are judged on a yes or no basis. In quality control after inspection, a part is declared as defective or not defective. We assume the parts come from a population which has fixed percentage of good and defective items, and this percentage remains the same as we withdraw samples to be tested.



Let the occurrence of selecting a good part is ‘success’ and its non occurrence as ‘failure’. Let ‘p’ denote the probability of success and ‘q’ denote the probability of failure such that p + q = 1. The number of successes in n trails

may be 0, 1, 2, 3, r……n and obviously a chance variate. In our case, n being the number of parts in a batch being tested. The probability of 'x' successes in 'n' trails is given by $p_n(x) = p^x q^{n-x} \frac{n!}{x!(n-x)!}$

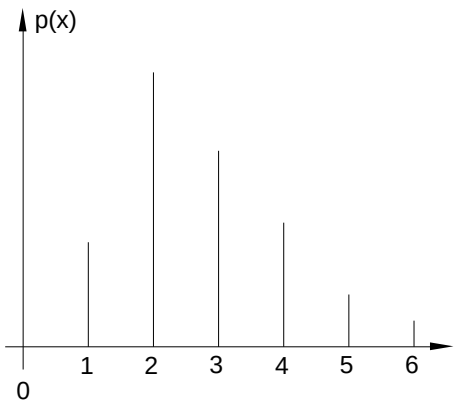


$P_n(x)$ is the probability of getting x good parts in a batch of n. It is also written as $p(x) = nC_x p^x q^{n-x}$. This is the probability mass function of Binomial distribution. The mean of the distribution $(\bar{x}) = n.p$ and variance $(\sigma^2) = n.p.q$. The probability of getting 'x' or less good parts in a batch of n is given by $C_n(x) = \sum_{x=0}^{x} p_n(x)$

In practical applications, we are often in the position of not knowing the true valve of 'p' and we try to estimate it from data on a finite size sample.

Poisson distribution

Poisson distribution is the discrete version of the exponential distribution. Poisson distribution is used to model the number of independent events that occur in a fixed amount of time or space. This distribution finds application in a wide variety of situations like number of typographical errors in a page, number of defects or flaws in a square meter of cloth, number of alpha particles emitted by a radio active substance etc.



If λ represents the average number of occurrence of an event, the probability of occurrence of x events is given by $p(x) = \frac{e^{-\lambda}\lambda^x}{x!}$; for x=0, 1, 2, 3……2τ>0. The mean and variance of Poisson distribution are equal mean $(\bar{x}) = \lambda$ and σ^2. The cumulative probability distribution is given by

$$C(x) = \sum_{i=0}^{x} e^{-\lambda}.\lambda^{i}$$

Ex: The surface defects noticed in each 10m length of cold drawn bar stock in number is given below. What is the probability of being or beyond the specification limit of 3 defects per 10m length?

Defects / 10m	Nos.	Defects / 10m	Nos.
0	40	4	1
1	9	5	1
2	4	6	1
3	2		

The average number of defects is calculated as given below

Defects	No.	Total
0	40	(0) (40) = 0
1	9	(1) (9) = 9
2	4	(2) (4) = 8
3	2	(3) (2) = 6
4	1	(4) (1) = 4
5	1	(5) (1) = 5
6	1	(6) (1) = 6
	58	**38**

The average number of defect

$\lambda=\frac{38}{58}=0.655$ $\qquad e^{-\lambda}=e^{-.655}=0.519$

$$P(0)=\frac{(0.519)(0.655)^{0}}{0!}=0.519$$

$$P(1)=\frac{(0.519)(0.655)^{1}}{1!}=0.339$$

$$P(2)=\frac{(0.519)(0.655)^{2}}{2!}=0.111$$

$$P(3)=\frac{(0.519)(0.655)^3}{3!}=0.024$$

∴.Probability of being at specification limit of 3 defects / length p (3)=0.024

Probability of being beyond the specification limit = p(x>3)

ie p(x>3) = 1-p(x≤3)

= 1 – [p(0)+p(1)+p(2)+p(3)]

= 1 – [0.519+0.339+0.111+0.024]

= 1 – 0.993

∴ P(x>3) = 0.007

Sampling distribution :

The major problem in statistics is relating the population and the samples that are drawn from it. There are two main aspects in this viz : i) What the population tell us about the behaviour of samples drawn from it, ii) What does the sample or series of samples tell us about the population from which the samples came? We discuss the first question.

Suppose we have population of 300 items, we take all possible samples of n = 10 and determine the mean of each sample $\bar{x}$. This would give population means of $\bar{x}$'s, one for each sample of 10. We call this as sample statistic. Similarly, we can have sampling distribution for s^2 or for any other sample statistic. Such distributions are called as "sampling distributions".

Distribution of sample menas.

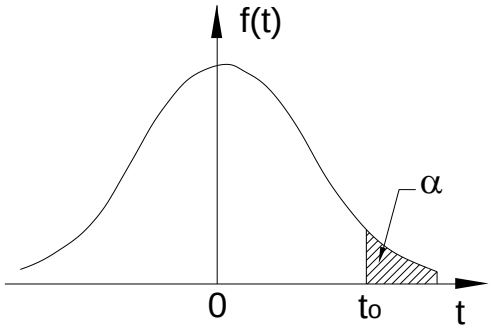
The mean of a sample provides an unbiased estimate of the mean of the population from which the sample was drawn. The sample values of the mean will be normally distributed about the population mean μ. The mean of the distribution of $\bar{x}, \mu\bar{x}$ is equal to the mean of the population of x's if the population is very large. (i.e. n > 30) The standard deviation of the sampling distribution is called the standard error of the mean and it is equal to $\sigma_{\bar{x}} = {}^{\sigma}/_{\sqrt{n}}$. We can use this to establish the standard variable 'z' , which is normally distributed with mean 'o' and standard deviation 1.

$$z=\frac{\bar{x}-\mu}{\sigma\bar{x}}=\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$$

Student's 't' distribution.

In many situations we may not know the standard deviation (σ.) of the population. In such cases, we cannot substitute 's' for 'σ' in the equation of the previous para and assume that the statistic is normally distributed unless 'n' is greater than 30. However, if x is approximately normally distributed and n < 30, the sampling distribution for the mean is the student's 't' distribution. The statistic 't' is given by $t = \frac{\bar{x}-\mu}{s/\sqrt{n}}$ for ϑ = n-1 degree of freedom.

The 't' – distribution is symmetrical about the line t = 0 and maximum ordinate is at t = 0. There is a different curve for each value ϑ. As ϑ increases, the distribution of 't' approaches normal distribution. The number of d.f of a statistic denoted by ϑ is defined as 'n' (independent observations) in the sample minus the number of population parameters which must be estimated from the samples. In the present case's' is the only statistic and hence ϑ =(n-1). The 't' distribution is available in tables where for a given ϑ and probability α, the t_0 value can be found. This gives p (t > t_o = α.



Chi-square distribution.

The distribution of sample variances s^2 from a normal population with variance σ^2 is given by chi-square (χ^2) distribution.

$$\chi^2 = \frac{\vartheta s^2}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2}$$

The chi-square statistic, χ^2 can also be computed for a given set of data by computing the observed frequency of data in each class interval and the expected frequency for the same class interval, predicted by the theoretical distribution as to $\chi^2 = \sum^k \frac{(f_o - f_e)}{f_o}$ where f_o : observed frequency and f_e : expected frequency, k : number of class intervals. If $\chi^2 = o,$ then the observed and theoretical frequencies agree exactly, whereas if $\chi^2 > 0$ they do not. The larger the value of χ^2 the greater is the discrepancy. The Chi-square

values are tabulated and available as published data. The χ^2 statistic is tabulated by d. f Vs (1-α) or the significance level. This is extensively used to establish goodness of fit test in data analysis.

F – distribution

Let $N_1(\mu_1,\sigma_1)$ and $N_2(\mu_2, \sigma_2)$ are two normally distributed populations with mean μ_1 and μ_2 and variances σ_1 and σ_2 respectively. If we take two independent samples of sizes n_1 and n_2 from the two populations respectively with variances of samples as $s^2_1$and s^2_2 The F-distribution tell us whether the two samples come from populations having equal variances. The larger two sample variance must be in the numerator. For instance S, is larger, then

$F_{\vartheta_1\vartheta_2} = {s_1^2}/{s_2^2}$ where ϑ_1=(n-1) and ϑ_2=(n_2-1) are the degrees of freedom. Simple calculation of s_1 and s_2 and comparison of s_1 and s_2 can be misleading. The values of F-distribution is available in the form of table for different values of ϑ_1 and ϑ_2 and chosen value of 'α', the level or significance.

SAMPLE DESIGN

Sampling

All items of interest to investigate such as individuals or of their attributes or results of operations, whether can be numerically specified are termed as 'population' or 'universe' in statistics. A part or a small portion selected from the population is called sample and the process of such selection is called "sampling". Sampling is resorted to when either it is impossible to enumerate the whole population or when it is too costly to enumerate in terms of money and time. To serve a useful purpose sampling should be unbiased or representative.

The aim of sampling is to get as much information as possible ideally the whole information about the population from which the sample is drawn. In particular, given the form of parent population, we would like to estimate the parameters of the population or specify the limits within which the population parameters are expected to lie with a specified degree of confidence. It is, however, to be clearly understood that the logic of the theory of sampling is the

logic of induction. That is, we pass from particular (sample) to general (population) and hence results will have to expressed in terms of probability.

Sample design.

A sample design is a definite plan for obtaining a sample from a given population. It refers to the technique or the procedure the researcher would adopt in selecting items for the sample especially the size of the sample. Sample design is determined before data are collected. There are many sample designs from which a researcher can choose. Some are relatively more precise and easy to use than others. A researcher must select / prepare a same design which should be reliable and appropriate for his research study.

While developing a sampling design, the researcher must pay attention to the following aspects.

- the type of universe, whether finite or infinite and also clearly define the set of objects which constitute the universe.

- sampling unit should be decided before selecting sample. For example the unit may be state, district or a zone etc. or specific items or individuals

- source list : known as sampling frame from which sample is to be drawn. If source list not available the researcher has to prepare it.

- size of sample : refers to the number of items to be selected from the universe to constitute a sample. While deciding the size of sample, researcher must determine the desired precision as also an acceptable confidence level for the estimate.

- Parameters of interest: In deciding sample design one must consider the question of specific population parameters which are of interest such as average or other measures.

- Budgetory constraint : Cost considerations from practical point of view have a major impact relating to not only the size but also the type.

- Sampling procedure : The researcher should decide the type of sample and the technique to be used in selecting the items, which gives smaller sampling error for given sample size and cost.

Sampling errors are the random variations in the sample estimates around the true population parameters. They occur randomly and equally likely to be on either direction, their nature happens to be compensatory type and the expected value of such errors happens to be equal to zero. Sampling error is the sum total of frame error, chance error and response error. Sampling error decreases with increase in sample size and it happens to be a smaller magnitude in the case of homogeneous population. Sampling error can be measured for a given sample design and size. The measurement of sampling error is usually called the "precision of the sampling plan". While selecting a sampling procedure, researcher must ensure that the procedure couses relatively small sampling error and helps to control the systematic bias in a better way.

Types of sample Designs

There are different types of sample designs based on two factors viz., the representation basis and the element selection technique. On the representations basis, the sample may be probability sampling or it may be non–probability sampling. Probability sampling is based on the concept of random selection whereas non–probability sampling is non-random' sampling. On element selection basis, the sample may be either unrestricted or restricted. When a sample element is drawn from a large population, then the sample so drawn is known as **"**unrestricted sample"' whereas all other forms of sampling are covered under the term "restricted sampling". Thus, the sample designs are basically two types viz., non-probability and probability sampling.

Representation basis / **Element basis**	**Probability sampling (1)**	**Non – probability sampling (2)**
1) Unrestricted sampling	Simple random sampling	Haphazard sampling or convenience sampling
2) Restricted sampling	Complex random sampling such as cluster, stratified, systematic etc.	Purposive sampling such as quota, judgment sampling.

Non-probability sampling

Non – probability sampling does not afford any basis for estimating the probability that each item in the population has a chance of being included in the sample. This sampling is also known as purposive, deliberate and judgement sampling. In this, items for the sample are selected deliberately by the researcher using his choice. In such a design, personal element has a great chance of entering into selection of the sample. The investigator may select a sample in such a way that it yield results favourable to his point of view and if that happens, the entire inquiry may get vitiated. Thus, there is a danger of bias entering into this type of sampling technique.

The experience of the investigator, his impartiality and capability of taking sound judgement may result in selecting a sample which is tolerably reliable. Sampling error in this type of sampling cannot be estimated and the element of bias, great or small, is always there. This type of sampling technique is usually adopted in small inquiries and researches by individuals due to the relative advantage of time and money inherent in this type. Quota sampling is an example of non-probability ones. Here, the interviewers are simply given quota and left their discretion. Quota samples are essentially judgement samples and inferences drawn on their basis are not amenable to statistical treatment in a formal way.

Probability Sampling

Probability sampling is also known as ‘random sampling’ or ‘chance sampling’. Under this, every item of the universe has equal chance of being included in the sample. The results obtained from random sampling can be assured in terms of probability. That is, we can measure the errors of estimation or significance of results obtained from a random sample. This method is much superior over the deliberate sampling design. In brief, the implications of a random sampling (or simple random sample) are:

- each element in the population has an equal probability of being selected into the sample and all choices are independent of one another.
- each possible sample combination has an equal probability of being chosen.

Suppose a population has a finite size of 6 numbers and we want to select sample size of 3 from it, then there are 6C3 = 20 possible distinct sample of the required size. If we choose any one of the samples in such a way that each has the probability of 1/20 of being chosen, we will then call this a random sample.

The method of obtaining a random sample can be simplified in actual practice by the use of random numbers table. Suppose we are interested in taking a sample size of 10 units from a population of 5000 units. We can number each item from 3001 to 8000. we can then select from the random number table random numbers which are not less than 3001 and not greater than 8000. A portion of that is given below.

2052	6641	3992	9792	7979	5911
3170	5624	4167	9525	1545	1396
7203	5356	1300	2693	2370	7483
3408	2769	3563	6107	6913	7691
0560	5246	1112	9025	6008	8126

If we decide to read the table numbers randomly from left to right, starting from the first row itself, we obtain the following numbers: 6641, 3992, 7979, 5911, 3170, 5624, 4167, 7203, 5356, and 7483. Which are >3000 and <8000. The units bearing the above serial numbers form the required random sample.

Complex random sample design

Probability sampling under restricted sampling techniques may result in complex random sample designs. They may also be called as “mixed sampling designs” as many such designs represent a combination of probability and non-probability procedures in selecting a sample. Some of the popular sampling designs are briefly discussed below.

Systematic Sampling

If in a sampling, we select every i th item on a list, then this constitutes systematic sampling. An element of randomness is introduced in this by using random numbers to pick up the starting unit. For instance, if a 4% sample is

desired, the first item would be selected randomly from the first 25 units, and thereafter every 25th unit would be automatically included in the sample. Though this is not a random sample in strict sense, it is often treated as a random sample.

Stratified Sampling

If a population from which a sample is to be drawn does not constitute a homogeneous group, stratified sampling technique is generally applied in order to obtain a representative sample. In this method, the population is divided into several sub-populations that are individually more homogeneous than that of the total population, and then we select items from each stratum to constitute a sample. If each strata is of different size, then proportional allocation of number of items to be selected from each strata is made. Within each strata simple random sampling approach is adopted.

Cluster Sampling

In cluster sampling the total population is divided into a number of relatively small sub-divisions which are themselves clusters of still smaller units. Then some of these clusters are randomly selected for inclusion in the overall sample. Suppose we want to estimate the proportion of defective parts in a population of 20000 units, packed in 400 cases, each combining 50 parts, we would consider the 400 cases as clusters and randomly select 'n' cases and examine all the parts in each randomly selected cases. If clusters happen to be some geographic subdivisions, then the cluster sampling is known as "Area sampling".

Multistage Sampling

This sampling is a further development of cluster sampling and is applied in big inquiries extending to a large geographical area, say the entire country. The first stage is to select a large primary sampling unit such as states in a country. Then we may select certain districts and do the survey. This represents a two stage sampling design with ultimate sampling units being cluster of districts.

Sample size determination in Simulation Experiments

When we use simulation to study a stochastic system, we represent one or more of the variables in the model by probability distributions from which we draw samples. Since these samples are randomly drawn we have some degree of imprecision in the result, which is highly influenced by the choice of sample size. As we use the information furnished by the simulation experiment as the basis for decision regarding the operation of the real system, we want this information to be as accurate and precise as possible or at least we want to know the degree of imprecision present. It is therefore essential that a statistical analysis be conducted to determine the required sample size.

The sample size may be determined in either of two ways: 1) prior to and independently of the operation of the model; ii) during the operation of the model and based upon the results generated by the model.

Prior determination approach

It is frequently possible to justify the use of certain forms of prior analysis based upon the knowledge of the model. Many forms of analysis are based upon the assumption that the responses of the model are independent and normally distributed. These assumptions are justified because of the central limit theorem from probability theory. It is usually sufficient that the response is the additive sum of a large number of contributing effects. The significance of control limit thereon lies in the fact that it permits us to use sample statistics to make inferences about population parameters without knowing the nature of population, other then we get from sample.

In the most straight forward case, where we can invoke the central limit theorem and assume no auto correlation, we can take a confidence limit approach to determine the sample size required for estimating the parameters to a specified level of precision. These parameters are population mean, standard deviation etc..

Suppose we wish to determine an estimate of $\bar{x}$ of the true population μ, such that, $P\,[(\mu-d) \leq x \leq (\mu+d)] = (1 - \propto)$ where $\propto$ is the sample mean, μ is the population mean and $(1 - \propto)$ is the probability that the interval $\mu \pm d$ contains $\bar{x}$, The problem is to determine the sample size such that the above equation holds.

If our assumptions of normality are valid, we can show that, $n=\left[\frac{\sigma z^{\alpha}/2}{d^2}\right]^2$ where $Z\frac{\propto}{2}$ is the two tailed standardized normal statistic for the probability we seek. In many cases, we must either guess at the value of σ or run a short pilot experiment. If we have some idea of what the highest and lowest possible responses of the system might be (feasible range) we can assume the range is approximately equal to 4 σ to estimate 'σ'.

If the variance is unknown, then do a pilot run or take a preliminary sample and obtain an estimate of the variance (s^2), then compute the total number of observations necessary. Thus, when σ is unknown, take a sample and estimate using 't' distribution, $n=\frac{t^2s^2}{d^2}$ where t = tabulated t value for the desired confidence level and d. f of initial sample, d = half width of desired confidence interval and s^2 is the estimated variance from sample.

Use of automatic stopping rule

In this approach the confidence intervals for output values are determined during a simulation run and then terminate the execution, when a predetermined confidence level has been reached. Thus, we can avoid the inefficiencies of runs that are either too long or too short. One of the approaches to incorporate automatic stopping rule is to run the simulation in two stages. First run a sample size 'n' use the results to estimate n* by one of the methods previously described. If n* <n, the run is over, otherwise, extend the run by (n* - n).

TESTING OF HYPOTHESES

Hypothesis: What it is?

A hypothesis may be defined as a proposition or a set of propositions set forth as an explanation for the occurrence of some specified group of phenomena. They are either asserted merely as a provisional conjecture to guide some investigation or accepted as probable in the light of established facts. Quite often a research hypothesis is a predictive statement, capable of being tested by scientific methods that relates an independent variable to some dependent variable.

Very often we make decisions about populations based upon sample information. Such decisions are known as “statistical decisions”. While attempting to make such decisions, it is usual to make assumptions or guesses about the population involved. Such assumptions, which may or may not be true are called “statistical hypotheses”, and in general they are statements about the probability distribution of the population.

Experimentation provides a method of hypothesis testing. The researcher define a problem and then propose a hypothesis. Test results either confirm or disconfirm the hypothesis. The confirmation or rejection is always stated in terms of probability rather than certainty. The ultimate purpose of experimentation is to generalize the variable relationship so that they may be applied outside the laboratory to a wider population of interest.

Null hypothesis and Alternative hypothesis

This is used in the context of statistical analysis. If we are to compare method A with method B about its superiority and if we proceed on the assumption that both the methods are equally good, then this assumption is termed as the “null hypothesis”. As against this, we may think that the method A is superior or the method B is inferior, we are then stating what is termed as “alternative hypothesis”. The null hypothesis is generally symbolized as Ho and the alternative hypothesis as H2. Suppose we want to test the hypothesis that the minimum yield strength for accepting a component made of steel is μ_o, then we would say that the null hypothesis symbolically expressed as,

$H_0 = \mu_{H0} = \mu_0$ (μ_{H0} hypothesis mean) or it could be simply written as

$H_0 : \mu = \mu_0$

If our sample data do not support the null hypothesis, we conclude that something else is true. What we conclude rejecting the null hypothesis is known as alternative hypothesis. In other words a set of alternatives to the null hypothesis is known “alternative hypothesis”. We may consider three possible alternatives as

i) $H_1: \mu \neq \mu_0$ ii) $H_1 : \mu > \mu_0$ iii) $H_1 : \mu < \mu_0$

Alternative hypothesis is usually the one which we wishes to prove and the null hypothesis is one, which we wishes to disprove or reject.

Level of significance

This is a very important concept in the context of hypothesis testing. It is always some percentage (usually 5% or 1%), which should be chosen with great care, thought and reason. If we choose a 5% significance level, then this implies that Ho will be rejected when the sampling result (observed evidence) has a less than 0.05 probability of occurring if Ho is true. In other words, the researcher is willing to take as much as 5% risk of rejecting the null hypothesis when it (H_0) happens to be true. Thus, the level of significance is the maximum value of the probability of rejecting Ho, when it is true and is usually determined in advance before testing the hypothesis.

Type I and Type II errors:

In the context of testing of hypothesis, there are two types of errors one can make. We reject Ho, when Ho is true and we may accept Ho when in fact Ho is false or not true. Rejecting Ho, when it is true is known as "Type-I error" and accepting Ho, when it is not true is called as "Type II error". In other words Type – I error means rejecting of a hypothesis which should have been accepted and Type – II error means accepting the hypothesis which should have been rejected. Type – I error is devoted by '$\propto$' (alpha) and known as $\propto$- error, also called the "level of significance of the test". The type – II error is denoted by 'β' (Beta) known as β-error. These errors are shown in the decision table.

State of nature	**Decision**	
	Accept H_o	**Reject H_o**
H_o (True)	Correct decision	Type – I Error ($\propto$)
H_o (False)	Type – II error (β)	Correct decision

If we fix Type – I error at 5% (assuming level of significance as 5%), it means that there are about 5 chances in 100 that we will reject Ho, when Ho is true. We can control Type – I error by fixing it at a lower level. But with a fixed sample size 'n' when we try to reduce Type – I error, the probability of committing Type – II error increases. Both cannot be controlled simultaneously. A trade-off

between the two types of errors has to be made taking into account, the cost or penalties attached to both type of errors.

In the example quoted earlier, if we conduct test with 10 specimens to get the mean value of yield strength for steel to be used in the construction of building (μo), Type –I error to reject Ho when infact is true would mean that steel would be shipped back to the supplier with resulting financial loss and penalty to the builder due to delay in construction. On the other hand, a Type – II error, accepting Ho, when it is false, would mean that steel of insufficient strength would be used to construct the building resulting in poor quality and failure of the building leading to loss. Thus, a Type – II error should be minimized in engineering situations.

Two tailed and one tailed tests

In the context of hypothesis testing, these two terms are quite important. The two tailed test is appropriate when the null hypothesis, Ho is some specified value and the alternative hypothesis, H1 is a value not equal to the specified value, which means, it may be more or less than the specified. Symbolically, the two tailed test is appropriate when we have

H_0 : μ = μo and H1 : μ ≠μO which may be

μ > μo (or) μ < μo

Thus, in two tailed test there are two rejection regions, one on each tail of the curve as shown in the figure.

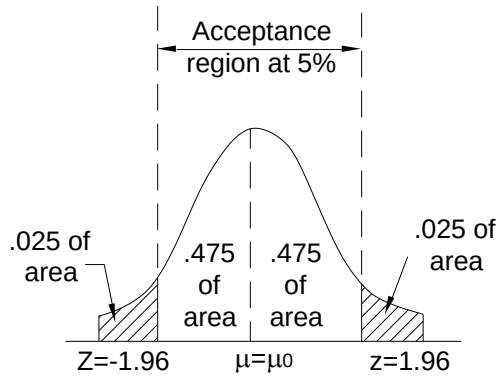


Fig. 1. Reject H0 if the sample mean $\bar{x}$ falls in either of the region. (Two tailed test at 5%)

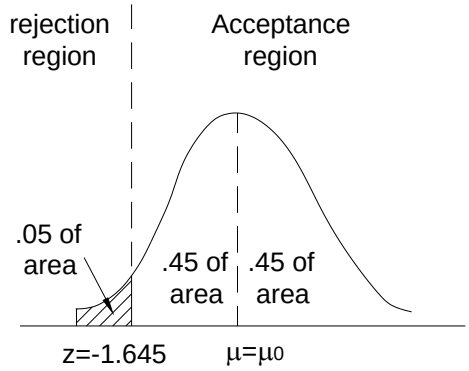


Fig. 2. Reject if the sample mean $\bar{x}$ falls in either of the region. (Two tailed test at 5%)

If the significance level is 5% and two tailed test is to be applied, the probability of the rejection area will be 0.05 (equally split on both tails) and the

acceptance region will be 0.95 of the area as shown in fig.1. If our sample mean deviate from, μo in either direction, then we shall reject the null hypothesis, but if the sample does not deviate significantly from μo, we shall accept the Ho.

One tailed test is used when the population mean is either lower or higher than some hypothesized value. For instance, if our Ho: μ=μo and H1: μ<μo, then we are interested in what is known as left tailed test (wherein there is one rejection region on the left tail) as shown in fig.2. In case our Ho : μ= μo and Ho: μ > μo, we are then interested in what is known as right – tailed test and the rejection region will be on the right tail of the curve.

It should be remembered that accepting Ho on the basis of sample information does not constitute the proof that Ho is true. We only mean that there is no statistical evidence to reject it, but we do not certainly say that Ho is true.

Steps Involved in Testing of Hypotheses.

The various steps involved in testing hypothesis include,

i. Making a formal statement of the null hypothesis Ho and alternative hypothesis H1.
ii. Selection of significance level, Generally in practice either 5 % or 1 % level is adopted for the purpose.
iii. Determination of appropriate sampling distribution
iv. Selecting a random sample and compute the sample characteristic
v. Determine the critical region for rejecting the null hypothesis, Ho.
vi. Compare the probability. If the calculated probability is $\leq \propto$ value in the case of one at ailed test and $\propto/2$ in the case of two tailed tests then reject the null hypothesis.

standard text books on statistics do give details of type of hypothesis with conditions and what test statistic to be used with the formula and criteria for refection of H_o. One can refer to them to get more information on this.

The tests of hypothesis (to test significance) can be classified as i) Parametric tests or standard tests of hypothesis; and ii) Non-parametric test or distribution free tests. The parametric tests usually assume certain properties of the parent population from which we draw samples. Assumptions like observations come from normal population, sample size is large, and assumptions about population parameters such as mean, variance etc. must hold

good before parametric tests can be used. In case, the researchers do not want to make such assumptions, they can use non-parametric methods to test hypothesis.

The important parametric tests are i) Z-test ii) t-test, iii) X^2-test and iv) F-test. All these tests are based on the assumption of normality ie., the source of data is considered to be normally distributed. In some cases it may not be normally distributed, but still we use it on account of the fact that we mostly deal with samples and that the sampling distributions closely approach normal distribution.

The Z - test is used for judging the significance of several statistical measures, particularly the mean. The relevant statistic used 'Z' is to be calculated and compared with table value (to be read from table showing area under normal curve) at a specified level of significance $\propto$. This is more frequently used by researchers.

The t-test is based on t-distribution and is appropriate to compare the significance of difference between the means of two samples in the case of small samples (ie $n < 30$). The X^2 test is based on X^2 distribution and is a parametric test used for comparing a sample variance to a theoretical population variance. F-test, which is based on F-distribution, is used to compare the variance of two independent samples. This test is also used in the context of Analysis of variance (ANOVA) for judging significance of more than two samples.

Ex: A manufacturer of switch assures a mean life of 2000 Hrs. with a standard deviation of 200 Hrs based on their testing for a large number of samples over a period of time. Later on the company made a small change in the spring. To determine whether this has changed the life of the switch, a sample of 100 switches tested to give a sample mean as 1960 Hrs and standard deviation to 180. Is there a change in the product?

Let us assume the assured life of 2000 Hrs represents the population mean μ, 200 Hrs represent the population standard deviation.

The null hypothesis, $H_o : \mu_o = 2000$

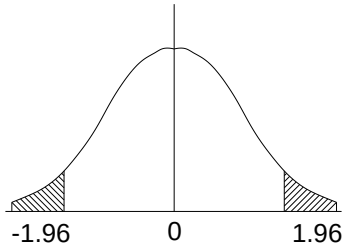
Alternate hypothesis, $H_1 : \mu \neq \mu_o$

Significance level is not given, we assume…..= 5% or .05

The sample size (n)=100.

As the sample size >30, we assume normal distribution σ is also known ie. σ = 200

As the alternative hypothesis H_1 is $\mu \neq \mu_o$, it may be more than or less than μ o and hence use two tailed test.

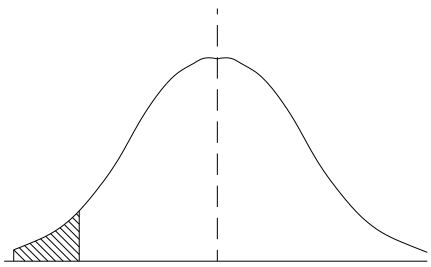


The critical region is $z < z\propto/_2 and > z\propto/_2$ from standard normal table find values corresponding to $-Z_{.025}$ and $Z_{.025}$ The value are – 1.96 and 1.96.

The calculated of value of $Z = \bar{x} - \mu = \frac{1960-2000}{180/\sqrt{100}} = \frac{-40}{18} = -2.22$

As the calculated value falls in the critical region, the null hypothesis is rejected and we conclude that the change in spring has significantly altered the life of the switch. There are 5 chances in 100 that we have rejected Ho, when it is true, we can minimize this error by selecting ∝ = 1% = 01. Now the critical region is z <$-z_{.005}$ and >$Z_{.005}$. The corresponding table values are - 2.58 and 2.58. Since the calculated Z of - 2.22 falls outside of the critical region, we accept Ho at1% significance level.

In reality we are concerned only with, whether the change in spring reduced the life of the switch; it is not a serious consequence if the life was increased. Suppose, we form alternative hypothesis, $H_1 = \mu < \mu_o$. that is $H_1 = \mu < \mu_o$ =2000. In that case we have to deal with one tailed test and the critical region for rejecting H_o is $Z<Z_{-.01}$ or Z <-2.326. Since Z=-2.22, we do not reject the null hypothesis, Ho.



Limitations of Tests of hypothesis

There are several limitations that a researcher should keep in mind while using tests of hypothesis. They include.

- It should not be used in a mechanical fashion. The test of hypothesis is not decision making itself; but they are only aids to decision making. Therefore proper interpretation of statistical evidence is important to intelligent decision making.

- The test results simply indicate whether the difference is due to fluctuations of sampling or because of other reasons.
- When the test result shows that a difference is statistically significant it simply suggests that the difference is probably not due to chance.
- Statistical inferences based on the significance tests cannot be said to be entirely correct evidences concerning the truth of the hypothesis. The error in inferences is higher with small samples compared to larger samples.

ANALYSIS OF VARIANCE

What is ANOVA?

Analysis of variance (ANOVA) is a statistical technique used when multiple sample cases are involved. The significance of difference between the means of two samples can be judged either through Z-test or t-test. But, when more than two samples are involved, the ANOVA technique enables us to perform the significance test simultaneously. It is an important tool in the analyst hand. Professor R.A. Fisher developed a very elaborate theory of ANOVA and its use in practical applications. Later on Prof. Snedocor has greatly contributed to its further development.

The essence of ANOVA is that the total amount of variation in a set of data is broken down into two types, that amount which can be contributed to chance and the amount which can be contributed to specific causes. There may be variation between samples and also within sample items. ANOVA consists in splitting the variance for analytical purposes. Using ANOVA technique, one can investigate any number of factors, which are said to influence the dependent variable. One may as well investigate the differences among various categories of within each of these factors which may have large number of possible values.

For example, the output of a process may be classified by machines and operators. From this cross classification, it could be determined whether the mean qualities of output of various operation differed significantly. Similarly, it could be determined whether the mean qualities of outputs of various machines differ significantly. This study may help the researcher as how to improve the

uniformity in quality of output whether by improving the operations or standardization of machines.

We make two estimates of population variance viz., one based on between sample (treatments) variance and the other within sample (error variance) variance. The two estimates of population variance are compared using F-test computing F-ratio as

$$F=\frac{\text{Estimate of population variance based on between sample variance}}{\text{Estimate of population variance based on within sample variance}}$$

This value of F is than compared with the F-limit for given degree of freedom (from tables). If the calculated F value is equal to or exceeds F – limit value, it is concluded that there are significant difference between the samples or treatment means.

One – way analysis of variance

In one–way analysis, a single factor is considered at different levels (treatments) and each treatment having certain number of observations. Consider a single factor at 'm' levels or treatments, each treatment containing 'n' observations. Let the jth observation of the ith treatment is denoted by x_{ij}.

Treatment	Observations	Totals	Mean
1	x_{11} x_{12} …. X_{1j} ….x_{1n}	x_1	$\bar{x}_1$
2	X_{21} x_{22} …. X_{2j} ….x_{2n}	X_2	$\bar{x}_2$
i	X_{i1} x_{i2} …. X_{ij} ….x_{in}	X_i	$\bar{x}_i$
m	X_{m1} x_{m2} …. X_{mj} ….x_{mn}	X_m	$\bar{x}_m$

$x_i = \sum_{j=1}^{n} x_{ij}$; $\bar{x_i} = {}^{x_i}/_{n}$ for i = 1,2 ….m.

The "dot" subscript notation implies summation over the subscript it replaces. Therefore the grand mean $\bar{x}$.

Is given by $\bar{x}.. = \frac{\sum_{i=1}^{m}\sum_{j=1}^{n} x_{ij}}{m.n}$

The total variability of the observations described by the total sum of squares (SST) is given by

SST = $\sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{x}..\right)^2$

The total sum of squares can be partitioned into treatment sum of squares or sum of squares between samples (SSTr) and on error sum of squares (SSE) or within sum of squares. That is,

SST = SSTr + SSE, where

SSTr = $n\sum_{i=1}^{m}(\overline{x_{i.}} - \bar{x}..)^2$ and $SSE = \sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{x}..\right)^2$

The SST r is based on the difference between the treatment means and the grand mean, while SSE is based on the difference between an observation within a treatment and the treatment mean. To get the mean square (MS) of SST, SSTr and SSE, divide each quantity by its appropriate degrees of freedom. For example, the d.f for total variance is equal to the total number of items minus 1 ie (m.n – 1). The d.f for SSTr is (m -1) and for SSE it is m (n-1). The one way ANOVA table will be as given below

Source of variation	Sum of squares	Degrees of freedom	Mean square (Ms)	F-ratio
Between treatments (m)	$n\sum_{i=1}^{m}(\bar{x}_{i.} - \bar{x}_{..})^2$	m-1	$\frac{SST_r}{(m-1)}$	$\frac{Ms\ Between}{Ms\ With\ in}$
Samples (SST_r) within the treatments (SSE)	$\sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{x}i.\right)^2$	m(n-1)	$\frac{SSE}{m(n-1)}$	
Total sum of squares (SST)	$\sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{x}..\right)^2$	(mn-1)		

To test the null hypothesis that the means of ‘m’ treatments are equal, we compare the variance based on the variation between treatment means ie MS for SSTr with the variance based on variation within this means ie MS for SSE by computing F-ratio as shown in the last column of the above table. If the F-ratio calculated is larger than the F statistic for F.., m-1, m (n-1) taken from table for level of significance ‘x’ we reject the null hypothesis and conclude the variable x is influenced by the level of treatments.

Ex: Suppose a factor is given three levels of treatment and for each treatment, there are four observations as given in the table, find whether the factor levels are significant.

Levels	observations	Total	mean
A	6 7 3 8	24	6
B	5 5 3 7	20	5
C	5 4 3 4	16	4

$x_{..} = 60, \bar{x}_{..} = \frac{60}{12} = 5$

Total sum of squares (SST) = $[(1^2 + 2^2 + 3^2 + 5^2) + (0^2 + 0^2 + (-2)^2 + 2^2 + (0^2 + 1^2 + (-2)^2 + (-1^2)] = 32$

Sum of squares within Treatments (SSE) = $[(6-6)^2 + (7-6)^2 + (3-6)^2 + (8-6)^2]$
$+ [(5-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2]$
$+ [(5-4)^2 + (4-4)^2 + (3-4)^2 + (4-4)^2] = 24$

Sum of squares between Treatments (SSTr) = $4[(6-5)^2 + (5-5)^2 + (4-5)^2] = 8$

Source of variation	**SS**	**D f**	**M.S**	**F - ratio**	**$\propto$ = 5% F - limit**
Between treatments	8	(3 -1) = 2	8/2 = 4.00	4/2.67 = 1.5	F .05, 2,9 = 4.26 (table)
Within treatments	24	3(A-1) = 9	24/9 = 2.67		
Total	32	2-1) = 11			

The calculated value of F is 1.5, which is less than the table value of 4.26 at 5% level of significance with d f ϑ_1, = 2, ϑ_2, = 9. Hence, the null hypothesis of no difference between means could not be rejected. The difference may be due to chance.

Two – way Analysis of variance

Two – way ANOVA technique is used when the data are classified on the basis of two factors. For instance in the case of process output, two factors considered could be operators and machines. Such a two way design may have repeated measurements of each factor or may not have repeated values. The technique is little different in the case of repeated measurements where we also compute interaction variations.

i) When repeated values are not there.

As we do not have repeated values, we cannot directly compute the sum of squares within treatments (SSE) as we had done in the case of one-way analysis. We have to calculate this residual or error variation by subtraction, once we have calculated the sum of squares for total variance (SST) and for variance between treatments (SSTr). ie, SSE = (SST – SSTr – SSB) where SSB is the sum of square between columns (blocks). The total sum of squares (SST) is given by

SST = SST_r + SSB + SSE

$SST = \sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{x}..\right)^2$ d.f: (m.n–1)

$SST_r = n\sum_{i=1}^{m}(x_i - \bar{x}..)^2$ d.f: (m-1)

$SSB = m\sum_{j=1}^{n}\left(x_{.j} - \bar{x}..\right)^2$ df: (n-1)

$SSE = \sum_{i=1}^{m}\sum_{j=0}^{n}\left[\left(x_{ij} - \bar{x_i}\right) - \left(x_{.j} - \bar{x}..\right)\right]^2$ df: (n-1)(m-1)

where m = number of treatments (rows) and n = numbers of columns (blocks). The AVOVA table can be set up as below:

Source of variation	**Sum of squares**	**d f**	**Mean square (MS)**	**F - ratio**
Between block (columns) SSB	$m\sum_{j=1}^{n}\left(x_{.j} - \bar{x}..\right)^2$	(n – 1)	$\frac{SSB}{(n-1)}$	$\frac{MS\ of\ SSB}{Ms\ of\ SSE}$
Between treatment (rows) SSTr	$n\sum_{i=1}^{m}(x_{i.} - \bar{x}..)^2$	(m – 1)	$\frac{SSTr}{(m-1)}$	$\frac{MS\ of\ SSTr}{Ms\ of\ SSE}$
Residual error SSE	SST-SSTr-SSB	(n -1) (m -1)	$\frac{SSE}{(n-1)(m-1)}$	
Total SST	$\sum_{i=1}^{m}\sum_{j=1}^{n}\left(x_{ij} - \bar{x}..\right)^2$	(m. n – 1)		

The MS residual or residual variance provides the basis for the F – ratios. The MS residual is always due to the fluctuations in sampling and hence serves as the basis for significance test. Both the F – ratios are compared with their corresponding table values for a given d f and at a specified level of significance. As usual if it is found that the calculated F – ratio concerning variation between columns (blocks) is equal to or greater than its value, then the difference between

block means is considered significant. Similarly, the F – ratio concerning variation between treatment means (rows) can be interpreted.

ii) When repeated value are there

In the case of two – way design with replication we can obtain a separate independent measure of inherent or smallest variations. By introducing replication, we are able to separate out the variation due to interaction effects from that due to random error. An interaction effect is present if the observed values for different levels of one factor are altered by the presence of other factor. The sum of squares due to error within each cell (sample for each treatment) is determined by first determining the subtotal sum of squares (SSS) and then sum of square within cells computed by subtracting SSS from SST. Ie, SSWC = (SST – SSS). Finally the interaction sum of squares is given by,

SSI = [SSS – (SSTr + SSB)]

Suppose, the number of columns (blocks) is n, and the number of treatments 'm', and within each treatment the number of replications is 'r', then,

$SST = \sum x_{ij}^2 - \frac{T_{..}^2}{m.n.r}$; where T is the cell subtotal and T.. is the grand total

Treatment sum of squares (between rows) is,

$SST_r = \sum \frac{Ti_{.}^2}{n.r} - \frac{T_{..}^2}{m.n.r}$; where $T_{.i}$ is the sum of column total of cell totals

The block sum of squares (between columns) is,

$SSB = \sum \frac{T_{.j}^2}{m.r} - \frac{T_{..}^2}{m.n.r}$;Where $T_{.j}$ is the sum of each column

$SSB = \sum \frac{T^2}{r} - \frac{T_{..}^2}{m.n.r}$

Sum of squares within cells,

SSWC = [SST – SSS]

The interaction sum of squares is given by

SSI = SSS – SSTr – SSB.

Once the computations are made, the results could be in the ANOVA table as given below.

Source of variation	S S	D f	M. S.	F - ratio	5% F - limit
Between blocks (columns)	SSB	(n – 1)	$\frac{SSB}{(n-1)}$	$\frac{MS\ of\ SSB}{MS\ of\ SSWe}$	
Between treatments	SSTr	(m – 1)	$\frac{SSTr}{(m-1)}$	$\frac{MS\ of\ SSTr}{MS\ of\ SSWe}$	
Within cells (error)	SSWc	m. n (r – 1)	$\frac{SSWe}{m.n(r-1)}$	–	
Interaction	SSI	(m-1) (n-1)	$\frac{SSI}{(m-1)(n-1)}$	$\frac{MS\ of\ SSI}{MS\ of\ SSWe}$	
Total		(m. n. r – 1)	–		

To check whether there is a significant interaction the two factors, the F – ratio for interaction is checked against the tabulated value.

DESIGN OF EXPERIMENTS

Importance of Experimental Design

The experimental studies generally provide a logical and systematic method for answering questions "what will happen if this is done, when certain variables are carefully controlled or manipulated?" Infact deliberate manipulation is a part of the experimental method. In an experiment, the researcher measures the effect of an experiment intentionally. Experimentation not only provide a method for hypothesis testing but also permit drawing of inferences about causality. As a result of experiments, relationship between data and the unknown in the universe is established.

The greatest benefit can be gained from statistical analysis, when experiments are planned in advance so that data are taken in a way that will provide the most unbiased and precise results commensurate with the desired expenditure and time spent. The major benefit from statistically designed experiment are,

- more information per experiment will be obtained than with unplanned experimentation.
- provides an organized approach to the collection and analysis of information

- credibility of the conclusions of experimental programme since the variability and sources of experimental error are made clear by statistical analysis.
- Ability to discover interactions between experimental variables.

The experimental design approaches are assuming greater significance due to the strong emphasis shown to improve quality. The technique developed in Japan by Genichi Taguchi employ experimental design methods in a unique way to produce robust designs.

Randomization

If an experimental programme involve large number of tests, it is necessary to randomize the order in which the specimens or combination of factors are selected for testing. Thus, by randomization any one of the many specimen involved in the experiment has an equal chance of being selected for a given test. By this approach, bias due to uncontrolled second order variables is minimized. For example in any extended testing programme errors can arise over a period of time owing to subtle changes in the characteristics of testing equipments or in the proficiency of the operator of the test. Similarly, when test specimens are prepared from a large forgings or ingots, the possibility of variation of properties in the forgings must be considered.

Variables and factors

Any experimental investigation starts with preliminary familiarization of experiments. This is more aimed at defining the problem. Once the goals of investigation are well defined, the large number of variables which are possible should be reduced a few important ones. The statistical design will be very useful. As the experiment moves to optimization stage, the number of variables should be reduced further. Hence, again statistical design provide effective means.

The output of an experiment run is the data termed as response variables. Response variables can be classified as quantitative, qualitative or quantal. A quantitative response, which is measured by a continuous scale, is the most

common and easiest to work with in statistical analysis. Qualitative responses like luster, odour etc. can be ranked on an ordinal scale, eg: worst (o) to best (1). Quantal or binary response produce one or two values, go or no-go ie. Pass or fail.

Factors are experimental variable that are supposed to be controlled by the experimenter. An important part of planning of experimental program is identifying the significant variables that affect the response and deciding how to exploit them in the experiment. Factors may be independent in the sense that the level of one effect factor is independent of the level of other factors. However, two or more factors may interest with one another ie. The effect on the response of one variable depends upon the levels of other variables.

In addition to primary variables that are under the control of experimenter, there are other variables which may not be strictly under experimenter's control. The use of randomization of test runs remove this unconscious bias. Also, the use of experimental blocks produce relatively homogeneous test conditions and minimize this. When an experiment is run with blocking, the effect of background variables is removed from the experimental error.

Planning of experiments

We should realize that not all primary factors may be capable of variation with equal facility. Complete randomization may also be impractical often the final experiment is a compromise between information that can be obtained and the cost involved in that. In developing the experimental design, the physical reality gets precedence over strict adherence to statistics.

It is important to make some initial estimate of the overall repeatability before embarking on major experimental test programme. The experience of previous experiments may also be useful in this. In case the pilot experiments indicate large variability in response then primary variables may not have been properly identified. It is not necessary to conduct statistically designed experiments all the way through completion. Infact there are advantages in conducting experiments in stages, taking into account the advantages of information gathered in early results.

The statistically designed experiments may be classified as under.

i) Blocking Designs

The Blocking designs use blocking techniques to remove the effect of background variables from the experimental error. The common type of blocking designs include,

Randomized block and Balanced incomplete block	: Remove the effect of singly extraneous variables
Latin square and Youden square	: Remove the effect of two extraneous variables
Gracco - Latin square and Hyper Latin square	: Remove the effect of three or more extraneous variables.

ii) Factorial designs

Factorial designs are experiments in which the levels of each factor is combined with the levels of all other factors to conduct experiment simultaneously. This is an important class of experiment. In factorial we have a) Full factorial designs b) Fractional factorial designs and c) Use of orthogonal arrays.

iii) Response Surface designs

The response surface designs are used to develop empirical functional relation between the factors (independent variables) and the response (output variables).

Full factorial Designs

A full factorial design is one, in which we control several factors and investigate their effects at each of two or more levels. The experimental design consist of making an observation at each of all possible combinations that can be formed for the different levels of factors. Each different combination is called a “treatment combination”. The approach in a factorial experiment is much different than in traditional experiment, in which all factors but one are held constant. The simplest and most common type of factorial design is one that uses two levels, i.e., 2^n factorial design.

To illustrate the point, let us consider three factors A, B, C each at two levels, one at low level indicated by a (-) sign and the other at higher level by a(+) sign. We are interested in the result of the experiment due to change in factor 'A' alone, B-alone and C-alone. These three potential output changes are called "main effects". There are also first order effects due to joint change in factors A and B, A and C and B and C. These interactions are denoted by AB, AC and BC respectively. A second order interaction effect due to simultaneous changes in A, B and C is denoted by ABC. This experiment is a three factor 2^3 design.

The main effect of a factor is the change in response produced by a change in the level of the factor.

Main effect of $x_i = \frac{\sum responses\ at\ high\ x_i - \sum responses\ at\ low\ x_i}{Half\ the\ number\ of\ runs\ in\ experiment}$

Thus, the main effect of factor X_B is,

Main effect of $x_B = \frac{(Y_b+Y_{ab}+Y_{bc}+Y_{abc})-(Y_1+Y_a+Y_c+Y_{ac})}{4}$

Similarly, other main effects X_A, X_c, can be calculated.

The effect for the interaction of B & C is giaven by,

BC interaction = $\frac{(Y_1+Y_a+Y_{bc}+Y_{abc})-(Y_b+Y_{ab}+Y_c+Y_{bc})}{4}$

Using Yate's notation, the three factor 2^3 factorial design table will be as given below.

Yate's std order	**Run No.**	**Level of Factors**						
		X_A	**X_B**	**X_c**	**$X_A X_B$**	**$X_A X_c$**	**$X_B X_c$**	**$X_A B_c$**
1	1	–	–	–	+	+	+	–
d	2	+	–	–	–	–	+	+
b	3	–	+	–	–	+	–	+
ab*	4	+	+	–	+	–	–	–
c	5	–	–	+	+	–	–	+
ac*	6	+	–	+	–	+	–	–
bc*	7	–	+	+	–	–	+	–
abc@	8	+	+	+	+	+	+	+

*: First order interaction @: Second order interaction.

Ex: A heat treatment involves three processes A, B, C, each treatment set at two levels (a high and low). The tensile strength is measured in KP_a (is, response 'Y') as indicated in the table below.

Treatment	Response	X_A	X_B	X_C	X_{AB}	X_{AC}	X_{BC}	X_{ABC}
(1)	1041	-	-	-	+	+	+	-
a	1014	+	-	-	-	-	+	+
b	1014	-	+	-	-	+	-	+
ab	696	+	+	-	+	-	-	-
c	1255	-	-	+	+	-	-	+
ac	1096	+	-	+	-	+	-	-
bc	1193	-	+	+	-	-	+	-
abc	1089	+	+	+	+	+	+	+

The main effect of

A = ¼ [(1014 + 696 + 1096 + 1089) – (1041 + 1014 + 1255 + 1193)]

= ¼ [3895 – 4501] = – 606/4 = – 152.0.

The minus sign indicates that the lower level of factor 'A' resulted in higher values of the response.

To test the significance of the effect of factor 'A', we can perform a paired t-test, by comparing the mean value of factor A at its high level within mean at low level.

$\bar{X}_L = 1126$

$\bar{X}_H = 974$

$\bar{d} = \bar{X}_L - \bar{X}_H = 1126 - 975 = 152$

$Also\ \bar{d} = \frac{\sum d}{4} = \frac{608}{4} = 152$

$\bar{X}_L = 1126, \bar{X}_H = 974, \sum d = 608$

Low A	High A	Difference d	Deviation From d
1041	1014	27	-125
1014	696	318	166
1255	1096	159	7
1193	1089	104	-48

$$S^2 = \frac{\sum(deviation)^2}{(n-1)} = \frac{(-125)^2 + (166)^2 + (7)^2 + (-48)^2}{(4-1)}$$

$S^2 = 15178$; $\therefore S = 15178 = 123.2$

we assume Null hypothesis as, there is no difference in mean due to factor – A.
ie, Ho :μ = o and H1 : μ ≠ o

The t – statistic can be obtained from $t = \frac{\bar{d}}{s\sqrt{n}} = \frac{152}{123.2/\sqrt{4}} = 2.47$

∴ Calculated t = 2.47; ϑ = 4 – 1 = 3

For a two tailed test, $t_{0.10}$ = 2.35. Therefore, the null hypothesis is rejected at a 10% significance level. Similarly the pained t – test can be used to check the significance of interaction effects. The precision of statistical analysis can be improved by replication.

Fractional factorial design

Because of cost and time limitations, we may wish to investigate an experimental model with less than full factorial design. One type of partial layout is called fractional factorial design. This type is usually used when four or more factors are considered. The number of treatment combinations in a factorial design increases with increase in number of factors. If n = 4, with 2 levels, the number of combinations will be, 2^4 = 16, but if n = 8, 2^8 = 256 and so on. If the number of main effects will be 'n', the first order interaction will be $n(n-1)/2$, second order will be $\frac{n(n-1)(n-2)}{(2)(3)}$

Considering a 2^6 factorial experiment, there are six main effects: since n = 6;
First order interaction: n (n-1) / 2 = 15,
Second order interaction: n (n-1) (n-2) / (2)(3) = 20;
Third order interaction: n (n-1) (n-2) (n-3) / (2)(3)(4)(5) = 15:
Fourth order interaction: n (n-1) (n-2) (n-3) (n-4) / (2)(3)(4)(5) = 6;
Fifth order interaction: = 1 and sixth order interaction = 1.

If we neglect higher order interaction effects, we can design an experiment that is some fraction of the total factorial design. For example, if we consider one half replicate of 2^3 factorial design, then N = r (2^n) – ½ (2^3) = 4. We could split the treatment contributions into two blocks of the previous example to study the main effects of A, B, & C.

Experiments using orthogonal arrays

In full factorial experiments, we conduct experiments under all combinations of factor levels. For example, if an experiment consists of three factors at four levels each, we need to conduct $4^3 = 64$ experiments to try all combinations. Obcourse, the interaction effects could also be studied in this. Fortunately, in most practical situations, the interaction effect is significantly less or absent, and additive models provide excellent approximation. In this context, use of the concept of orthogonal arrays provide a powerful tool for frugally exploring relations that involve many factors.

Designed experiments are matrix experiments, which consist of a set of experiments where we change the settings of various input parameters (factors) from one experiment to another. Orthogonal arrays are special matrices which allow the effect of several parameters to be determined efficiently. Dr. Genichi Taguchi introduced the concepts of Robust design to find optimum values of design parameters or process variables wherein he made the use of orthogonal arrays extensively.

As the name suggests, the columns of the matrix of the array are mutually orthogonal. Here, the orthogonality implies that for any pair of columns, all combination of factor levels occur and that they occur equal number of times. This is called the balancing property and its means orthogonality. In order to preserve the benefit of using an orthogonal array, it is important that all experiments in the matrix be performed. If experiments corresponding to one or more rows are not conducted, or if their data are missing or erroneous the balancing property and, hence the orthogonality is lost. In some situations, incomplete matrix experiments can give useful results, but analysis of such experiments is complicated.

Taguchi has tabulated 18 basic orthogonal arrays, which are called **"Standard orthogonal arrays",** which are available in many standard texts. The process of fitting an orthogonal array to a specific project has been made easy by the standard orthogonal arrays and the graphical tool called '**linear graphs'** developed by Taguchi to represent interactions between pairs of columns in the

array. Before constructing an orthogonal array, one must define the following. i) Number of factors to be studied ii) Number of levels of each factor iii) Specific 2 factor interactions to be estimated iv) Specific difficulties if any in the experiment.

The first step in constructing an orthogonal array to fit a specific case study is to count the total degrees of freedom that tells the minimum number of experiments that should be performed to study the main effects of all control factors and chosen interactions. In many problems one of the 18 standard tabulated orthogonal arrays can be directly used to plan the matrix experiment. Orthogonality is not lost by keeping empty one or more columns of the array. In orthogonal array experimentation, one may choose not to estimate any interactions among control factors, but in situations where it is needed linear graphs makes it easy to plan.

The first four standard orthogonal arrays and the L9 (3^4) Orthogonal array of Taguchi is given below for reference.

Standard Orthogonal Arrays (First four)

Orthogonal array	**Number of rows**	**Maximum No. of factors**	**Maximum No. of columns of these levels**			
			2	**3**	**4**	**5**
L4	4	3	3	-	-	-
L8	8	7	7	-	-	-
L9	9	4	-	4	-	-
L12	12	11	11	-	-	-

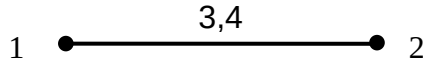
The array L_9 (3^4) has a rows and four 3 level columns. Similarly array L_{18} (2^1 3^7) has 18 rows, one 2 level Column and seven 3 level columns. The Number of rows of an array represents the number of experiments.

L_9 (3^4) orthogonal array

Expt. No.	Columns			
	1	**2**	**3**	**4**
1	1	1	1	1
2	1	2	2	2
3	1	3	3	3
4	2	1	2	3
5	2	2	3	1
6	2	3	1	2
7	3	1	3	2
8	3	2	1	3
9	3	3	2	1

Linear graphs represent the interaction graphically. It make it easy to assign factors and array. The columns of an array represented by 'data' and the connecting line, the interaction of two columns represented by dots is contained in (confounded with) the column represented by the line.

Linear graphs for L_9



The first step in constructing an orthogonal array for a problem is to decide the total degrees of freedom, which tells the minimum number of experiments to be performed to study all chosen factors. Suppose, in a problem we have four factors, each at three levels, the total number of d.f is 9 (that is, for overall mean = 1, for 4 factors 4 (3-1) = 8. ∴ total is 8 + 1 = 9). A minimum 9 experiments required. The use of matrix method with orthogonal array is explained using a case example. In this example the interaction effects of factors is not taken into account. Ex: Consider a project where we are interested in determining the effect of four process parameters: temperature (A), pressure (B), settling time (C), and cleaning method (D) on the formation of certain surface defects in a chemical vapor deposition on silicon wafers. Three parameter settings are chosen to cover the range of interest. The factors and the chosen levels are given in table 1. The starting values are indicated by an underscore. Our goal is to determine the best settings for each parameter so that the surface defect formation is minimized.

Table - 1

Factors	**Levels**		
	1	**2**	**3**
A. Temperature o^c	T_o - 25	$\underline{T_o}$	T_o + 25
B. Pressure	P_o - 200	$\underline{P_o}$	P_o + 200
C. Settling time (method)	$\underline{t_o}$	t_o + 8	t_o + 16
D. Cleaning method	$\underline{None}$	CM_2	CM_3

There are four factors at three levels each. Therefore the total degrees of freedom for this case is 9 including the grand mean. i.e. 4(3 - 1) = 8+1 = 9. Therefore L9 (3^4) array of Taguchi is chosen.

The matrix experiment selected for this project is given in table. 2. It consists of 9 individual experiments corresponding to the 9 rows. The four columns of the matrix represent the four factors as indicated in the table. The entries in the matrix represents the levels of the factors. Thus, the experiment 1 is to be conducted with each factor at the first level. This implies the experiment 1 has factor levels as (T_o – 25) temperature, (P_o – 200) for pressure, (t_o) minutes settling time and no cleaning. Similarly for other experiments. For experiment 4, temperature is at level 2, pressure at level 1, settling time at level 2 and cleaning at level 3(CM3). The experimental settings for experiment 4 can also be referred concisely as A_2, B_1, C_2, D_3.

The matrix experiment of table 2 is the standard orthogonal array L_a of Taguchi. As the name suggests, the columns of this array are mutually orthogonal. For any pair of columns, all combination of factor levels occur and they occur equal number of times. In this matrix for each pair of columns there exists 3x3 = 9 combination of factor levels and each combination occurs precisely once. For columns 1 and 2, the 9 possible combination of factor levels are (1,1); (1,2); (1,3); (2,1); (2,2); (2,3); (3,1); (3,2); and (3,3) occur in experiments (or rows) 1,2,3,4,5,6,7,8 and 9 respectively.

(Table – 2)

Expt. No.	**Column no. & factor assigned**				**Observation η (dB)**
	1 (A)	2 (B)	3 (C)	4 (D)	
1	1	1	1	1	η_1 = - 20
2	1	2	2	2	η_2 = - 10
3	1	3	3	3	η_3 = - 30
4	2	1	2	3	η_4 = - 25
5	2	2	3	1	η_5 = - 45
6	2	3	1	2	η_6 = - 65
7	3	1	3	2	η_7 = - 45
8	3	2	1	3	η_8 = - 65
9	3	3	2	1	η_9 = - 70

Suppose for each experiment we observe the surface defect count per unit area at three locations on 3 silicon wafers, so that there are nine observations per experiment. Let us define the following formula for

summary statistic, η_i for i th experiment.

η_i = -10 $\log_{10}$ (mean square defect count for except i) where the mean square refers to the average of the squares of the 9 observations in the experiment i. We refer to the η_i calculated using the above formula as the observed η_i. The observed η_i for the 9 experiments is shown in the last column of table 2. The objective of minimizing surface defects is equivalent to maximizing η. The summary statistic η.is called signal – to – noise (s/n) ratio.

In order to estimate the effect of the four process parameters from nine experiments, first the overall mean values of η for the experiment is obtained as,

$$m = \frac{1}{9}\sum \eta_i = \frac{1}{9}[\eta_1 + \eta_2 + \ldots\ldots + \eta_4]$$

The effect of a factor level is defined as the deviation it causes from the overall mean. Let us examine how the experimental data can be used to evaluate the effect of temperature at level A_3. Temperature was level A3 for experiments 7.8, and 9. The average S/N ratio for these experiments, devoted by m_A is, $m_A = \frac{1}{3}[\eta_7 + \eta_8 + \eta_9]$

The effect of temperature at level A_3 is given by (m_{A3} – m). From the table 2, it may be noticed that for experiments 7, 8, 9, the pressure level takes values 1,2 & 3 respectively. Similarly for these three experiments, the levels of settling time and cleaning method also take values 1, 2 & 3. So the quantity m_{A3} represents an average η when the temperature is at level A3, where the averaging is done in a balanced manner over all levels of each of the other three factors.

The average S/N ratio for levels A_1 and A_2 of temperature, as well as those for various levels of other factors can be obtained in a similar way. Thus for example $m_{A2} = \frac{1}{3}[\eta_4 + \eta_5 + \eta_6]$ represents the average S/N ratio for temperature at level 2, and $m_{B3} = \frac{1}{3}[\eta_2 + \eta_5 + \eta_8]$ is the average S/N ratio for pressure at level B_2. Because the matrix experiment is based on orthogonal array, all the level averages possess the same balancing property described for M_{A3}.

By taking the numerical values of η listed in Table 2, the average η for each level of the four Factors can be obtained as listed in table 3. For Example, the average value for B_2 is 1/3[-10 -45 -65] = -40

Factor	Level		
	1	2	3
A-Temperature	-20	-45	-60
B- Pressure	-30	-40	-55
C-Settling time	-50	-35	-40
D-Cleaning method	-45	-40	-40

(table – 3)

As already stated the primary goal is to determine the best or optimum level of each factor. The optimum level for each factor is the level that gives the highest value for ŋ in the experimental region. In this case, our goal is to minimize the surface count. Since -log is a monotone decreasing function, it implies that we should maximize ŋ. Note that ŋ = -20 is preferable to ŋ = -45 because -20 is greater than -45. From table -3, we can determine the optimum levels for each factor as the highest value of ŋ. The best temp setting is A_1, best pressure is B_1 and best settling time is C_2 and the best cleaning method is either D_2 or D_3. We can, therefore conclude that settings A_1 B_1 C_2 D_2 or A_1 B_1 C_2 D_3 would give the highest ŋ or the lowest surface defect count.

MULTIVARIATE ANALYSIS

Classification of Multivariate Techniques

All statistical techniques which simultaneously analyse more then two variables on a sample of observations can be categorized as multivariate techniques. The basic objective underlying multivariate techniques is to represent a collection of massive data in a simplified way. In other words, multivariate techniques transform a mass of observations into a smaller number of composite scores in such a way that they may reflect as much information as possible contained in the raw data obtained from a research study.

There are two types of multivariate techniques one type containing both dependent and independent variables and the other type for data containing several variables without dependency relationship. In the former category, we have techniques like multiple regression analysis, multiple discriminant analysis, and multivariate analysis of variance and cananical analysis. In the latter

category, we include techniques like factor analysis, cluster analysis, multi dimensional scaling (MDS) and latent structure analysis.

In this section we will cover only regression analysis, correlation and curve fitting which will be more useful for engineers.

Curve fitting

On many occasions, the researcher has to deal with two or more variables, which are functionally related and he would like to determine the relationship. Some times it is straight forward and easily discovered, but more often the functional relationship is complex and completely unknown. The first step in deriving an equation to approximate the desired relationship is to collect the data showing the corresponding values of the variables under consideration. Suppose in a chemical process, we consider the yield of the chemical process as a function of the amount of catalyst added to the action. We may let y as the yield and x equal to the amount of catalyst added.

If we take a sample of n data as (X_1, Y_1) $(X_2, Y_2)\ldots$ (X_n, Y_n) and plot it on a rectangular coordinate system, we will get a scatter diagram as shown in the figure 1 & 2. From the scatter diagram, it is often possible to vsualize a smooth curve to approximate the data. It may be a straight line as shown in fig. 1 or a non-linear curve as in fig 2.

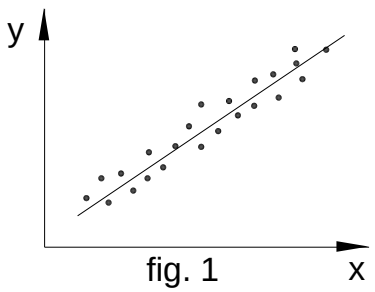


fig. 1

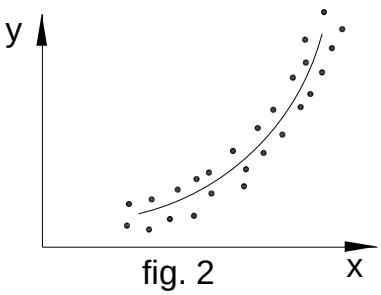


fig. 2

The problem of finding equations for the approximating curves that best fit the given sets of data is called curve fitting. The researcher should select the curve to be fitted. Several common types of approximating curves and their equation are given below.

i) $y = a_o + a_1 x$: Straight line

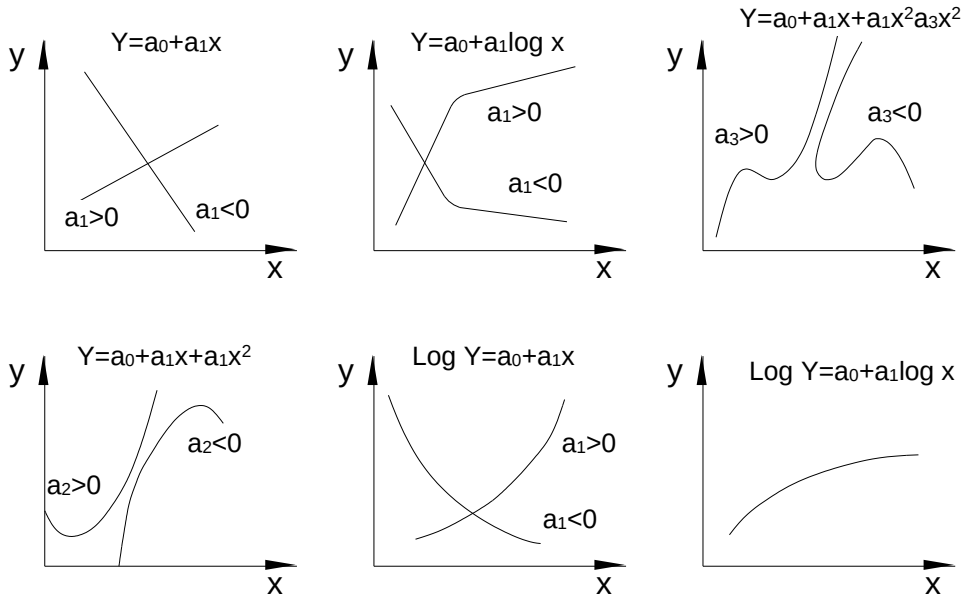
ii) $y = a_1 x + a_2 x^2$: Parabola or quadratic curve

iii) $y = a_o + a_1 x + a_2 x^2 + a_3 x^3$: Cubic curve

iv) $y = a_o + a_1 x + 0_2 x^2$: n^{th} degree curve

v) $y = ab^x$ or $\log y = a_o + a_1 x$: exponential curve

There are many other curves. We can examine the scatter diagrams and compare results with general shapes of curves given by different equations. Fig below shows the shape of some of the curves and more could be found in standard text books.



Sometimes it is useful to examine scatter diagrams of transformed variables using special graph paper, in which one or both scales are calibrated logarithmically. These are called semi log or log – log graph paper respectively For example, if the scatter diagram of log y Vs x shows a linear relationship, we could use equation, $y = a\ b^x$ or $\log y = a_0 + a_1 x$ for log y Vs x experimental curve. Likewise if we plot logy Vs $\log^x$ and if it appears linear, we can use a cubic logarithmic curve, $\log y = a_0 + a_1 \log x$.

To determine the best fit, one can decide intuitively or using a rules or french curves and more it on the scatter diagram until pass through the centre of the points. The method most often used is fitting a curve by eye, that is we want

to minimize the deviations of the points from the prospective line. Mathematically this method is known as “method of least squares”.

Simple regression and Correlation

The mathematical technique of fitting a set of data points to a chosen curve according to the least square criterion for best fit is called regression analysis. The correlation analysis gives us some indication of how well the data points fir or cluster around the equation so derived. The general form of the curve to be fitted must be chosen by the analyst after examination of the data and the scatter diagram.

Suppose that two variables x and y are measured over a range of values and we wish to obtain a simple analytical relation for y as a function of x, the simplest one is a linear relation $y = a_0 + a_1x$. The problem is one of finding the best linear function for the data gathered. The mathematical model for the population is, $y_i = \alpha_0 + \alpha_1 x_i + \varepsilon_i$ in which x_i, y_i are the I th observations and α_0 and α_1 are the population values of intercept and slope regression coefficients, ε_i is the error.

The sample equation is, $Y = a_0 + a_1 x$ in which y is the predicted value of dependent variable and a_0 and a_1 are the sample estimates of regression coefficients. The assumptions that underlie the method of least squares are,

- the errors are normally distributed with mean of zero and constant variance σ_e^2
- the errors are independent of each other.

The method of least squares establishes the line through the data such that the sum of squares of the vertical deviations of observations from the line is smaller than for any other line that could be drawn through the data.

$$\varepsilon_i = a_0 + a_1 x_i - y_i \text{ and } \Sigma \varepsilon_i^2 = \Sigma (a_0 + a_1 x_i + y_i)^2$$

In order to get the values of parameter a_0 and a_1 that result in best value of $\Sigma \varepsilon_i^2$

$$\frac{\partial E_i^2}{\partial a_0} = 2\Sigma(a_0 + a_1x_i - y_i) = 0$$

$$\frac{\partial \varepsilon_i^2}{\partial a_0} = 2\Sigma(a_0 + a_1x_i - y_i)x_i = 0$$

By separating and rearranging we get,

$$\Sigma y_i = a_0 n + a_1 \Sigma x_i$$

$\Sigma x_i y_i = a_0 \Sigma x_i + a_1 \Sigma x_i^2$

Solving these two equations we get,

$$a_1 = \frac{n\,\Sigma xy - \Sigma x\,\Sigma y}{n\,\Sigma x^2 - (\Sigma x)^2} = \frac{\Sigma\,(x - \bar{x})(y - \bar{y})}{\Sigma(x - x)^2}$$

$$a_0 = \frac{\Sigma y - a_1 \Sigma x}{n} = \bar{y} - a_1 x$$

$$= \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n\Sigma x^2 - (\Sigma x)^2}$$

The standard error of estimate of y for the data is,

$$SEE = \left[\frac{\Sigma(y_i - \hat{y}_i)^2}{n-2}\right]^{½}$$

$$= \left[\frac{(y_i - a_0 x_i - a_1)^2}{n-2}\right]^{½}$$

Correlaion

It should be noted that fitting of a straight line (or any other curve) to the data does not mean that the physical data or relationship are best described by that equation. Mathematical operations always establish the value of parameters that give the best least squares fit to the equation form chosen. To evaluate whether the data are a ‘good fit’ to our line or equation, we need the concept of “correlation”. It tells us how close the data points cluster around the curve or line. While regression defines a proposed relationship between the variables correlation tell us how good that relationship is.

A high correlation between variables shows that they change their values in a related manner, but this does not prove or imply a cause and effect relationship. Regression analysis assumes that there is a cause and effect relationship between the dependent and independent variables, correlation makes no such assumption. The important point is that the researcher makes an assumption or hypothesis about cause and effect relationship between the variables, when he does a regression analysis. This assumption may or may not be true. Correlation tells him how closely his data points cluster around the curve he has hypothesized. It does not tell him whether his basic assumption of cause and effect is valid. However, correlation analysis does tell him whether the data are consistent with his hypothesis.

Correlation coefficients will range from -1 to +1. A coefficient of -1 means perfect negative correlation. A coefficient of zero means no correlation and a coefficient of +1 means perfect positive correlation. Very seldom we get any of these three values and we get some fraction that we must then test for statistical significance. For a simple regression problem, the correlation coefficient 'R' is given by.

$$R = \frac{n\Sigma xy - \Sigma(\mathrm{x})\Sigma(\mathrm{y})}{[n\Sigma \mathrm{x}^2 - (\Sigma \mathrm{x})^2][n\Sigma \mathrm{y}^2 - (\Sigma \mathrm{y})^2]^{1/2}}$$

The total variation of y is defined as $\Sigma(y - \bar{y})^2$ - ie, the sum of the squares of the deviations of y from the mean $\bar{y}$. The ratio of amount of this variation, which can be explained by our regression equation, to the total variation observed is called the "coefficient of determination", which is given by the square of the correlation coefficient (R^2).

Ex: The observed sample data is given below. Develop the regression equation and also find the correlation coefficient.

X:	0	2	2	3
Y:	1	4	3	5

X	y	x^2	y^2	xy
0	1	0	1	0
2	4	4	16	8
2	3	4	9	6
3	5	9	25	15
7	13	17	51	29

n= 4

Σx = 7

$\therefore \bar{x} = \frac{7}{4} = 1.75$

Σy = 13

$\therefore \bar{y} = \frac{13}{4} = 3.25$

$$a_1 = \frac{n\,\Sigma\, x\, y - (\Sigma\, x)\,(\Sigma\, y)}{n\,\Sigma\, x^2 - (\Sigma\, x)^2}$$

$$= \frac{4(29) - (7)\,(13)}{4(17) - (7)^2} = \frac{116 - 91}{68 - 49} = 1.316$$

$a_0 = \bar{y} - a_1\,\bar{x} = 3.25 - (1.316)\,(1.75)$

$= 0.947$

∴ The regression equation is

$Y = 0.947 + 1.316x$

Coefficient of correlation (R) is

$$R = \frac{n\sum xy - (\sum x)(\sum y)}{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]^{1/2}}$$

$$= \frac{4(29) - (7)(13)}{[4(17) - (7)^2][4(51) - 13^2]} = \frac{25}{[(19)(35)]^{1/2}}$$

$$= \frac{25}{25.8} = 0.969$$

The coefficient of determinations (R^2) is,

$R^2 = (0.969)^2 = 0.939$

This implies that 93.9% of the variation of y as x changes can be explained by our relationship, y = 0.947 + 1.316 x.

Multiple Regression and Correlation

When the dependent variable of interest 'is influenced by more than one independent variable and we want to study this relation, the statistical tool of multiple regression is often useful. The most common and mathematically tractable form of this method requires that the model be assumed be linear in the coefficients that will be adjusted to minimize the sum of squares of all the residuals. Mathematically the relation must be expressible in the form

$Y = a_0 + a_1 f_i (x_i) + a_2 f_2 (x_2) + a_3 f_3 (x_3)$……

The functions $f_i (x_i)$ can be any form; they need not be linear in x.

The best values of the coefficients 'a' will again be found by using a least square criterion. We first decide as best as we can on the form of the model (ie the $f_i (x_i)$ functions) and then gather data consisting of observation of y and x_j. We then define the residual at each observation equal to the difference between the measured 'y' and the model predicted y. We add the squares of all the residuals, forming the total error (E). To minimize 'E', we choose the a' s so as to make all the partial derivatives of E with respect to each 'a' equal to zero. Suppose, the simple multi regression equation is,

$Y = a_0 + a_1 x_1 + a_2 x_2 + \ldots\ldots\ldots a_n x_n$

$E = \Sigma (y_j - a_0 - a_1 x_{1j} - a_2 x_{2j})^2$

$\frac{\partial E}{\partial a_0}$= - 2Σ ($y_j$ - a_0 - a_1 x_{1j} – a_2x_{2j}) = 0

If there are 'm' coefficients to be found, we need atleast 'm' observations to be able to find 'a' s. Usually we have more observations then coefficients, therefore, we need to use least squares methods of linear algebra to solve these equations and get the values of a' s.

Computer software packages are available to perform the multivariable calculations indicated above. 'Microsoft Excel' is one such package. A computer programe for multiple regression and correlation analysis will typically include ANOVA of the regression. This will consist of total, multiple regressions, residual sum of squares, degrees of freedom and as well as residual mean squares. The total sum of squares is an expression of the total amount of variability among the 'y' values $(y_j\bar{y})$, the regression sum of squares expresses the variability among the y values attributable to the regression being fit $(\hat{y}_j - \bar{y})$ and the residual sum of squares tells us about the variability of 'y' still remaining after fitting the regression $(y_j\hat{y}_j)$. The ANOVA table will be,

Source of variation	Sum of (SS) squares	d.f	Mean square (MS)
Total	$\Sigma(y_j - \bar{y})^2$	(n – 1)	
Regression	$\Sigma(\hat{y}_j - \bar{y}_j)^2$	m	$\frac{\text{regression SS}}{\text{regression d f}}$
Residual	$\Sigma(y_j - \hat{y}_j)^2$	(n –m – 1)	$\frac{\text{regression SS}}{\text{regression d f}}$

n = total no. of data prints (ie total number of 'y' values; m = number of independent variables m the regression model)

If we assume 'y' to be functionally dependent on each of the x' s, then we are dealing with multiple regression. If no such dependence is implied, then any one of the (M = m+1) variables could be designated as 'y' for the purpose of utilizing computer program, this being the case of "multiple correlation".

The ratio $R^2 \frac{\text{regression SS}}{\text{total SS}}$ or equivalently $R^2 = 1 - \frac{\text{residual SS}}{\text{total SS}}$ is the "coefficient of determination" for multiple regression or correlation or the "coefficient of multiple determination". In the regression situation, it is an

expression of the total variability in ‘y’ attributable to the dependence of ‘y’ on all the X_i’s as defined by the regression model fit to the data. In the case of correlation R^2 may be considered to be the amount of variability in any one of the ‘M’ variables that is accounted for by correlating it with the other (M – 1) variables. The square root of the coefficient of determination is referred to as “multiple correlation coefficient”, ie. $R = \sqrt{R^2}$

Non – linear regression models

In many engineering problems, we may encounter with non – linearity. For instance, a second order model with two independent variables is given by:

$$Y = a_0 + a_1 x_1 + a_2 x_2 + a_3 x_3^2 + a_4 x_1 x_2 + a_5 x_2^2$$

Such models is needed to defect non – linearity and second order effects. Consider another example.

$Y = a_1 x_1 x_2 + a_2 x_3^3 + a_3 \sin \chi_4$.In this case, the independent variables are x_1, x_2, x_3 and x_4. To make this to fit into linear multiple regression model, transformation could be made by defining new variables as, $x_a = x_1, x_2$ and $x_b = x_3^3$. Similarly in the case of another example, $y = a_1 m_1^{x1} m_2^{x2} m_n x^n$, the equation could be modified as $\log y = \log a_1 + x_1, \log m_1 + x_2 \log m_2 + \ldots\ldots + x_n \log m_n$, Standardized computer programs are usually used to determine the parameters of this type of equations.

When the analyst use the ‘canned’ statistical programmes in computer based data reduction, due care has to be exercised. One should understand what the statistical package really does. The user should take time and trouble to read the back up documentation for the program thoroughly, if necessary consult if experts, and then use it.

UNIT - IV
ANALYSIS AND INTERPRETATION OF DATA

RESEARCH DATA ANALYSIS

Introduction

After collecting data, the researcher has to accomplish the task of drawing inferences followed by report writing. To draw meaningful inferences, the analyst has to carryout proper analysis of data gathered so that his interpretation is on the desired lines. If this is not done properly, misleading conclusions may be drawn and the whole purpose of doing research may get vitiated. It is only through interpretation that the researcher can expose relations and process that underlie in his findings. Generally, tools available for analysis include statistical, graphical and numerical methods. Many of the techniques have already been covered in the previous chapter.

A general approach to data analysis may include

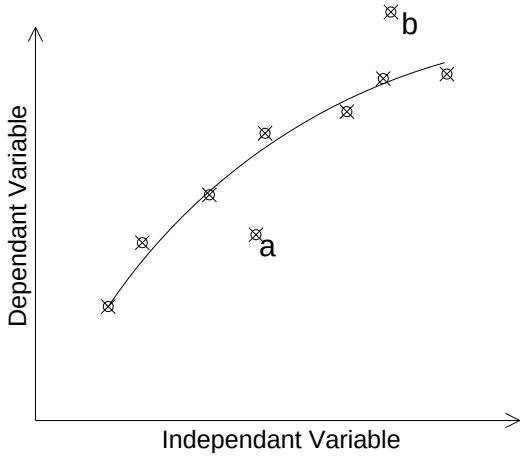
- Examine the data for consistency
- Perform necessary statistical and other analysis, where appropriate
- Estimate the uncertainties in the results
- Anticipate the results from theory
- Correlate the data.

Data checking

Prior to, and perhaps as a part of data analysis we should continue to be on the watch for invalid data, using whatever data checking procedures are appropriate to our study. Weeding out ‘bad’ data needs to be continuing effort at all stage of the study. Undetected spurious data, once they get into a sophisticated data analysis procedure, can bias results and interpretation in many subtle ways. Tools for rejection of spurious data include ‘treatment of outliers’ and use of ‘running graphs’.

The researcher must do preliminary screening of data before carrying out the desired analysis. The purpose of such screaming is to detect and discord any data points which are questionable, so that they do not improperly bias the analysis. These decisions must be made prudent y because the wild points, usually called as ‘outliers’ may actually be valid data documenting unexpected but potentially important behavior.

The outlier may crop up due to various factors such as misreading of instrument, voltage fluctuations, poor connection etc. However, if one takes more readings it becomes easier to detect outlier data that might occur. If we could estimate the standard deviation from data values (s), then any value beyond ±3s from average would reasonably a candidate for rejection. Though this is only a guideline, we cannot avoid the responsibility of exercising our best engineering judgement. If readings are rejected, we should make note of this and explain the basis of our judgement.



A different context of our outlier decisions is found when we plot a graph using data of independent and dependent variable as shown in the figure. Points ‘a’ & ‘b’ are both outliers, however point ‘a’ is more easily judged as wild than point ‘b’. At the ends of a curve, a wild point could possibly be accurate one, which shows the beginning new regime of behavior rather than errors of some kind. Unless we have clear evidence of mistakes, such points should be tested by extending of the range of experiment so that these questionable points become interior rather than exterior points.

Plotting running graphs is a reliable method of detection of “odd behavior”. As we gather data points, we immediately plot appropriate graphs. Compared with a table, a graph is much more sensitive and a reliable dector. By keeping these running graphs updated as we add each reading, we can often detect problems and malfunctions quickly. Such graphs are usually not a part of our final data sheet and will usually done on a rough note book. Here, we need to plot the actual data wherever possible, rather than related quantities that might be the computed results of our study.

For example if we are recording water level in a stand pipe and the discharge rate of an orifice at the bottom of the stand pipe for the purpose of producing a graph of flow rate versus pressure drop for the orifice, our running graph could use raw water level data in cms rather than require us to take the time to compute the pressure in Pascal from the water level.

The data checking methods actually used will depend on the nature of experiment or study. However a few general suggestions can be made.

An obvious, but often too expensive approach is simply to repeat part of all those runs which appear questionable. If this can be justified, it will be most convincing if any random effects are allowed to be different from the original run when we replicate it. The decision as whether to undertake a replication and how comprehensive it should be will depend on the degree of doubt we have about the original run and how important the questioned results are to the success of the study.

When we graphically display data relating to two variables, the extra polatation of the curve to the origin may be a useful checking method. Efficiencies of machines plotted against load on the machine should extrapolate to zero at zero load. When we study properties of mixture of components, these properties should extrapolate to the properties of a component when the percentage of other components goes to zero. We should always look for and use any such known relations to check the behaviour of data.

Another useful tool employs conservation or balance principles to check whether data is consistent with itself. Conservation of energy tells us that if we have measured all the inputs and outputs of energy to a process operating at a steady state, then the sum of all the inputs must equal to the sum of all the outputs. If our calculation of these inputs and output show a discrepancy, we may have some faulty data.

Conservation of mass can be used in a similar manner, if we are measuring material inflow and outflow rates. In practice, such balance checks may not be possible or totally convincing if there are significant unmeasured losses such as leaks in the system. In general, we should consider carefully all the available theoretical information about our study and make use of those aspects that allow us to partially or totally check the validity of our data. This should be done before we undertake any large scale overall data analysis associated with the objective of our study / research.

Statistical Analysis

When we assume some randomness in our data, then the analysis method involve statistical considerations. If our study involves establishing the numerical value of some measurable quantity, we must choose the percentage of the mean value we wish our confidence interval to be and the confidence level we want. We may also need an estimate of the coefficient of variation. If no data have yet been taken, the estimate of coefficient of variation must be based on past experience or lacking this outright guess work. Using the statistical tools, we can also determine how large a sample is needed to maintain the desired level of accuracy.

If we have a set of raw data, data analysis consists merely to determine the sizes of confidence intervals for the chosen confidence levels. For example, if we have a sample size of 10, mean value of 6.8 and standard deviation of 0.47, the 95% confidence interval is 6.8 ± 0.33. Suppose our best estimate of the measured value is 6.8, we are 95% sure that it is somewhere between 6.47 and 7.13.

Suppose we are trying to decide whether one material or process is better than an alternative in terms of having a different average value or standard deviation. Comparison of the average values be made by using confidence intervals or the difference between two mean values or use of t – test. For comparing two standard deviations, F – ratio test could be used to reach decisions as to whether the two values are about the same or significantly different. The various statistical tools have been covered in the previous chapter.

In the case of a problem finding a relation between a single independent variable and a dependent variable, we can use simple regression analysis technique. Preliminary data analysis should certainly include plotting of scatter diagram and then choose an appropriate equation to fit the data. If we have a physical theory which predicts a definite mathematical relation between the two quantities, we can check whether the graph is consistent with the theory. If visual evaluation indicate that the data could follow the predicted straight line, parabola, sine wave, etc. then we can use the regression software to find the specific curve that best fit the data.

If we have a theoretical prediction for the type of curve to be expected, but the data show a large scatter, then the interpretation becomes less clear cut. If inspite of larger scatter, we can still visually discern a trend in the data, which does not contradict the theory, then of course we should get a best fit regression, using the theoretically predicted type of curve. We can then graph the residuals for this fit to see if they are largely random, or if we can detect a trend. If no trend is noticed, we may have found the best model. If a trend is noticed. its nature may give us some clues about how to adjust the form of the model to get a better fit. A final choice of the model is based on criteria such as visual goodness of fit, numerical value of R^2, numerical values of individual residuals relative to the practical importance of different regions of the curve.

When we want to study the relation between one dependent and two independent variables, we can resort to data analysis using multiple regression techniques. The logical starting point of the modeling process is the theoretical predication of the type of functional relation to be expected is available. Regression methods can be used to find the best - fit - surface in the case of two independent variables adopting the usual criteria for evaluating the quality of fit. As usual, data with large noise make the choice of the best model more difficult.

If we have no theoretical prediction to guide us, three dimensional plots of raw data are possible but are almost useless for the purpose of choosing a model form, if there are many points and / or there is much scatter.

Multiple regression models not only enable us to study the main effect of independent variables but also the interaction effect. For example consider the regression equation,

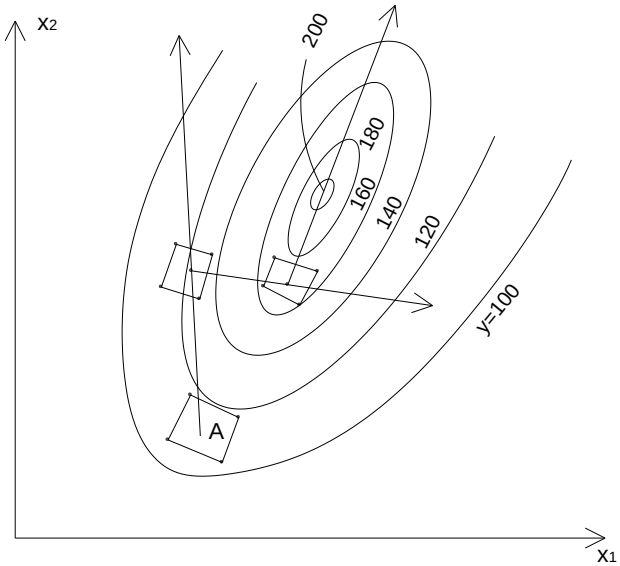
$$Y = a_o + a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_1 x_2 + a_5 x_2 x_3 + a_6 x_1 x_3 + a_7 x_7 x_1 x_2 x_3$$

The main effect of variable x_1 would be the model term $a_1 x_1$ while the term $a_4 x_1 x_2$ would be called an interaction between $x_1 x_2$. That is, interaction refers to a situation where the effect on the dependent variable of changes in one independent variable depends upon the value of some other variable.

When we study models with more than a few independent variables the concepts of factorial or fractional factorial experiment and orthogonal arrays become useful. They are not only important in experiment planning but also helpful in data analysis.

The significance of difference between two process means can be analysed using either z – test or the t – test. But, when we want to analysis more than two, the ANOVA techniques enable us to conduct simultaneous test and it is an important analysis tool.

Response surface methodology (RSM) is a powerful statistical procedure that employs factorial analysis and regression analysis for the determination of optimum operating conditions on a response surface. It uses a two step procedure aimed at rapid movement from the current operating position into the central region of the optimum. This followed by the characterization of the response surface in the vicinity of the optimum by a mathematical model. The basic tolls used in RSM are two level factorial design, method of least squares model and its simpler polynormal forms.



The analysis should bring forth the meaning of our data. The meaning will have its own detailed interpretation for each specific example. For instance the study involves estimation of average value of a parameter or the variability of the parameter, that can be used to study a system's performance. The performance parameter tell us whether overall design goals have been met. Average values are used to calculate the nominal value of the parameter, the standard deviations used to estimate variability around the average. Comparison of mean or average values can be used to evaluate alternative designs, process, materials etc.

Simple linear regression and multiple regression models used to find relations between process outputs such as quality, cost, yield and the process inputs which affect these outputs. Once such relations have been identified, we can use them to guide adjustments to the process aimed at improving operations. Operating points which not only maximize some desirable output but also make it less sensitive to changes in input values. This can be used to relax specifications on component parts or materials, reducing costs still maintaining quality. The

meaning of data which defines a process model thus lies in our better understanding of how the process works.

Graphical data analysis

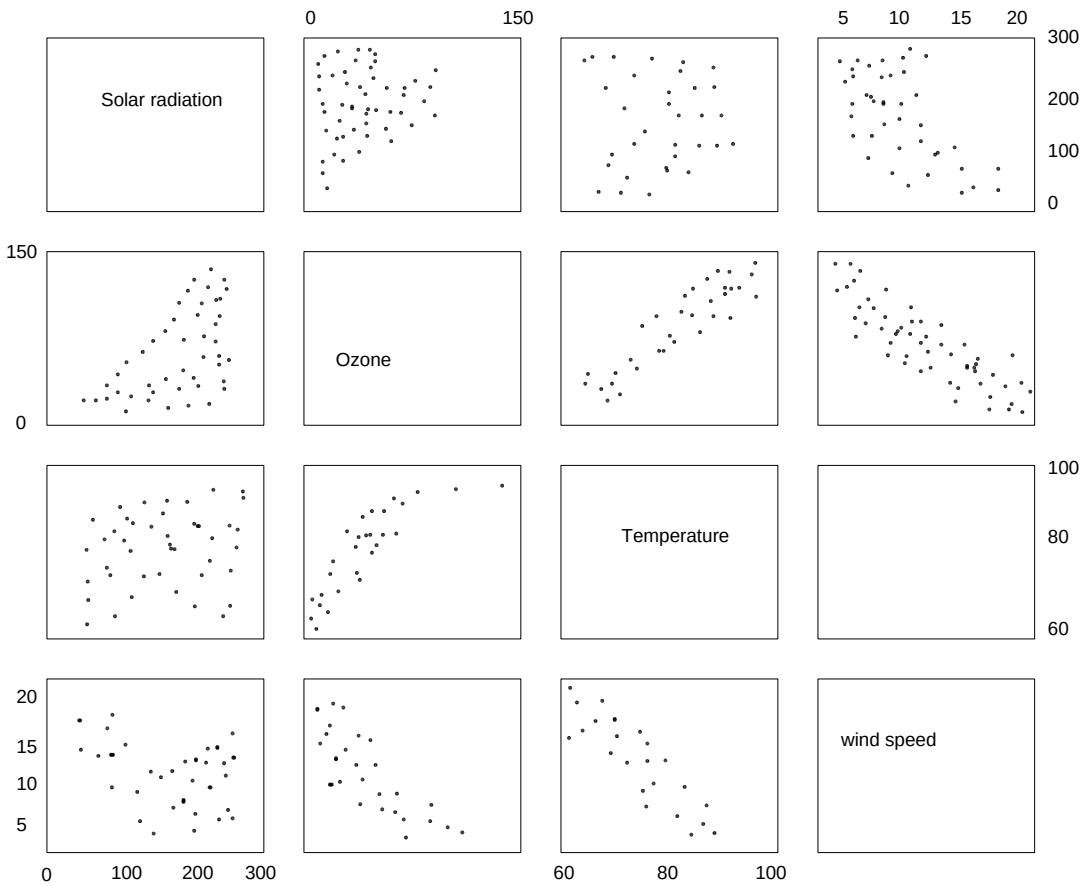
Though interactive software for personal computers makes regression methods easy, graph plays useful role for simpler problems. Most of these methods are intended for exploring the relations between two variables by using a simple x, y graph. To speed up the work, special graph paper with various kinds of distorted scales (log – log, linear – log, gaussian probability etc.) is often used. When such paper is not available, one can get the equivalent effect by using linear scale paper not plotting the raw data, but rather the desired function (such as logarithm) on the graph. This is infact what the most modern computerized versions of these methods do, using regression analysis to define the best line.

Let us consider the relation of the form $y = cx^a$. Taking logarithm on both sides $\log y = \log c + a \log x$ which is equivalent to $y = b + m x$. We can graphically implement this transformation by plotting Y (log y) and X (log x) on the linear graph paper or y and x on log – log paper. We fit by eye the best straight line to our data and then decide by visual judgement whether the fit is acceptable. If the fit is acceptable, we get numerical values for the fitting constants ‘c’ and ‘a’ by graphical measurements. If we were using computer, the best line and its equation would automatically decided by the regression procedure. If the fit is not acceptable, we need to study its faults to get some clues as to the next type of relation to try.

A fitted curve that is best for the transformed values may not be best for actual values in the least square fitting process, so we always need to look at the fit quality for the actual values before deciding whether the fit is acceptable. This can be done by introducing gaussian random noise to the y values and compare the noisy data (YG + NOISE) and the fit to the noisy data (YGFIT) using actual x and y values rather than the logarithm used in the fitting process. By doing this we can ensure whether the use of logarithms has created any problem in our case.

When one dependent variable with multi independent variables are involved computerized methods using regression analysis are generally preferred. However, a sequence of graphs of the dependent variable versus each of the independent variables will of course reveal which independent variable is dominating and what the shape of this dependency looks like. Scatterplot matrix for multivariable graphical data analysis is generally used for this type of analysis.

Scatterplot matrix is applicable to models with any number of variables. If the total number of variables (independent and dependent) is k, then one can make ordinary x, y graph between any two of them, there being a total of k (k – 1) /2 such graphs. The idea of making such graphs is not novel; the contribution of the scatterplot matrix concept lies in how the graphs are organized and presented. The example shows how Cleveland used the actual air pollution data to explain this graphical data analysis method.



Measurements of ozone concentration, wind speed, solar radiation and air temperature were made for a period. These data are plotted in scatterplot matrix

form. In this the graphs are arranged in rows and columns such that each row or column has all the graphs relating a certain variable to all the others. In our case, the upper row has graphs of solar radiation plotted against each of the other three variables. The third column from the left plots solar radiation ozone, and wind speed against temperature. All the other rows and columns are similarly interpreted. Combining some qualitative knowledge of pollution chemistry with information from the various graphs allows one to reach some useful conclusions about this pollution process. Some of the commercial statistical software packages do provide this facility. It is called the “draftsman plot”.

Numerical data analysis

Most data analysis methods employ a combination of statistical, graphical and mathematical (numerical) tools. While a software can be written to perform any calculations we might need, the scope of such methods is essentially unlimited. The time frame and effort may be the constraints. But, now a days a large number ready made dedicated software packages are available to help the researchers and analysts. They are user friendly and menu driven.

The statistical software packages commercially available can easily perform much statistical analysis such as computation of means, standard deviations, correlation coefficients, t-test, analysis of variance, analysis of covariance, multiple regression, factor analysis and various non-parametric analysis. These are just a few of the programs and sub programs that are available in many packages.

One can use conventional text based languages such as ‘e’ to write the application programme, or higher – level more user friendly languages such as Lab VIEW. The Lab VIEW uses graphic software modules called virtual instruments (VIs) that have front – panel user interface and block diagram programs. By connecting icons for these modular VIs in a block diagram, one can easily create, modify, combine and exchange VIs to form more sophisticated VIs. A compiler is used to generate machine code so that VIs execute on a computer at speeds comparable to compiled ‘C’ programs.

The routines available includes broadly as indicated below.

- Fourier and spectral analysis

- Digital filters and windows
- Wave form analysis
- Wave form generation
- Statistical analysis
- Vector and matrix algebra
- Numerical analysis

In every new edition new features are being in corporated. Depending upon the need, the analyst can choose and analyze the date gathered.

MATLAB is another important computational software extensively used by researchers for analysis. The MATLAB is available for a large number of computer systems and has great degree of platform independence. The MATLAB environment provides the user with an interactive work space in which to enter commonds and view results. With MATLAB one can

- Define variables, enter their numerical values and perform calculation interactively
- Create vectors and matrices and perform operations on them using the many built-in MATLAB functions
- The calculated results can be visualized by plotting them on the screen and print a hard copy
- The built-in MATLAB programming language can be used to create a calculation procedure consisting of a number of related steps.
- The text editor can be used to write and save a script files, which contains a specific series of MATLAB statement that one want to use in the future.
- The script files can be use to define functions that are of use in future work.
- The tool boxes of functions developed by others can be employed and made available as script files.

Online help is available within the MATLAB environment. A menu provides a table of contents as well as an index. Every new version has many new features incorporated.

INTERPRETATION OF RESULTS

Meaning of Interpretation

Interpretation refers to the task of drawing inferences from the collected facts after an analytical, experimental or simulation study. It is a search for broader meaning of research findings. The task of interpretation has two major aspects viz:

i) The effort to establish continuity in research through linking the results of a given study with those of another.

ii) The establishment of some explanatory concepts.

In one sense, interpretation is concerned with relationships within the data gathered, partially overlapping analysis. It also extends beyond the data of the study to include the results of other research, theory and hypothesis.

The interpretation is always related to the objective and scope of the study envisaged. It is a device through which the researcher has a better comprehension and understanding of factors that seam to explain the observations of the researcher in the study. Also, it provides a theoretical conception, which can serve as a guide for the future researchers.

Need for interpretation

The interpretation is essential for the simple reason that the usefulness and utility of research findings lies in it. Moreover, it is being considered as a basic component of research process because of the following reasons.

i) It is through interpretation that the researcher can well understand the abstract (theoretical) principles that lie beneath his findings. Through this, he can link up his findings with those of other studies, having the same abstract principles, thereby can predict concretely. Fresh inquiries can test these findings later on. This way the continuity in research can be maintained.

ii) Interpretation leads to the establishment of explanatory concepts, methods that can serve as a guide for future research studies. It opens up intellectual adventure and stimulates the quest for more knowledge.

iii) The researcher can better appreciate only through interpretation why his findings are what they are and can make others to understand the real significance of his research work.

iv) In some exploratory type of research studies, the interpretation of findings often results into hypothesis of further experimental research study. Hence, the interpretation is involved in the transition from exploratory to experimental research. Since an exploratory study does not have a hypothesis to start with, the findings of such a study have to be interpreted on a 'post – factum' basis. In such a case the interpretation is technically described as 'post – factum' interpretation.

Correlation with Scientific facts

Research involves a combination of analytical, experimental or simulation work. The theoretician strives to explain or predict the results of experiments on the basis of analytical models which are in accordance with fundamental physical principles that have been well established over the years. Suppose the outcome of the experiment do not fit into the scheme of existing physical theories, the researcher may start doubting his experimental data and then at appropriate theories. In some cases, the earlier findings are modified to take into account the results of the new experimental data, after ensuring the validity of the data. In any event all physical theories must eventually rely upon experiment for verification. Hence, there has to be consistency of results achieved.

The researcher must give reasonable explanation of the relations which he has found and he must interpret the relationship in terms of underlying scientific processes or principles. He should try to find out the thread of uniformity that lies under the surface layer of his diversified research findings. Any findings based on theoretical, or experimental study should be explained with the science involved. Infact, this is the technique of how generalization should be done and concepts be formulated.

Extraneous information, if collected during the study, must be considered while interpreting the final results of research study, for it may prove to be a key factor in understanding the problem under consideration.

A statistical analysis is only appropriate when measurements are repeated several times. If this is the case, we have to make estimates of parameters such as average, standard deviation etc., In those cases where the uncertainty of data is to be prescribed by statistical analysis, calculations should be performed using appropriate distribution. Justification for using a particular distribution should form part. Also, this may be used to determine level of confidence using appropriate level of significance. The number of measurements made should be justified using confidence levels and confidence interval.

Use of regression techniques for analysis and interpretation presupposes that the investigator has carefully reviewed the theory appropriate to the subject and try to gather necessary scientific information that will indicate the trends the results may take. Important dimensionless groups, pertinent functional relations and other information may lead to fruitful interpretation of data. Interpretation of interaction effect between factors indicated by the regression model should be backed by the theoretical knowledge about the behavior of interacting factors pertaining to the subject under consideration.

Modelling whether mathematical or simulation is an art and science. The researcher develops a model of an existing system or its part or a hypothetical system which he conceives for his study. It depends upon the objective and scope of the study. The modeler ignores many of the actual features of the system under study and abstract from the real situation only those aspects that make up an idealized version of the real system. Thus, models are simplifications and abstractions of real situations. In that process the researcher makes many assumptions to simplify his model.

The rationale behind the various assumptions in developing the model has to be explained based on the results / test of small experiments, experience of other researchers quoted in their published work, expert's views, logical reasoning by the researcher. Each assumption has to be justified as how it does not affect the model's result for the purpose of study undertaken.

Sometimes we assume linear relationship between two variables even though we may suspect or even know that the true relationship is curvilinear. We assume that at least over a limited range of variation such an approximation will be satisfactory. For, instance, an electrical engineer working with the models of

circuits in which the values of resistors, capacitors, etc. are assumed to have constant values – a simplification because we know that electrical characteristics of many of these component vary as a function of temperature, humidity, age etc. The mechanical engineer works with models in which gases are perfect, pressures are adiabatic, and conduction is uniform. For majority of practical cases, such approximations are good enough and yield usable results. All that the researcher has to do is to justify his explanation based an conceptual and other scientific knowledge he commands.

Before attempting to draw conclusions based on the output results of experiment or a simulation model. the researcher should establish the validity by comparing with real world results and expert's opinion. If there are certain limitations, that has to be explained and how this has not affected his findings and conclusions. The experience of other researchers may also be quoted to support his justifications.

While analyzing results and making interpretations consideration has to be given to aspects such as accuracy vs precision, and error vs uncertainty. While accuracy refers to the agreement between a measurement and true or correct value, precision refers to the repeatability of a measurement. Error refers to the disagreement between true or accepted value whereas uncertainty of a measured value is an interval around that value such that any repetition of the measurement will produce a new result that lies within this interval.

Depending upon the nature of the study, experiment and objective etc. the researcher has to do analysis applying the aspects mentioned in the previous para. But, however if anybody wants to establish error, he must know the correct value to discuss error. But we do not know what the true value is ahead of time, especially when the central objective is to discover new things. An experienced scientist assumes the experiment is not in error. It is the only choice available. Later research attempts by other researchers to repeat the result will hopefully reveal problems if any, but the first time around there is no such guide. But, this does not mean, the researcher can simply assume his approach is correct. He should explain as how at each step of his study, he has exercised checks with available scientific background of knowledge.

Precautions in Interpretations

A researcher must remember that even if the data are properly collected and analyzed, wrong interpretation would lead to inaccurate conclusion. Therefore, it is necessary that the task of interpretation be performed with patience and in an objective manner. The following guidelines may be useful to the researcher for correct interpretation.

i. At the outset, the researcher must invariably satisfy himself that a) the data are appropriate, trust worthy and adequate for drawing inferences b) the data is homogeneous and, c) proper analysis has been made through statistical methods.

ii. One should be cautious about the errors that can possibly arise while interpreting results. Errors can arise due to false generalization and / or due to wrong interpretation of statistical measures such as application of findings beyond the range of observations, identification of correlations with causation and the like.

iii. Another pitfall is the tendency to affirm that definite relationships exist on the basis of confirmation of a hypothesis. Infact, the positive test results accepting the hypothesis must be interpreted as “being in accord” with the hypothesis rather than as “confirming the validity of the hypothesis”. The researcher should remain vigilant about all such things so that false generalization may not take place. He should be well equipped with and must know the correct use of statistical measurers for drawing inferences concerning his study.

iv. The researcher should be aware that the task of interpretation is very much intertwined with analysis and cannot be distinctly separated. Hence, he should consider the task of interpretation as a special aspect of analysis and accordingly must take all precautions that one usually observes while going through the process of analysis viz; precautions concerning reliability of data, computational checks, validation and comparison of results.

v. The researcher should be conscious that his task is not only to make sensitive observations of relevent occurrences, but also to identify and discord the factors that are initially hidden to the eye. This will enable

him to do his job of interpretation on proper lines. One should avoid broad generalizations as most research is not amenable to it, because the coverage may be restricted to a particular time, a particular context and conditions. Such restrictions, if any, must invariably by spelt out and the results must be framed within their limits.

vi. The researcher must keep in mind that "ideally in the course of a research study, there should be constant interaction between initial hypothesis or premise, empirical observations and theoretical conceptions. It is exactly in this area of interaction between theoretical orientation and empirical observation that opportunities for originality and creativity lie". One should pay special attention to this aspect while engaged in the task of interpretation.

ACCURACY, PRECISION AND ERROR ANALYSIS

Defnition of terms

Accuracy

Accuracy refers to the agreement between measurement and true value. For instance if a clock strikes twelve when the sun is exactly overhead, the clock is said to be accurate. The measurement of the clock (twelve) and the phenomena it is meant to measure (location of sun at zenith) are in agreement. With regard to instruments, accuracy is defined as the ability of the device to respond to true value of a measured variable under the reference conditions. It is measured as a percentage of full scale reading or percentage of true value or percentage of scale span.

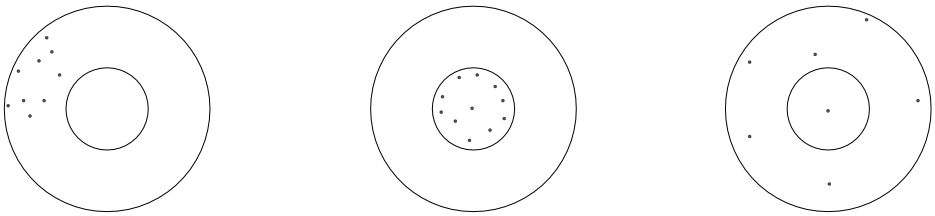
Precision

Precision refers to repeatability of measurement. It does not require us to know the correct or true value. If each day for several years a clock reads exactly 10:17 AM when the sun is at zenith, this clock is very precise. The true meaning of noon is not important because, we only care that the clock is giving a repeatable result. As regards measuring devices, it is defined as the degree of exactness for which a device is designed or intended to perform. The precision is composed of two characteristics viz: conformity and significant figures.

Suppose, the true value of a resistor is 28,34,267 ohm, and a multimeter constantly measures this as 2.8MΩ, the error created by the limitation of the scale reading is known as precision error and has good conformity. While conformity is necessary, it not a sufficient condition for precision. Significant figures convey the actual information regarding the magnitude and the measurement precision of quantity. For example, if a voltage is specified as 240 V, its value should be closer to 239 V or 241 V. There are 3 significant figures. Suppose if it is specified as 240.0V, then it may be closer to 239.9V or 240.1V. Thus, there are four significant figures.

Often the terms precision and accuracy are used interchangeably. But actually there is a difference between these two terms. Consider the measurement of a known pressure of 100 MPa with pressure gauge. The six readings taken are 103, 104, 102, 103, 102 and 104 MPa. The average value is 103 MPa and the maximum deviation from the average reading is ± 1 MPa in 100 MPa. Thus, the scale of the instrument can be calibrated to read ± 1 MPa. But if we consider with accuracy, the readings are not accurate. The accuracy of the instrument is only (104 – 100)／100 ie 4%. Thus, the precision is ± 1% but the accuracy is 4%.

A precise measurement may not be necessarily be accurate and vice versa. This statement is illustrated by an analogy from shooting practice on a target as shown below.



High precision poor accuracy | High precision & high accuracy | Poor precision & average accuracy

Repeatability and reproducibility

Repeatability may be defined as the closeness of agreement among the number of consecutive measurements of the output for the same value of input

under the same operating conditions. It may be specified in terms of units for a given period of time.

Reproducibility may be defined as the closeness of agreement among the repeated measurements of the output for the same value of input under the same operating conditions over a period of time. Perfect reproducibility implies that there is no drift in the output over a long period of time. Both the reproducibility and repeatability are a measure of the closeness with which a given input may be measured again and again.

Extending this concept to the research study, if the researcher has planned his study properly, designed his experiment or model with utmost care and dexterity thereby making it robust, reproducibility and repeatability of results under identical conditions by another researcher, who wants to use one's model or experiment may not pose any problems. This is an essential feature of a research work.

Error Analysis

Definition and Classification of error

The knowledge we have of the physical world is obtained by conducting experiments and making measurements. In doing this it is crucial to understand that all measurements of physical quantities are subject to uncertainties. It is never possible to measure anything exactly. It is good, of course to reduce the error as small as possible, but it is always there. In order to draw valid conclsions the error must be indicated and dealt with properly.

If the result of a measurement is to have meaning, it cannot consist of measured value alone. An indication of how accurate the result is, must also be included.

Though error refers to the disagreement between a measurement and the true or accepted value, in reality the true value is not known with certainty. Therefore, error has to do with uncertainty in measurement that nothing can be done about. The result of any physical measurement has two essential components i) a numerical value giving the best estimate possible of the quantity measured, and ii) the degree of uncertainty associated with this estimated value.

For example, a measurement of the width of a table would yield a result such as 95.3 ± 0.1 cm.

If a measurement is repeated, the values obtained will differ and none of the results can be preferred over others. Although it is not possible to do anything about such error, it can be characterized. For instance, the repeated measurements may cluster tightly together or they may spread widely. This pattern can be analyzed systematically.

As already stated, errors can be broadly classified as systemic error and random error.

Systemic errors are errors which may tend to shift all measurements in a systematic way so that their mean value is displaced. This may be due to such things as incorrect calibration of equipment, consistently improper use of equipment or failure to properly account certain effect. While large systematic errors can and must be eliminated in a good study, but small systemic errors will always be present. In science we need several independent confirmations of experimental results because the equipment behavior may be affected at different places thereby introducing systemic errors.

Random errors are errors which fluctuate from one measurement to the next. The results they yield are distributed about some mean value and this can occur for a variety of reasons such as

- Lack of sensitivity in equipment
- Due to noise ie extraneous disturbances which cannot be taken Into account.
- Due to imprecise definition

Random errors displace measurements in an arbitrary direction whereas systemic errors displace measurements in a single direction. Random errors are unavoidable and must be lived with.

Measuring error

As the random error scatter around the mean value, statistical analysis tools can be used to estimate the uncertainty associated. The uncertainty of a measured value is an interval around the value such that any repetition of the measurement will produce a new result that lies within this interval. Uncertainties

may also be stated with a probability. In this case the measured value has the stated probability to lie within the confidence interval.

If an experiment is repeated say 'n' times, to get x_1 , x_2 ..... x_n, then the average mean value (x) is calculated as $\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n}$.Doing this way give a result with less error than any of the individual measurements. The average gives a best estimate of the true value of x, provided n→∞. There are several ways to indicate the error.

The maximum error could be specified as $\Delta X_{mzx} = (X_{max} - X_{min})/2$ and no measurements should even fall outside $\bar{x} \pm \Delta x_{max}$.The probable error Δx_{prob}, specifies the range $\bar{x} \pm \Delta x_{prob}$ which contains 50% of the measured values. The average deviation which is the average of all deviations from the mean is yet another measure. $\Delta x_{av} = \sum_R {|x_R - \bar{x}|}/{n}$. For Gaussian distribution of data, about 58% will lie within $\bar{x} \pm \Delta x_{av}$.

Because of the law of large numbers, the assumption that random errors are normally distributed (Gaussian distribution) is valid. So, it is a common practice to quote error in terms of the standard deviation of a Gaussian distribution fitted to the observed data. This is the way a researcher should quote error in reports.

While the standard deviation of any Gaussian distribution is given by $(\sigma) = \left[\frac{\sum(x_1-\bar{x})^2}{n}\right]^{1/2}$ the true standard deviation is $(\sigma_x)=[\Sigma (x_I - x)^2/n-1]^{1/2}$ The reason why we divide by (n-1) is to get the best estimate of the standard deviation. The true mean value of x is not being used to calculate the variance, but only the average of the measurements as the best estimate of it. Thus $(x_1 - x)^2$ as calculated is always little less than $(x_1 - \bar{x}\,\mathrm{true})^2$, the quantity really wanted. In the theory of probability, it has been shown that this under estimate is corrected by using (n-1) instead if 'n'.

If one more measurement of x is made, as per Gaussian property, it would have 68% probability of lying within $x \pm \sigma_x$. This means that about 30% of all experiments will disagree with the accepted value of more than one standard deviation (ie 1 σ).

We are also interested in the error of the mean, which is smaller than s_x if there are several measurements. An exact calculation yields, $(\sigma_x) = \left[\frac{\sum(x_1-\bar{x})^2}{n}\right]^{1/2}$ for the standard error of the mean. The researcher should report for this series of n measurements of $x\ as\ \bar{x} \pm \sigma_{\bar{x}_1}$ where $\sigma\bar{x} = \sigma x/\sqrt{n}$. The meaning of this is that if the n measurements of x were repeated, there would be a 68% probability the new mean value would lie within $x \pm \sigma_x$ (that is between $X \pm \sigma_x\ and\ \sigma_{\bar{x}}$ .This also implies that there is a 32% probability that it will fall outside this range. That means out of 100 experiments of this type, on the average, 32 experiments will obtain a value which is outside the standard errors.

Ex: the following measurements were made in 9 repetition of the experiment. 90, 95, 100, 110, 90, 115, 85, 120, 105. Compute the error estimation.

For the 9 measurements, $\bar{x}$ = 910/9 = 101, Σx = 910

$$\sigma_{\bar{x}}{}^2 = \frac{1177}{9} = 131 \qquad \sum(x_1 - \bar{x})^2 = 1177$$

$$\sigma_x = 11$$

The probability that one more measurement of x will lie within 100 ±11 is 68%. The value to be reported for this series of measurement is $\frac{X \pm \sigma_x}{\sqrt{n}} = 100 \pm {}^{11}/_{3}$ or 100 ± 4. If one were to make another series of nine measurements of x there would be 68% probability the new mean would lie within the range 100 ± 4.

Propagation of Errors

In many cases, the result of an experiment will not be measured directly, rather it will be calculated from several measured physical quantities (each of which has a mean value and an error). What is the resulting error in the final result of such an experiment?

If A and B are two measured quantities with errors ΔA and ΔB respectively, the combined errors in the worst possible case will be,

Z = A +B = (A + ΔA) + (B + ΔB) = (A+B) + (ΔA+ ΔB)

Z = A -B = (A - ΔA) + (B - ΔB) = (A+B) - (ΔA+ ΔB)

This could happen only if the errors in the two variables were perfectly correlated. (ie, if the two variables were not really independent).

In case the variables are independent then sometimes the error in one variable will likely to cancel out some of the error in the other and so, on the average, the error in z will be less than the sum of the errors in its parts. A reasonable way to try to take this into account is to treat the perturbations in z produced by perturbations in its parts as if they were 'perpendicular' and added according to the Pythagorean theorem.

$$\Delta Z = \sqrt{(\Delta A)^2 + (\Delta B)^2} \qquad \text{– (i)}$$

That is, if A = 100 ± 3 and B = 6 ± 4, then Z = 106 ± 5 since $\sqrt{3^2 + 4^2} = 5$.

Using this idea a general rule can be derived. Suppose there are two measurements A and B, and the final result is Z = F (A, B) for some function of F. If A is perturbed by ΔA, then Z will be perturbed by $\left(\frac{\partial F}{\partial A}\right)$ ΔA. Similarly for perturbation of AB in B, the effect of that in Z will be $\left(\frac{\partial F}{\partial B}\right)$ AB. Combining these as in the previous case,

$$\Delta z = \left[\left(\frac{\partial F}{\partial A}\right)^2 (\Delta A)^2 + \left(\frac{\partial F}{\partial B}\right)^2 (\Delta B)^2\right]^{1/2} \qquad \text{– (2)}$$

In the example of Z = A+B considered above, $\left(\frac{\partial F}{\partial A}\right) = 1$ and $\left(\frac{\partial F}{\partial B}\right) = 1$. So, this gives the same result as before. Similarly if Z = A – B, then $\left(\frac{\partial F}{\partial A}\right) = 1$ and $\left(\frac{\partial F}{\partial B}\right) = 1$,which also gives the same result.

If Z = (AB) then $\left(\frac{\partial F}{\partial A}\right) = B$ and $\left(\frac{\partial F}{\partial B}\right) = A$.

$$\text{So, } \Delta Z = [B^2 (\Delta A)^2 + (A^2 (\Delta B)^2]^{1/2} \qquad \text{- (3)}$$

$\therefore \frac{\Delta Z}{Z} = \frac{\Delta Z}{AB} = \left[\left(\frac{\Delta A}{A}\right)^2 + \left(\frac{\Delta B}{B}\right)^2\right]^{1/2}$ Therefore the fractional error is the square root of the sum of squares of the fractional errors in its parts. For example,

$$(100 \pm 0.3)(6 \pm 0.4) = 600 \pm 600\left[\left(\frac{0.3}{100}\right)^2 + \left(\frac{0.4}{6}\right)^2\right]^{1/2} = 600 \pm 40$$

$$\therefore (Z + \Delta Z) = AB + AB\left[\left(\frac{\Delta A}{A}\right)^2 + \left(\frac{\Delta B}{B}\right)^2\right]^{1/2}$$

If a variable z depends on one or two variables A and B, which have idependent error (ΔA and ΔB) then the rule for calculating the error in Z is tabulated below.

1. $Z = A + B \qquad (\Delta Z)^2 = (\Delta A)^2 + (\Delta B)^2$
2. $Z = A - B \qquad (\Delta Z)^2 = (\Delta A)^2 + (\Delta B)^2$
3. $Z = AB \qquad \left[\frac{\Delta A}{Z}\right]^2 = \left[\frac{\Delta A}{A}\right]^2 + \left[\frac{\Delta B}{B}\right]^2$
4. $Z = \frac{A}{B} \qquad \left[\frac{\Delta Z}{Z}\right]^2 = \left[\frac{\Delta A}{A}\right]^2 + \left[\frac{\Delta B}{B}\right]^2$
5. $Z = A^n \qquad \frac{\Delta Z}{Z} = n.\frac{\Delta A}{A}$
6. $Z = \log A \qquad \Delta Z = \frac{\Delta A}{A}$
7. $Z = E^n \qquad \frac{\Delta Z}{Z} = \Delta A$

IDENTIFICATION OF LIMITATIONS

It is imperative on the part of researcher to identify limitations of the study he has carried out and identify scope for further work while he do data analysis and interpretation. In particular the following aspects are worthy of consideration.

i) The researcher must spell out the constraints of his experimental setup, resulting in limitations on the outcome of his study. The constraints may be due to resource availability or time frame. The researcher should indicate as how the outcome of the experiment will get improved if such constraints are addressed.

ii) A model is developed for a specific purpose and the scope is mainly limited to the objective of the study. The researcher might have abstracted reallife situation for simplicity. He should spell

out as how improving abstractions improve the outcome of the model.

iii) Based on error analysis, the researcher should identify the short commings in his study and how by improving replications or otherwise the error could be reduced.

iv) If there is a significant gap between the findings of the researcher and accepted theoretical outcome, he should explain the inadequacies of his model or experiment as the case may be and how removal of such Inadequacies will improve the result.

v) Based on the outcome of the research work, if generalization could not be made, the limitations of his study, which are responsible for this must be highlighted.

vi) Any relevant aspect or question the study has not covered, the reasons there of should be identified.

USE OF OPTIMIZATION TECHNIQUES

Concept of Optimization

Optimization is the act of obtaining the best result under given circumstances. An engineer has to make decisions at several stages, when he design, construct and maintain an engineering system. The ultimate goal of all such decisions is either to minimize the effort required or to maximize the desired benefit. Since, the effort required or the benefit desired in any practical situation can be expressed as a function of certain decision variables, optimization can be defined as the process of finding the conditions that give the maximum or minimum value of a function.

The optimum seeking methods are also known as mathematical programming techniques and are generally studied as a part of operation research (O.R). The operations research is a branch of mathematics concerned with the application of scientific methods and techniques to decision making problems and with establishing the best optimal solutions.

One way of classifying O.R methods is as i) mathematical programming techniques ii) stochastic process techniques and iii) statistical methods. Techniques like Linear programming, Non-linear programming, Dynamic

programming, Network etc. form part of mathematical programming techniques. The stochastic process techniques include Markov process, Queuing theory, Simulation, Reliability etc. Regression analysis, pattern recognition, Design of experiments, factor analysis etc. form part of Statistical methods.

Statement of an optimization problem

An optimization problem can be stated as,

$$\text{Find } X=\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{which minimize } f(x)$$

Subject to the conditions or constraints,

$g_j(x) \leq 0$, for j = 1, 2 ……m

$h_j(x) = 0$, for j = 1, 2 .....p

$x_i >, 0$, for i = 1, 2 .....n

where x is a 'n' dimensional vector called the 'design vector' or 'design factors'.

f (x) is the objective function

$g_j(x)$ is inequality constraints and

$h_j(x)$ represent equality constraints

The number of variables 'n' and the number of constraints 'm' and / or 'p' need not be related in anyway. The above problem is a constrained optimization problem.

In the case of unconstrained optimization problem, it will have only the objective function as,

$$\text{Find } X=\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{which minimize } f(x)$$

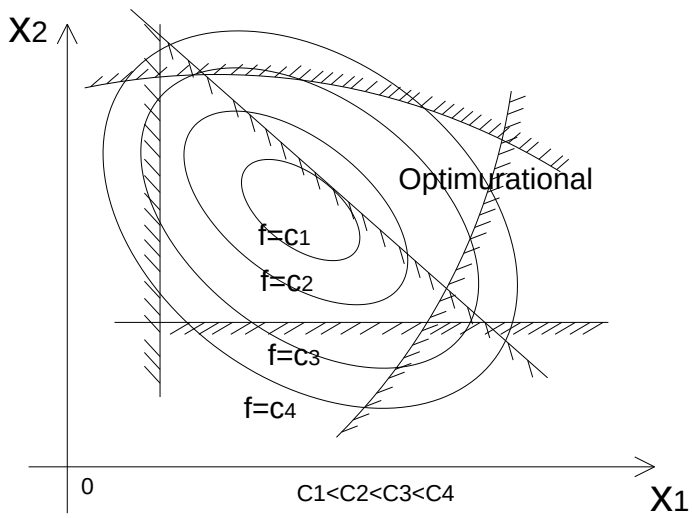
The functions f (x), $g_j(x)$ and $h_j(x)$ must depend upon some or all of the design variables, only then they are valid for the problem. The number of independent equality constraints must be less than or at the most equal to the number of design variables (is $p \leq n$). If all the functions f (x), $g_j(x)$ and $h_j(x)$ are linear in design variables, then the problem is called a linear programming problem. If any of these functions is non - linear the problem is called Non – linear programming problem (NLP).

The Optimal design point

In an optimization problem with only inequality constraints g_j (x) ≤ 0 ; j = 1,2….m, the set of values of x, that satisfy the equation g_j (x) = 0 in the design space is called a constraint surface. The constraint set for a design problem is a collection of all feasible designs. Mathematically 'S' is a collection of designs points satisfying all the constraints,

S = {x | g_j (x) = 0; j = 1, 2 …..m, hj (x) = 0; j = 1, 2……p}

Thus, S represents a set of feasible designs and is called as "feasible region". Feasible region shrinks when more constraints are added and expands when some constraints are removed. Likewise the locus of all points satisfying f (x) = C (constant) forms a hyper surface in the design space and for each value of 'C', there corresponds a different member of a family of surfaces. These surfaces are called "objective function surfaces".



Once the objective function surfaces are drawn along with constraint surfaces, the optimum point can be determined without much difficulty as shown in the figure for two design variables case. But, however once the number of variables exceeds two, the constraint and objective function surface become complex even for visualization. The problem has to be solved purely as a mathematical problem.

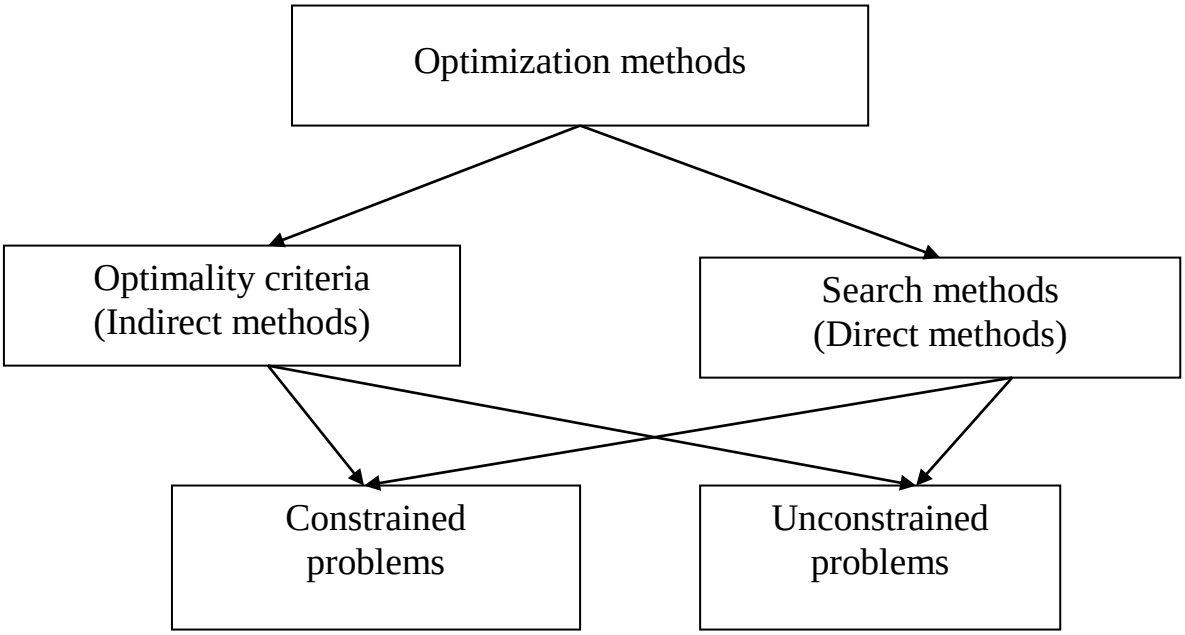
Optimization Techniques

The following three steps are adopted to transcribe a verbal statement of the design problem to a mathematical formulation,

- Identify and define design variables (factors)
- Identify the objective function and express it in term of design variables.
- Identify constraints and develop expressions for each constraint in terms of design variables.

It must be recognized that the overall process of designing system in different fields of engineering or purpose is roughly the same. Perhaps, the statement of the problem can contain terminology specific to the domain under consideration. Once the problems of different fields are stated using standard notation, they all look alike.

The optimization techniques can generally be classified as indicated below



Optimality criteria are the conditions a function must satisfy at its minimum point. Minimization techniques seeking solutions to optimality conditions are often called indirect methods. These methods assume that the functions is differentiable twice with respect to the design variable and the derivatives are continuous. Differential calculus can be used to find the unconstrained maxima or minima. These methods are also known as ‘classical methods’.

The Direct or Search techniques are based on a different philosophy. Here, we start with an estimate of the optimum design, which satisfy the optimality criteria, thereafter, it is improved iteratively searching the design space for optimum points or the difference between two successive improvements is very marginal. Several search techniques like Exhaustive search, Interval halting, Fibonacci, Golden section methods are available for one dimensional search.

Scope of Search and Optimization techniques

In many research projects, the researcher may be interested to find the values of levels of independent variables (factors) that yield the optimum (maximum or minimum) values of the response or dependent variable. If the dependent and independent variables are both quantitative and continuous Response Surface Methodology (RSM) using steepest ascent (or descent) approach could be used. Even experimental studies gives the value of response variable only for the given set of input variables assigned by the researcher. From the response for the limited number of experimental results, he decides which is the best among them for the assumed conditions. But, it is not known whether they really represent optimum level of input factors.

It may so happen in certain studies, the researcher may suspect multiple peaks in the response surface, or there may be several saddle points. These may pose more difficulty in searching for a global optimum. The presence of multiple peaks in the response surface would imply that some fundamental changes in the response of the system occur under certain combination of factors.

Traditional methods

Many of the real life problems involve multivariable and non – linear relationship. Over a period of time many computer based solutions for multivariable non – linear problems have been evolved. Traditional search and optimization methods can be classified into two distinct groups viz: those requiring information about derivative values in determining search direction for optimization (indirect methods) and direct methods that rely solely on the evaluation of the objective function.

We first consider methods for unconstrained multivariable optimization. Newton's method is an indirect technique that employs a second order approximation of the function. This method has very good convergence properties, but it can be an inefficient method because, it requires the calculation of n(n+1)/2 second order derivatives, where n is the number of design variables. Therefore, methods that require the computation of only first derivatives and use information from the previous iterations to speed up convergence have been

developed. The DPF (Davidon, Fletcher and Powell) method is one of the most powerful methods used. Updates and improvements on this method were provided by Broyden, Fletcher, Goldfarb, and Shannon in what is called the BFGS method. It appears this algorithm has better rating with practitioners.

One of the earliest direct search methods was the method of Hooks and Jesves. Because, it does not use derivatives it is fast and is less sensitive to irregular or discontinuous functions. The method does have problem in highly constrained situations where a search can get stalled when the contour lines and the inequality constraint line have certain orientation. Other direct search techniques are the Simplex method (no relation to the technique in linear programming) and Powell's quadratic convergence method.

Optimization of non – linear problems with constraints is a more difficult area. The more general approach to this class of problems is called the Generalized Reduced Gradient method (GRG). The idea of GRG is to convert the constrained problem into an unconstrained one by direct substitution. However, with non – linear constraint equations this is not fessible by direct substitution and the procedures of constrained variation and Lagrange multiplier must be used.

Another approach is to successively linearize the constraints and objective function of a non–linear problem and solve using the technique of linear programming. Sequential linear programming (SLP) is the name of the method. The recent and perhaps the best method of solving non–linear problem is sequential Quadratic programming (SQP). Computer programs like GINO, NLP solver, and OPTISOLVE are available to perform non–linear optimization on micro computers.

Common difficulties with Traditional methods

Because of the non–linearities and complex interactions among problem variables often exist in optimization problems, the search space may have many optimal solutions. Many of them are local optimal solutions (sub optimal) having inferior objective function values. When solving these problems, if traditional methods get attracted to any of these local optimal solutions, there is no escape

from it. Some of the common difficulties with most of the traditional direct and gradient based techniques are,

- Convergence to an optimal solution depends on the chosen initial solution.
- Most algorithms tend to stuck to a sub optimal solution
- An algorithm efficient in solving one optimization problem may not be efficient in solving a different problem.
- Algorithms are not efficient in handling problems having discrete variables.
- Many algorithms cannot be efficiently used on a parallel computing machine.
- The designer need to know many optimization algorithms

The above discussion suggests that traditional methods are not good candidates for an efficient search and optimization algorithm. Evolutionary optimization techniques such as Genetic Algorithm, Particle Swarm Optimization (PSO) etc. can be applied to a wide variety of problems.

Evolutionary Optimization Techniques

Evolutionary techniques (EA) have now gained immense popularity in real world engineering search and optimization problems. To be economical and competitive, it is essential to choose an optimal solution out of many feasible solutions satisfying the required performance criterion such as cost, performance etc.

Evolutionary Algorithms (EA) are computerized search and optimization methods that work very similar to the principles of natural evolution. EA's intelligent search procedure finds the best and fittest design solutions, which are otherwise difficult to find using other techniques. EA's are attractive and easy to use and they are likely to find the globally best solution, which is superior to any other solution. They are used in many fields such as design of components, process design, pattern recognition, VLSI design, control systems etc.

Genetic Algorithm (GA)

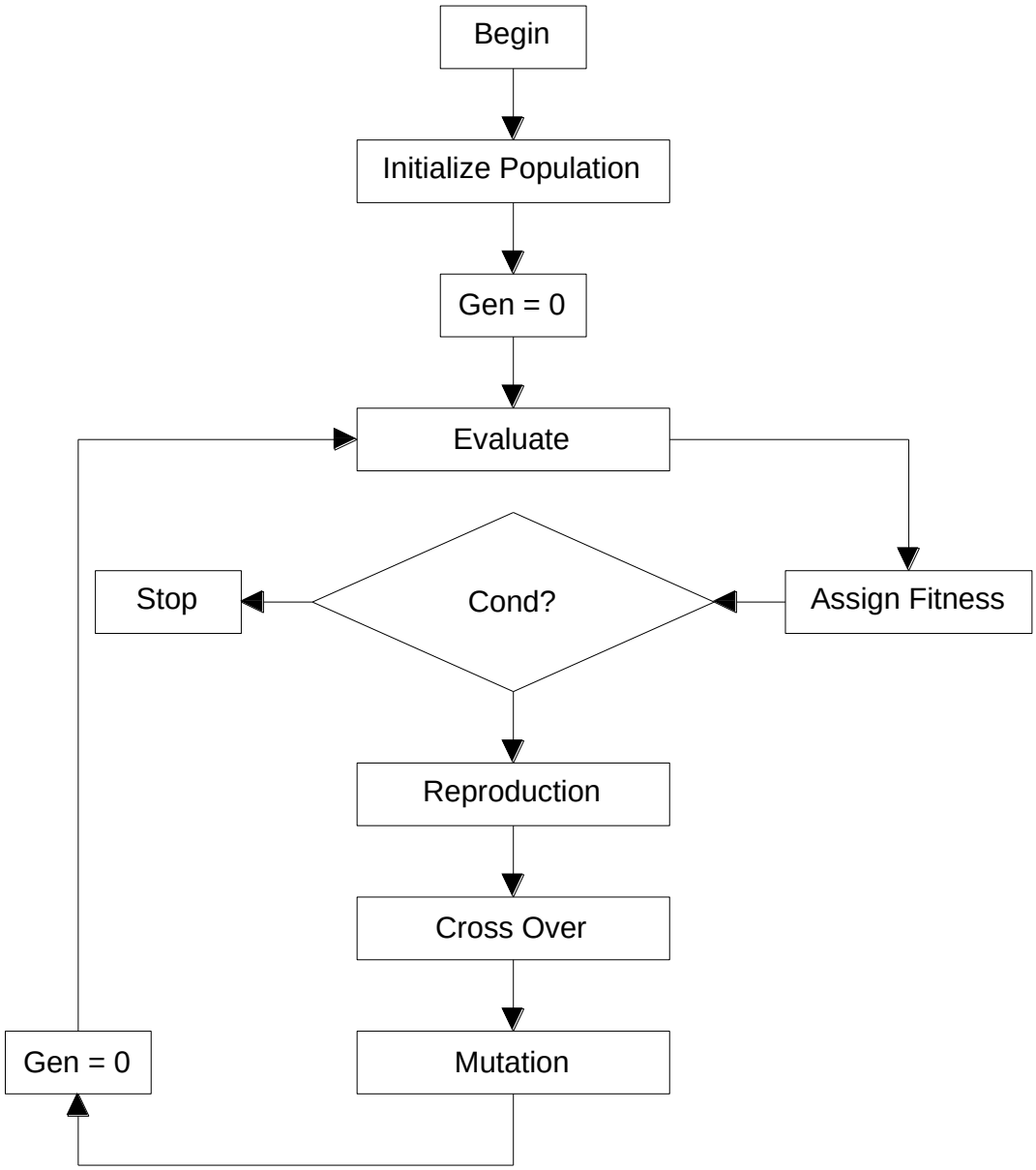
Genetic Algorithm is an evolutionary optimization technique, which works by mimicking the evolutionary principles and chromosomal processing in natural genetics. It is a search and optimization technique developed by Professor John Holland of University of Michigan in 1965. Over the last one decade and more, GAs have been successfully applied to a wide variety of problems, because of their simplicity, global perspective and inherent parallel processing.

A GA begins its search with a random set of solutions usually coded in binary string structures. Every solution is assigned a fitness which is directly related to the objective function of the search and optimization problem. Thereafter, the population of solutions is modified to a new population by applying three operators similar to natural genetic operators – reproduction, crossover, and mutation. A GA works literally by successively applying these three operators in each generation till a termination criterion is satisfied.

Working principle

As already stated GA is an iterative optimization procedure. Instead of working with a single solution in each iteration, it works with a number of solutions (collectively known as population) in each iteration. A flow chart of the working principle of a simple GA is given below.

In the absence of any knowledge of the problem domain, a GA begins its search from a random population of solutions. A solution in GA is represented using a string coding of fixed length. The GA process these strings in an iterative way. If a termination criterion is not satisfied three different operators–reproduction, crossover, and mutation–are applied to update the populat6ion of strings. One iteration of those three operators is known as a generation in the parlance of GA. Since the representation of a solution in GA is similar to a natural chromosome and GA operators are similar to genetic operators, the above process is named as genetic algorithm.



Representation

Binary code is adopted to code the variables in GA. The variables are of fixed length. For example the following is a string representing N population variables.

$$\underset{X_1}{\underline{11010}} \qquad \underset{X_2}{\underline{1001001}} \qquad \underset{X_3}{\underline{010}} \qquad \underset{X_N}{\underline{0010}}$$

Each staring created in GA either in the initial population or in the subsequent generations are assigned a fitness value, which is related to the objective function value. For maximization problems, a string's fitness can be equal to the string's objective function value. For minimization problems, the goal

is to find a solution having the minimum objective function value. The fitness function is calculated as the reciprocal of the objective function value ie,

$$\text{Fitness} = \frac{1}{1+f(X_1X_2\cdots\cdots\cdots X_N)}$$

Reproduction

Reproduction (selection) is usually the first operator applied on a population. Reproduction selects good strings in a population and forms a mating pool. A number of reproduction operators exist in GA. The essential idea is that the above average strings are picked up from the current population and duplicate of these are inserted in the mating pool. The commonly used method is the proportionate selection where a string in the current population is selected with probability proportional to the string's fitness. Thus the i th string is selected as $\frac{f_i}{\sum_{j=1}^{N} f_j}$ one way to achieve this proportionate selection is to use a roulette wheel with circumference of the wheel is marked for each string proportionate to the string's fitness. The roulette wheel is spun 'N' times and a string is selected based on the pointer indication.

Cross over

Cross over operator is applied next to the strings of the mating pool. A number of crossovers exist, but in almost all crossover operators, two strings are picked from the mating pool at random and some portion of the strings are exchanged between the strings as indicated below

Parent 1	0 0	0 0 0	0 0 1 1 1	(child 1)
		$\longrightarrow$		
Parent 2	1 1	1 1 1	1 1 0 0 0	(child 2)

The main purpose of the cross over operators is to search the parameter space. The other aim is that the search needs to be performed in a way to preserve the information stored in the parent string maximally, because these parent strings are instances of good strings selected using the reproduction operator.

Mutation

Cross over operator is mainly responsible for the search aspect of genetic algorithm, even through mutation operator is also used for this purpose sparingly. Mutation operator changes a 1 to a 0 and vice versa.

For example in 0 0 0 0 0 $\longrightarrow$ 0 0 0 1 0, fourth gene has been changed its value, thereby creating a new solution. The need for mutation is to maintain diversity in the population.

In the field of modern research genetic algorithms are a means of optimizing complete problems (ie problems with many relative optima, but only one absolute minimum) with very low effort. Purely numeric methods fail in this case, because examination of very small, limited areas requires enormous effort. Further it cannot be guaranteed that the absolute optima lies within the tasked area.

Particle Swarm Optimization (PSO)

Particles Swarm Optimization is another computational technique inspired by biological systems. Here, we consider another biological system – social system, more specifically the collective behaviours of simple individuals interacting with their environment and with each other. Some call it as swarm intelligence. Consider or scenario, where a group of birds are randomly searching food in an area. There is only one piece of food in the area, being searched. All the birds do not know where the food is, but they know how far the food is in each iteration. So, the best and effective strategy is to follow the bird which is nearest to the food. PSO learned from the scenario and used it to solve optimization problems.

The (PSO) is a stochastic optimization technique developed by Dr. Eberhart and Dr Kannedy in 1995, inspired by the social behavior of bird flocking or fish schooling as stated in the previous para. There are two commonly used versions of PSO viz i) global version, and ii) local version. Global version is faster but might converge to local optimum for some problems. Local version is bit slower but not easy to be trapped into local optimum. One can use global version to get quick result and use local version to refine search.

PSO is a population based search algorithm like GA and it is initialized with a population of random solutions, called 'particles' (birds). Each particle in PSO flies through the search space with a velocity that is dynamically adjusted according to its own and its companion historical behaviors. Each particle keeps track of its coordinates in the problem space which are associated with the best solution (fitness) it has achieved so far. The fitness value is also stored and this value is called 'pbest'. Another "best" value that is tracked by the particle swarm optimizer is the best value, obtained so far by any particle in the neighborhood of the particle, which is called "lbest"? When a particle takes all the population as its topological neighbors then the best value is a global best and is called "gbest".

The various steps involved in PSO is given below.

i) Each particle is considered as volume less (a point) in the 'n' dimensional search space. The ith particle is represented as $x_i = [x_{i\,1}, x_{i\,2}, x_{i\,3} \ldots\ldots x_{i\,n}]$

ii) The best previous (the position giving the best fitness value) of the ith particle is recorded and represented as $p_i = [p_{i\,1}, p_{i\,2}, p_{i\,3} \ldots\ldots p_{i\,n}]$

iii) The index of the best particle among all particles in the population is represented by the symbol 'g'.

iv) The rate of position change (velocity) for particle 'I' is represented by $V_i = [V_{i\,1}, V_{i\,2}, V_{i\,3} \ldots\ldots V_{i\,n}]$

v) The particles are manipulated according to the following equation $V_1 = V_{in} + C_1,^{*} rand\ (\,)^{*}(P_{in} - X_{in}) + C_2{}^{*} rand\ (\,)^{*}(P_{in} - X_{in})$ (1)

$$X_{in} = X_{in} + V_{in} \quad (2)$$

Where c_1 & c_2 are two positive constants, and rand ( ) & Rand ( ) are two random functions in the range (0,1).

vi) All particles in PSO are kept as numbers of the population through the course of the run (a run is defined as the total # of generations of the evolutionary algorithm prior to termination.

vii) It is the velocity of the particles which is updated according to its own previous position and the previous best position of its companions. The particles fly with the updated velocities. In other wards, at each generation, each particle in PSO can fly in a limited number of

directions which are expected to be good areas to fly towards according to the group's experience.

viii) In PSO a parameter called inertia weight is used for balancing the global and local search and equation (1) and (2) is changed to

$$V_{in} = W^{*}V_{in} + C_1{}^{*}rand\ (\ )^{*}(P_{in} - X_{in}) + C_2.^{*}\ rand\ (\ )^{*}(P_{in} - X_{in}) \qquad (3)$$

$$X_{in} = X_{in} + V_{in} \qquad (4)$$

ix) A large inertia weight facilitates a global search while a small inertia weight facilitates a local search. By linearly decreasing the inertial weight from a relatively large value to a small value through the course of the PSO run, the PSO tends to have more global search ability at the beginning of the run, while having more local search ability near the end of the run.

Thus at each step, the velocity (accderation) of each particle is changed towards its 'pbest' and 'lbest' location. Acceleration is weighted by a random term with separate random numbers being generated for acceleration towards 'pbest' and 'lbest' location. This procedure is repeated till the termination criteria is achieved thereby reaching the global optimum.

Comparison of PSO with GA

There are many similarities between PSO and GA. A comparison of these techniques is given below.

- Both start with a group of randomly generated population
- Both have fitness values to evaluate the population
- Both update the population and search for the optimum with random techniques.
- Both systems do not guarantee success.
- PSO does not have genetic operators to update but, particles update themselves with the internal velocity.
- PSO also have memory which is important to the algorithm.
- Compared to GA, information sharing mechanism in PSO is significantly different. In GAs, chromosomes share information with each other. The whole population moves like a group towards optima area while in PSO, only 'gbest' or 'lbest' gives out

information to others. It is a one-way information sharing mechanism.

UNIT - V
THE RESEARCH REPORT

PURPOSE OF WRITTEN REPORT

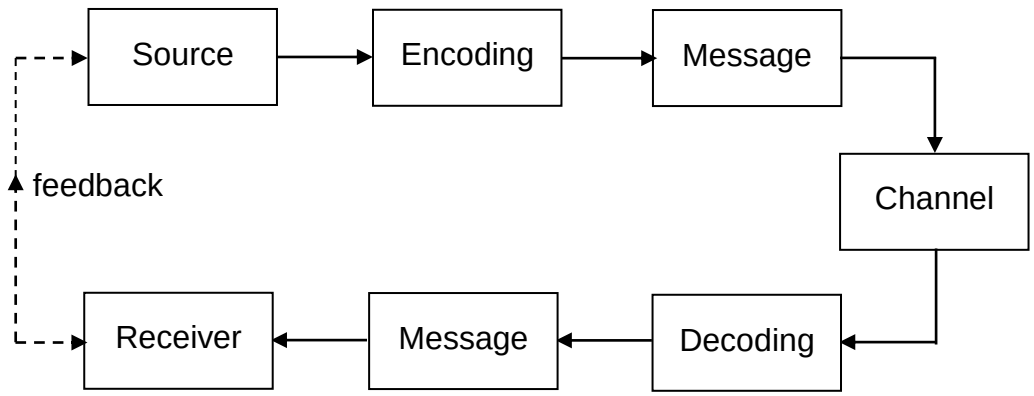
Introduction

A research scholar is expected to write and publish papers and articles in reputed journals and magazines. Often he/she does this as a part of the requirement of the research programme, especially doctoral programmes. In addition, the research study culminates in preparation of thesis based on the results of the study for evaluation by examiners. After evaluation, the scholar has to defend his thesis in the viva- voce examination. All these requires adequate knowledge and skill on the part of the scholar in preparing technical report for each need. Infact even the brilliant findings of a research study are of little value unless they are effectively communicated to the concerned audience.

This chapter deals with established practices, guidelines and rules which should be followed in the preparation of research report or paper. In addition, the ethical practices the researcher should adopt and how he can obtain patent for his original contribution are also covered.

Audience Analysis

Writing of a technical paper for publication or writing research thesis for evaluation by examiners is an exercise in written communication. The basic communication model is given in the figure below.



The written communication takes the form of papers, synopsis, brief memorandum reports, theses, technical proposals etc. In terms of the communication model, the source is the mind of the scholar or writer. The

process of encoding consists of translating the idea from the mind to words on paper. The channel is the pile of manuscript papers. Decoding the message depend upon the ability of the reader to understand the language and familiarity with the ideas presented in the report. The final receiver is the mind of the reader. Noise is present in the form of poor writing mechanics, incomplete diagrams, incorrect references, etc. Since, there is no direct feedback, the writer must anticipate the needs of the receiver and attempt to minimize the noise.

The first principle of written communication is to know the accidence, to whom the report is addressed, so that the researcher can anticipate and fulfill its needs. We have the natural tendency to write for ourselves rather than someone else and we must overcome this, therefore, we must actively force ourselves to analyse the audience and then tailor our presentation to their needs, not ours. This may not always be easy since the audience may be diverse or not familiar to us, but we must nevertheless do the best we can in appraising their needs and capabilities.

Audience for papers and articles

The distinction between a paper and an article may not be obvious or universally defined, yet it is worth to find subtle difference between them. By a technical paper, it is meant a document submitted to some technical society or industry group for printing in a journal and / or oral presentation at a conference. Such papers are usually reviewed by a group of referees who recommend acceptance or rejection with or without modification. An article is usually considered to be at a lover technical level than a paper, generally appears in a trade journal rather than society's transactions and is not subject to peer review.

The audience for paper published in reputed journals are widely read by researchers, students, scientists, professionals engaged in academic and in industry, planners, etc. As already stated, the editorial board of these journals do peer review by experts in the respective fields before publication. Sometimes, it takes more than a year to get a paper published. Moreover every journal has its style in the matter of presentation, length and depth of treatment. It is imperative, therefore the researcher should tune his paper to the requirements of the journal.

The researcher cannot assume too much of knowledge among the readers and make the text hard to understand.

As the researcher has worked in the area in which the paper is published, he is so certain concepts may seem to be too elementary or axiomatic to be explained. Even experts in the subject would welcome a clear exposition of ideas that are not well known in the discipline.

Quoting the statement of the Area Editor for the Journal Operations Research, relating to the area of manufacturing, service and supply chain management will enable the researcher to appreciate this. “We seek well written papers that span the full spectrum from theory to practice. Papers may be methodological contribution, contributed by developing and analyzing novel models motivated by current industrial problems or phenomenon; or make an empirical contribution by conducting and analyzing tests of hypotheses related to mathematical models, validating the behavior of systems or algorithms, or reporting the results of major implementation of specific methodologies. In all cases, however, the contribution must clearly be significant, relevant, and conceptually sound and must meet the rigorous standards of the journal”.

All these require proper understanding of the readership of the journal so that the researcher can orient his paper in the proper direction.

Audience for synopsis

Another important technical report, a research scholar submit on fairly completing his research work is synopsis. This summary report is submitted to the doctoral or research advisory committee which consist of 3 or 4 experts in the field. Eventhough they have been periodically review the work of the scholar, synopsis is the document they go through and accord approval to the scholar to write the thesis. Not only synopsis required for approval by research advisory committee, this document is also used by the university to get acceptance of the possible examiner to evaluate the thesis of the scholar.

The researcher has to understand the requirements of the synopsis and what the audience look for in that, so that he / she can meet the needs in the report. Synopsis is a brief summary of the research work carried out, indicating

the problem taken up for study, brief methodology, major findings and the researcher's original contribution.

Audience for research report

The scholar's thesis is the research report. The report concerns a problem or series of problems in the scholar's area of research and it should describe what was known about it previously. Further, it should describe what the researcher did to solve it and what he think his results mean and where or how further progress in the field can be made. Infact one should not carry over his ideas from undergraduate assessment; a thesis is not an answer to an assignment question. One important difference is that the reader of an assignment is usually the one who had set it and he / she knows the answer. The reader of a thesis do not know what the "answer" is. If the thesis is for Ph.D, the university requires that it should make some original contribution to human knowledge. Therefore the work of the researcher must discover something hitherto unknown.

The thesis examiners will be both foreigners and our nationals. They will be experts in the general field of thesis of the scholar but, may not be on the exact topic of the thesis. Infact, the researcher or scholar is the world expert. One should keep this in mind. The scholar should write to make the topic clear to a reader who has not spent most of the last three or four years thinking about it.

The thesis will also be used as a scientific report and consulted by future scholars of the same university, who want to know in detail about it. Theses are also consulted by people from other institutions and many universities store it in digital form as pdf files on a server. The advantage is that a scholar's thesis can be consulted much more easily by researchers around the world. In India, the University Grants Commission (UGC) has already made arrangement to get soft copies of theses from various universities in India for storing so that they can be referred by others easily. Now, it is mandatory that every scholar has to submit a soft copy to UGC after the viva-voce examination is over.

The thesis supervisor is expected to read each chapter of the thesis in its draft form so that he can make corrections, suggestions and comments. The supervisor expects the thesis to be as good as possible, because his / her

reputation as well as that of scholar is affected. So one has to take a positive attitude to all the corrections, comments which the supervisor gives. Each comment tells the way as how the scholar can improve his / her thesis.

PREPARING PAPERS FOR JOURNALS

When a researcher think of publishing a paper in a refereed journal, he cannot simply take the relevant portion of his thesis and send it to the editor. The researcher will have to prune down the extra fat and also edit it so that it is intune with the style of the journal requirement. Most often the scholar will have to rewrite and remove the foot notes and other obstructions for easy reading. Even the prose style may have to be different than the one adopted for the thesis. A wise approach would be to send an abstract to the editor and get the confirmation of his interest in the topic and his approval for the submission of paper.

On many occasions, the researcher publish paper in reputed journals based on his / her research work even before preparing thesis. Sometimes, this may be a requirement prior to submission of thesis. In that case the scholar have to prepare the paper keeping in mind the need of the journal to which he is planning to send the paper. Many technical journals require authors to follow a style manual provided by the journal. These style manuals may be quite specific about how papers for that journal are to be written.

An outline is very essential for preparing any papers for journal publication. A model outline is given below.

- Abstract
- Introduction
- Methodology
- Discussion of results
- Conclusions and findings
- Acknowledgement
- Tables
- Figures
- References.

The abstract section is the opening section of a paper which provides a brief summery of the paper. It briefly describes the objective of the study and the problem studied and how the solution arrived. The abstract should also highlight the major finding and how it is a contribution to the literature. The abstract enable the busy reader to determine its worth. The length should not be more than 250 to 300 words.

The introduction gives a brief background to the problem and how it is important. The problem can be stated as simply as possible and should be interesting. One should not over estimate the reader's familiarity with the topic at this stage. The introduction sets the stags in the some way like a summary but in greater detail.

In the methodology sections, depending upon the nature of the study the details should be covered. For example a study involving experiments, includes details such as how the data were obtained, equipments used and / or techniques employed.

The results and discussion part should cover how the results were obtained and what analysis were carried out. The data in the form of tables or figures may be placed at the end of the report and referred to by number in the results section. If the tables were limited and only one or two figures are there, can be included in the results and discussion section itself. Both the practices are prevailing. All arguments based on data are developed here.

The conclusion section states in a concise form as possible the conclusions drawn from the study and interpretations. In general this section is the culmination of the research work and it is directly related to the objective of the study.

Sometimes small appendixes are used for mathematical developments, sample calculations etc. that are not directly associated with the main body of the paper and if placed in the main body may impair the logical flow of thought.

Aspect regarding references will be covered under writing of thesis report.

PREPARATION OF SYNOPSIS OF RESEARCH WORK

A bench mark in the research study of scholar, especially for obtaining Ph.D., is preparation and submission of synopsis of his / her research work. As

already stated the synopsis required for assessment by the research advisory committee of the scholar and for preliminary acceptance by the probable examiner. Many Universities have prescribed detailed guidelines for the preparation of synopsis for the scholars to follow. A typical framework is given below.

- Introduction
- Motivation
- Objective (s) and Scope
- Description of the study
- Conclusions and Contributions
- List of Publications made
- References
- Proposed Content of the thesis.

It is better that the title of the synopsis must be the same as the title of the thesis, otherwise confusion may arise later. One should devote attention in the selection of title as it reflect the major contribution of the research work. It should not be too long and it should be crisp and convey the meaning properly.

Under the introduction section, the researcher briefly outline the technological / engineering / scientific relevance or significance of the research work to be reported in the thesis. The introduction can cover at the most a maximum of one page but preferably half a page. The scholar should be precise and include only the relevant background information. In addition, he can provide information on past works by giving appropriate references.

The scholar should further develop on the background material provided in the introduction section under motivation section. He should bring the state-of-the art in the chosen area of research and must clearly indicate the existing drawbacks and why further research is required to eliminate the drawbacks. The importance of research problem identified should be stressed based on the current status. Further, the researcher should enumerate those technical challenges he has to address to solve the problems posed in the study. This will exemplify the quality of the research work. The motivation section can cover one page at maximum or preferably half a page.

The next section to be covered is objective(s) and scope of the proposed research work. Here, the researcher should state clearly the questions for which answers are sought through the research. The conceptual, analytical, experimental and / or methodological boundaries within which the exercise has been carried out should be defined. Again this section could be restricted to half page if possible, otherwise a maximum of one page used.

In the section on description of the research study, the researcher should give brief but sufficient details regarding the research problem(s) and the solution methodologies adopted or developed by him / her. In case the study involves simulation and / or experiments, these should also be explained succinctly. The scholar should also explain as how he interpreted the results. If necessary, the scholar may introduce sub – sections appropriately to improve elegance. As this is the main body of the synopsis, a maximum of ten pages may be allotted for this.

Under the heading conclusions and contributions, the researcher should enumerate all major conclusions. He should clearly bring out the advantages arising out of the work to be presented in the thesis, highlighting original contributions of the research work. In addition, the researcher should also give the limitations of his research work, especially when he could not generalize results. In case, there are no conclusions to be drawn, then he should list the contributions of his work. Perhaps, he can change the title of the section as summary of the work.

The researcher should appreciate if the synopsis is written well, a prospective examiner will not hesitate to say ‘yes’ to review the proposed thesis. There is a feeling among the academicians that it is better to write the synopsis after writing the first draft of the thesis atleast.

It is imperative that the researcher should list the publications he made arising out of the research work in the next section captioned “list of publications made”. Here, he should give complete details using the standard format internationally adopted. Only the papers published or accepted for publication should be included. Even ‘papers under review’ may by included but not in any case ‘papers under preparation’. The listing sequence will be International Journal Publications first, followed by National Journal articles, then International

conference papers and lastly articles presented in National Conferences. If the researcher has obtained any patents arising out of his research work can be included in this section.

The next section is the reference section, Here, the references are listed in the same order as they are referred to in the synopsis, ensuring that all references listed in this section are properly referred in the text. It is better to restrict the number of references to ten or less. The order of reference will be i) books ii) patent iii) journal paper iv) papers presented and published in conference proceedings and v) web page. The specification for reference will be covered elaborately under thesis writing.

The last section of the synopsis will be the proposed contents of the thesis. In this section, the researcher should provide the chapter and section titles of the proposed thesis. Maximum of one or two pages may be allocated for this section.

Figures and tables must be embedded along the of the synopsis. They should appear immediately after the first time the figure or table is cited in the text. Every figure and table should be numbered properly. The figure should be legible and provide enough information so that it is “self contained”. Legends be made legibly so that they correctly describe the axes of a graph. In the case of units, adopt SI units. A synopsis can have a maximum of 16 to 18 pages in A4 size paper. The font and other details will be dealt with under thesis writing.

WRITING THE THESIS

Steps in writing a report.

Writing a research report is a major component of the research study. The research task remains incomplete till the report is written and presented. There are five operations involved in writing a high quality report, which can be easily remembered with the acronym POWER, where

P : Plan the writing

O : Outline the report

W : Writing the report

E : Editing the report

R : Rewriting the report

Plan the Writing

The planning stage of the report is concerned with assembling the data, analyzing the data, drawing conclusions from the data analysis and organizing the report into various logical sections. The planning of a report is usually carried out by considering the various facets of the work and providing a logical blend of the material. The initial planning of the report should begin before the work is carried out. In that way the planning of the work and planning of the report are woven together, which facilities actual writing operation.

The choice of a suitable title for the thesis should not be treated lightly. The need for a proper title was also stressed in the synopsis section. The researchers and other professionals mostly search for information based on the title and decide whether to pursue a particular reference. Therefore, we should try to come up with the most descriptive and concise title that we can. If possible the title should clarify whether the research work is theoretical, experimental or computer simulation and whether it is original or a review of previous work. The title should aid the literature searcher easily.

The researcher should also decide about the proper choice of a word processor he / she is going to use. The word processor software is a very useful tool due to its ability to move sections of the report by “cut and paste”, to insert or change words, to check spelling and grammar, and to increase the readability by changing the format or size of type or font etc. Commercial word processors have gradually become bigger, slower, less reliable and more awkward to use as they acquire more features. A researcher needs a word processor compatible with MS ward, but which allows him to type mathematical symbols and equations conveniently. One solution is LaTex, which is a powerful, elegant, reliable, fast and freely downloaded software. The web site is http: // www.latexproject.org//

As a part of planning, the researcher should also decide as how he / she is going to organize the files. One word processor file can be opened for each chapter and one for references. He can put notes in these files, as well as text. This gives lot of flexibility to swap between chapters if necessary. Apart from digital files, the scholar should have a physical filing system: a collection of folders with chapter numbers on them. All relevant materials collected from

different source can be put in the respective folders for easy reference as and when required while writing the thesis.

It is advisable that a scholar should sit down with his supervisor and prepare a time – table for writing the thesis. The time – table should contain list of dates for submission of chapters to the supervisor for correction and comments. This will structure the time of the researcher and provides targets. If he merely aim to finish the whole thing by some distant date, he may procrastinate more easily. If needed, the time – table can be put in the form of bar chart.

Outlining the report

An outline of the report actually consists of formulating a series of headings, sub – headings, sub – subheadings, etc., which encompass the various sections of the report. Figures, titles (to indicate which results go where) and some other notes and comments may also be included. Once the researcher prepare a list of chapters and under each chapter heading a reasonably complete list of things to be reported, he / she have struck a great blow against writer's block. When he / she sit down to type or write, his / her aim is no longer a thesis – a daunting goal – but something simpler. The aim of the researcher now is to write a paragraph or section about one of his / her sub – headings. May be, he can start to write with a easy one which gives self confidence.

As a part of preparation of outline, a table of contents, which may be tentative, should be prepared. This is a framework like a skeleton of a human body to which the material is stuffed. The content should enumerate all chapters, main heading, sub – headings etc. in a logical order. If main headings and sub – headings are numbered according to some scheme, it is better to keep such numbering simple. Many universities favour decimal numbering system. For chapter number, Arabic numeral such 1,2 are used with main heading numbered as 1.1, 1.2 etc. and sub – headings 1.1.2, 1.1.2 etc. But, however, one should not go to the extend of numbering like 1.2.3.3.8, which is awkward and confusing proliferation of sub – sections.

While formulating chapter main headings, sub – headings etc., the researcher should keep in mind the chapter outline he formulated. This leads to

a question as how one can make an outline of a chapter. A suggested approach is that the research can assemble all the figures he / she will use in it and put them in the order that the researcher would use it if he were to explain to someone what they all meant. Perhaps, the are researcher can rehearse it if necessary. Once he / she finds a logical order, the key words of the explanation may be noted down. These key words will provide a skeleton for much of the chapter outline.

Once an outline is prepared, the researcher should better discuss with the supervisor. This step is an important one as it provides a chance to get useful suggestions and comments from his supervisor. Based on this, the researcher can refine the outline. It is better that a copy of the agreed outline is given to the supervisor for reference when reading chapters which may probably present out of order. If the scholar has a co – supervisor, he should discuss the outline with him also before finalizing it.

Writing the report

This section forms the crux in writing a thesis. The various issues concerned here are,

- How much detail one should provide?
- How to make clear the contribution of the researcher?
- What are the issues involved in the mechanics or principles involved in expression?
- How to present graphs, charts etc.?
- Where to stop?

The short answer for the question, how much details is more than for a scientific paper. The thesis is mainly read by fellow researchers who are seriously doing research in the same area. They may be interested to find out exactly what the researcher of the thesis did. They may look for details of experimental arrangement, circuit diagram or a sub routine of the computer programme, or how he had applied a particular technique and in what context. Therefore the thesis should include workshop drawings, circuit diagrams, computer programmes as appendices. The researcher who had written the programme knows each line of the code, but for somebody else to understand it

properly, it should be extensively annotated. The researcher might have read the thesis of others when he / she was working on the research topic and so know the advantages of a clearly explained, explicit thesis and / or the disadvantages of a vague one. Personal experience is the best judge in this.

It is incumbent on the part of researcher to make clear distinction between his / her views and that of others. If a researcher use a result or observation that is not of his own, he must state wherein the scientific literature that result is reported. The only exceptions are cases where every researcher in the field already knows it: dynamics equations need not be followed by a citation of Newton, circuit analysis does not need a reference to Kirehoff. The importance of this practice in science is that it allows the reader to verify the researcher's starting position.

In some sciences results are built upon results which in turn are built upon results. They are vertical sciences. For instance Physics is a vertical science. Good references allows the reader to check the foundations of the researcher's additions to the structure of knowledge in the discipline. It helps the reader to trace back to a level so as to enable him to judge as reliable. Good referencing also tells the reader, which parts are additions by the researcher to that knowledge. In a thesis, written for the general reader who has little familiarity with the literature in the field, this should be especially clear. The researcher should eschew the practice of leaving out a reference in the hope that the reader will think that a nice idea or a nice bit of analysis, is that of the researcher. In today's digital world one can easily final out via the net.

Mechanics of writing

Style and tense

The text of thesis must be clear. Good grammar and thoughtful writing will make the thesis easier to read. Scientific writing has to be formal. Since scientific English is an international language, slang and informal writing will be harder for non – native English speaker to understand. The researcher should use short, simple phrases and words rather than long ones. But, however where it is essential to use a complicated sentence because the idea is complicated, the researcher can resort to this. The need is clarity in written communication.

Sometimes it is easier to present information and arguments as a series of numbered or bulleted points, rather than as one or more paragraphs. A list of points is usually easier to write. The researcher should be careful not to use this presentation too much. The thesis must be connected, and should contain convincing argument, not just a list of facts and observations.

An important stylistic choice is between the active and passive voice. The active voice ("S measured the temperature….") is simpler, and it makes clear what the researcher did and what was done by others. The passive voice ("the temperature was measured…..") makes it easier to write ungrammatical or awkward sentences. The researcher should be wary of dangling participles if he / she use the passive voice. The choice whether to use active voice or passive voice is a question of taste. In scientific writing, it used to be required that the author use the third person and passive voice. The idea was to make the writing sound scientific ie. objective and detached from any personal influences. To day many style manuals for high level technical journals encourage a less formal and more conversational tone by using active voice.

Authors like Ernest O. Doebelin recommend the use of active voice and first person as the usual form unless there are specific requirements to the contrary. Some of the Australian Universities also prefer active voice because, it is clearer, more logical and makes attribution simple. The arguments against the use of active voice is that many theses are written in the passive voice and some very polite people find the use of "I" immodest. It appears the choice of active voice or passive voice not only depends upon the choice of the researcher but also that of the university.

The choice of tense of verbs is often confusing to student writers. The following simple rules is usually employed by experienced writers.

i) Past tense is used to describe the work done in a laboratory or in general to past events (eg: Hardness readings were taken on all specimens)

ii) Present tense is used in Preference to items and ideas in the report itself (eg: It is clear from the figure.4 that strain energy is the driving force for recovery).

iii) Future tense is used in making predication from the data that will be applicable in the future. (eg: The market data given in table.2, indicate that the sales will continue to increase in the next ten years).

Organization of Sentences

Just as experienced readers expect an entire report or thesis to have a structure or organization there is also a more subtle and sub-conscious effect of the structure of individual sentences. Information is interpreted more easily when it appears most reader expect it to occur. This fact can be used to improve the sentences and paragraph structure.

A reader will generally expect the subject of a sentence to be followed fairly closely by the verb. When this does not happen, clarity suffers because, without verb, we don't know what the subject is doing and thus what the sentence is all about. Therefore, the researcher should avoid separating; the subject and verb too much; and watch for this problem when the proof read rough draft of thesis.

Another aspect deals with the materials that we place in the topic position (beginning of the sentence) and the stress position (and of the sentence). The reader will naturally expect that material at the beginning of sentence will provide linkage (looking backward) into the previous sentence and context (looking forward) in to the new sentence. Thus the topic position should be used for such transitional material which provides a smooth flow between sentences. Readers also expect a sentence to build emphasis at the end. Thus, the researcher should strive to place the next important material in that location, the stress position.

Suitable punctuation can create stress positions, allowing several pieces of stress worthy information to be clearly presented in a single sentence. This is most commonly done with a semi-colon, which creates essentially two 'sentences' each with its own stress position. Though such constructions lead to long sentences, which nevertheless are clear and easy to read some authors say that a sentence should not exceed 35 words. While lengthy sentences should not be the norm, length alone is not the problem. Even a 15 word sentence can be

quite unintelligible while a properly designed 40 word sentence can be perfectly clear.

Paragraph structure

The researcher should structure the paragraphs properly. Most writers prefer to begin a paragraph with a topic sentence. This sentence presents the idea to be developed in the rest of the paragraph. It is always better to have only one main idea in each paragraph; however simple ideas may require one or a few sentences leading to evenly short paragraphs. The researcher should then try to treat several ideas in one paragraph by showing some relation among them.

Complicated ideas can lead to overly long paragraphs. Long, unbroken sections of the text repel the reader's eye and overload the memory, so the researcher need to force paragraphs in such cases, even though he is still developing the main idea. The researcher can usually achieve this by using suitable transitional words at the paragraph break. It is better to have one or more paragraph breaks per page.

Choice of words

In addition to preparing a proper framework for the thesis, the paragraphs and sentences, it is vital that the words the researcher choose be the best possible. A researcher gets a good vocabulary mainly from wide general reading, but other tools are available for those who need to improve this background in a limited time. A good word processor will have a rather extensive thesaurus, and regular use of this will gradually improve the researcher's word choice. Some writing manuals have alphabetical lists of misused words which the researcher can always refer. The technical writer's Handbook by Matt Young is especially useful book for researchers in this.

While writing a thesis, the researcher should avoid use of vague words, unfamiliar words, needlessly complicated words, overworked words and phrases. At the time of proof reading the researcher should look for unnecessary words and delete them. A word becomes vague whan the researcher use it in the place of a correct specific word that is available. For instance, instead of using a specific word like hourly, daily, weekly where it is warranted, if the writer uses the

word ‘periodically’, then it becomes vague. Therefore, the word selection and context is very essential.

The researcher should not use words unnecessarily. For example a sentence is written as “Though the use of numerical integration, we are able to effect a solution of the equation”. Unnecessarily many words have been used in this. A better way to write this sentence is, “we solved the equation by using numerical integration”. Similarly if a word is very frequently used, it may lead to monotonous of reading, therefore, a researcher should use proper alliteration of words. A thesaurus will be of much use in this.

Misuse of words should also be avoided to the extent possible. For absolute words such as ‘perfect’, ‘essential’, ‘unique’ and ‘certain’ do not normally take modifiers. It is better to avoid phrases such as ‘very perfect’, ‘most essential’, ‘less unique’ etc.

Presentation of Graphs, figures and tables.

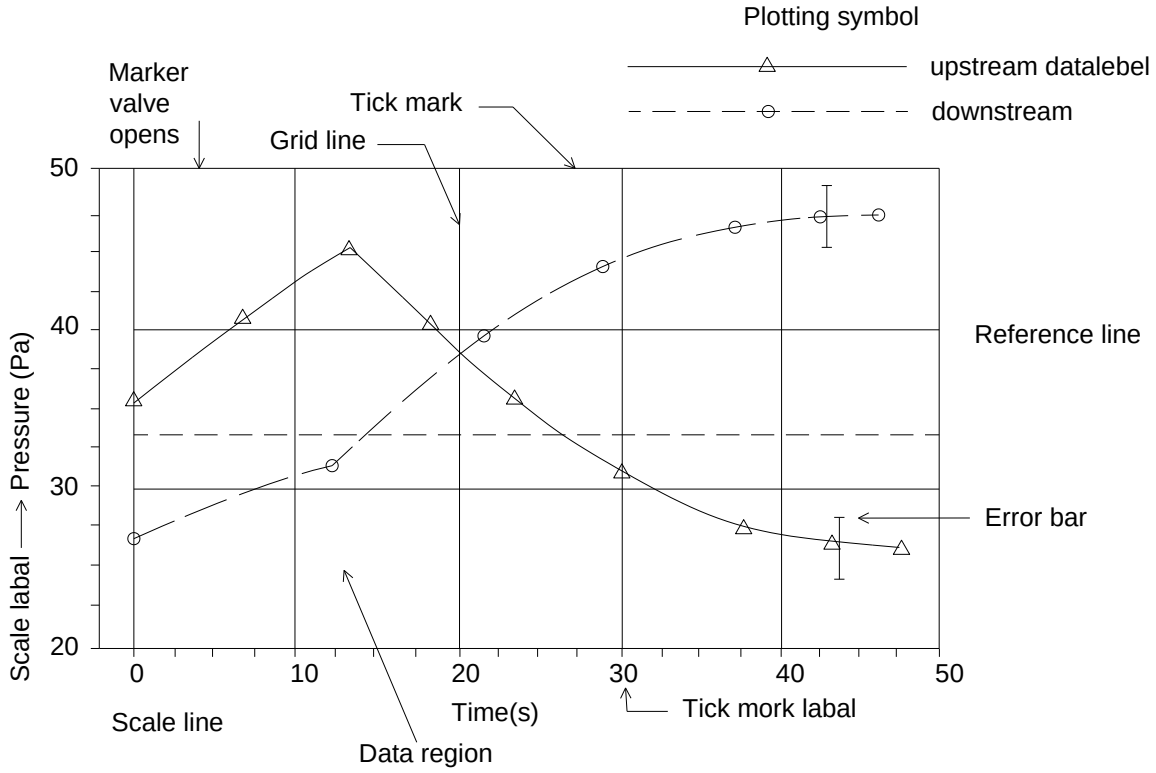
It may be necessary for the researcher to include graphs in the thesis, which can be of great aid in understanding. Reference material relating to graphs can be classified into two main types. One type concentrate on the principles guiding the direct visual communication between the graph and the reader. The other deals with methods used to analyse the meaning that may be hidden to the casual eye. The presentation graphs are used to present or display data for many purposes.

Nowadays computer based softwares are extensively used for preparing graphs of various kinds, therefore researcher should select a proper software for this. Scale lines (also called axes) form the horizontal and vertical borders of a rectangle called the data region. Instead of using one vertical scale line on the left and one horizontal scale line at the bottom, it is better to use two horizontal and two vertical lines to completely enclose the data region in a box. Along the scale lines tick marks may be placed to locate numerical values. The data region should be kept uncluttered as for as possible.

It is better to avoid gridlines, but if grid lines are felt necessary for some specific reason, they should be made light related to the bolder data curve. A key or legend has to be placed outside the data region to identify some feature of

data within the data region. Different symbols used for plotting each of several curves in the data region and this should also be included in the legend. Super imposition of curves can be made if it is necessary. But, if it is going to cluster, then they could be provided in several bands located next to each other, the curves are juxtaposed. When they are juxtaposed, the data labels are placed inside the data region in a location where they do not interfere with the curves or other features.

Depending upon the need, markers which are brief notes usually outside the data region with arrows pointing to specific locations on the scale line can be used to highlight certain aspects. The details are given in the figure below:



Error bars are 95% confidence intervals when data have been subject to statistical analysis which provides both expected valves and measures of uncertainty. The researcher may want to include in the graphs some indication of uncertainly in the plotted points. Error bars are the most common means of doing this and methods for superimposing them on the basic data plots are provided in many commercial software's. When error bar option is invoked proper mention should be made in the text of the report the precise meaning of the displayed error bars.

Many software packages provide three dimensional graphical displays, which may look impressive, but the researcher needs to be little critical because, the beauty of the display overshadow its real utility in communicating information. The two dimensional graphs even though multiple in number provide clearer and more detailed information. Therefore, the researcher has to be careful in choosing a particular display depending upon the context and clarity required.

Various types of bar charts such as clustered bar, stacked bar and pie charts are available for the researcher to use. Instead of providing simply bar or pie chart, if more details are needed, the researcher can add labels with exact quantity to each bar. One defect of pie chart is the poor angle-measuring capabilities of the human eye and mind, therefore, this can be easily overcome by simply labeling the pieces of pie with their percentage contributions.

The term figures include line drawings, computer solid modeling, photographs and text charts. It is better to use line drawings because they are by far the most common in reports and papers. Line drawings can be classified into pictorial drawings, schematic diagrams and block diagrams. Pictorial drawings can be two dimensional or three dimensional and are simplified picture of some real object or objects. Schematic diagram are more abstract than pictorial drawings, showing symbolic elements and their interconnection to make clear the configuration and / or operation of a system.

When making pictorial drawings, the researcher should give enough thought to the process and his / her illustration shows only the needed items and not more. The advantage of a drawing over photograph is that the researcher has complete control over which parts of the photographic view he has to choose and which he can leave out. A reasonably neat diagram can be drawn by hand faster than with graphics package and the researcher can sean it if wants an electronic version. Either is usually satisfactory.

A one bit (ie black and white), moderate resolution scan of hand drawn sketch will be bigger than a line drawing generated on a graphics package, but not huge. While talking about the size of files, one should note that photographs look pretty but take lot of memory. The major advantage is the person who draw the schematic diagram can think about the components to be depicted and the way in which the components of the system interact with each other. So

numerically small information content of line drawing may be much more useful information than that in the photograph.

Though line drawings usually are more informative than photographs, however, there are cases where photographs are desirable or essential. Surface details of metallurgical samples or failed specimens may be vital for proper interpretation. Many flow visualization methods use photographic records for data presentation and analysis. The researcher has to use line drawings pictorial drawings, photographs etc. optimally depending upon the situation.

Another important point for the information of researchers with regard to figures and photographs is as how to save them in the computer. In the digital version of the thesis, the researcher need not save ordinary photographs or other illustrations as bit maps, because they take up lot of memory and are therefore slower to transfer. Many graphics packages allow saving in compressed form as .jpg (for photos) or .gif (for diagrams) files. Further the researcher can save space / speed things up by reducing the number of colours. In vector graphics (as used for drawings) compression is usually not necessary.

The design of tables usually presents less difficulty for a researcher than with drawings and diagrams. Ruled tables are most common, but he / she can also use text tables and bulleted lists. A text table is usually short and is not numbered, titled or ruled into rows and columns; it is considered considered part of the text. For example,

TRANSDUCER TYPE	DC RESPONSE?	STATIC CALIBRATION
Piezoelectric	No	usually not

Bulleted lists generally have one column and not usually numbered. For instance,

TYPES OF DISPLACEMENT TRANSDUCERS

- Potentio meter
- LVDT
- Capacitance
- Eddy current
- Optical
- Pneumatic

Ruled tables are major tables which are titled, numbered and ruled into rows and columns. The following is an example of a ruled table.

TABLE 7.2

RESULTS OF THREE ALLOYING TREATMENTS

Property	Alloy 1	Alloy 2	Alloy 3
Hardness, Rockwell C	54	43	49
Specific weight kg / m^3	0.253	0.289	0.222
Tensile strength MPa	210	180	200

A better arrangement for this table could be

	Hardness Rockwell C	Specific weight Kg / m^3	Tensile strength MPa
Alloy 1	54	0.253	210
Alloy 2	43	0.289	140
Alloy 3	49	0.222	200

This arrangement make it easy to compare the various properties.

What to end writing the thesis?

The writing operation of thesis should be carried out in the form of a rough draft using the maximum technical and computational skill at the command of the researcher. The writing should be in conformity with the table of contents prepared and the general outline adopted. However, the researcher should not be worried about perfection at this stage especially fine details such as punctuation or sentence structure.

Drawing of a deadline for completing writing of rough draft is very useful. The researcher should hand in the thesis even if he thinks that he need one more draft of a chapter or to include someone's comment on a particular section or some other refinement. The researcher should not forget that a thesis is a very large work and that it cannot be made perfect in finite time. Inevitably there may

be things that the writer could have done better. So, the researcher has to set a deadline, when he prepared the time table and he has to strict to that.

Editing and Rewriting the thesis

Editing is the process of reading the rough draft and employing self criticism. It consists of strengthening the rough draft by analyzing paragraph and sentence structure, economizing on words, checking spelling in general, asking oneself the question “why”? Editing can be the secret of good writing. It is better for the researcher to ask himself embarrassing questions than to hear them from his technical readers, a supervisor.

As already mentioned elsewhere in this chapter, it is better that a researcher give his / her rough draft to his closest colleagues who could takes the pain of going through the draft and offer comments, which the researcher can consider and make amends. The supervisor is bound to go through each and every chapter of the thesis, suggest corrections, additions and deletions. Infact sometimes he may even suggest rewriting certain portions of the thesis, add some graphs and some tables.

The researcher might have published papers based on his / her work even before writing the thesis. When he sent papers, the editorial board of the journal normally subject the paper for peer review by experts. These experts offer critical comments, suggestions etc., which the researcher carries out and sent the paper again for publications. When a researcher write the thesis, all the comments made by peer review members should also be taken into consideration. If by chance any omission took place, he / she to has carry out this while editing the thesis.

It is quite likely the research advisory committee while reviewing the synopsis make certain observations or comments. This not only calls for editing the synopsis but also incorporate the suggestions made in the thesis. With regard to editing, it has been said that a best writer makes good use of both the ends of the pencil.

It is generally good practice to allow sometime, atleast a day to elapse after writing the rough draft before editing it. This allows the researcher to forget the logical pattern used in writing the thesis and appear more in the role of an

unbiased reader while editing. Many mistakes or week lines of thought that would normally escape unnoticed are thereby uncovered. The rewriting operation consists of retyping or rewriting the edited rough draft to put it in a form suitable for the evaluators.

The researcher must be aware that how much he / she reflect and how many times he edit it, there will be something that could be improved. There is no point hoping that the examiners will not notice. Experience reveals that examiners do suggest some correction or addition of points since they meticulously go through the thesis taking time. Similarly during defense of the thesis in a public viva – voce, suggestions and comments are likely to arise, which the research has to incorporate in his / her thesis. Therefore editing and rewriting takes place at several stages of the research work.

STRUCTURE OF THESIS

As stated in the section on thesis outline, the thesis structure will contain as how the various parts of the thesis should be arranged. It is a stand alone document. There is no single framework, which is suitable to all types of theses. Depending on the nature of study the list of contents and chapter headings will vary. The researcher should think about the plan of the chapters and decide what is best to report his work. He should make a detailed list of points that corresponds to heading and sub – headings or even the paragraphs of his thesis. He should also think about the logic of the presentation within each chapter. However, a general structure of the thesis may not vary much.

A suggested structure of a thesis is given below

Declaration / Bonafide certificate

Title page

Abstract

Acknowledgement

Table of contents

List of figures and tables

Definition of symbols used.

Introduction

Literature review

Core chapters

Findings and discussions

Conclusions and suggestions for future work

References

Appendices.

Declaration / Bonafide Certificate

The researcher has to make a declaration using the phrases recommended by the university / institution. Many universities have standardized the format for declaration. An example format runs like this," I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material, which to a substantial extent has been accepted for the award of any other degree or diploma of any university or institute of higher learning, except where due acknowledgement has been made in the text. "(Signature / Name / Date). It may be a bonafide certificate by the supervisor.

Title page

This page will contain the title of the of the thesis, author's name with an indication that it is being submitted for the degree of Ph. D / M.S in the concerned faculty / University and the date of submission. Again the format is prescribed by the university in many instances.

Abstract

Abstract is an important part of the thesis. This is the part that would be read the most and would find a place in many future journals and in Dissertation Abstracts. Therefore, this should be drafted with utmost care. It is better to write the abstract after the finalization of the draft thesis and, it may require several revisions. It should be a distillation of the thesis; a concise description of the problem(s) addressed, the method adopted to solve them, results and conclusions. An abstract must be self contained and no references included. In case a reference is necessary, its details should be included in the text of the abstract. It should be narrative and the number of words not to exceed 600 or the

limit prescribed by the university. Though it appears at the beginning, it is not an introduction and it is a resume of a thesis.

Acknowledgements

It is conventional to express gratitude to all those individuals or institutions that helped the researcher in the process (advice, scientific / technical / financial support, etc.) If the research work is a collaborative endeavor, the researcher should clearly indicate the contributions of individuals involved. A page may be devoted for this.

Table of contents

The table of contents should contain chapter headings, sub – headings and their respective page number in a chronological order. It is conventional to use Arabic numerals in numbering from the introduction page till the end of the thesis and Roman numerals for the pages before the introduction page. As the thesis will be used as a reference, the title of contents helps to locate matters easily.

List of figures and tables

These are provided to help the reader to locate these features when they are referred to in a text location remote from the feature location. That is, if figure 12 appears on page 23 but is later referred to on page 134, the reader should be able to quickly locate it. The list should use exactly the same captions as they appear below the figures. The same is applicable in respect of tables also.

List of symbols and abbreviations

Definition of symbols and abbreviations used in the thesis provide clarity to the reader. They include both word description and the proper units. However, inclusion of this section is optional and may depend on the necessity for providing such a list.

Introduction

This should embrace the significance of the research topic and its place in the overall scheme of things in the vast discipline of the researcher. The relevance of the study and the problem statement should be couched in a fascinating style and in a language that a non – expert in the particular area of the study can easily follow. The introduction should arouse curiosity and interest in the reader to go through the thesis. Infact, the researcher should not over estimate the reader's familiarity with his topic.

The purpose of introduction is to introduce the research project to the reader. It should contain clear statement of the objectives of research which provides background to the problem considered. A brief summary of other relevant research may also be stated so that the present study could be seen in that context. Hypothesis if any, methodology adopted may also very briefly stated. How observations and analysis were made forms part of introduction. In addition the limitations if any be narrated.

The introduction should be interesting so as to enthuse the reader to evince interest in studying the thesis. This section might go through several drafts to make it read well and logically, while keeping it short. If necessary revise it after the thesis is written.

Literature review

The literature review would encompass genesis of the problem in the mind of the researcher, the extent of knowledge already available about the problem. The literature review will have a fairly extensive list of significant papers or other works in the area along with the summary of their relevant findings. The researcher should link the background information with the proposed research problem. It was explained in detail in earlier chapter as how a researcher should go about doing a literature survey. What is required is maintenance of proper record either in the form of a spread sheet file or a word processor file by the researcher since this is done over a period of 2 to 3 years.

Some authors feel that an order of 100 documents (papers) will be reasonable. But, however, it is a matter of judgement by the researcher as how many papers he should scan through and how relevant are they before he

include them. Since the researcher is the world expert on the (narrow) topic of his thesis, the literature review will demonstrate his expertise. A word of caution is that the researcher should not omit any relevant paper from which he has extensively borrowed ideas.

Core Chapters

The structure of the core varies widely with the discipline and the nature of the topics. It may not be single chapter wherein all materials are input by the researcher. For convenience and easy readability the various chapters or sections under the core category has to be evolved. One cannot put forward a pattern to be followed universally.

The researcher should eschew the tendency to mechanically compile the papers, which he had published in journals based on certain aspects of his / her research work. There are many disadvantages in this approach. The journal articles cannot substitute the independent writing of thesis by the researchers. The papers would have suffered the constraints of length and detail to agree with the norms of the concerned journals. Writing thesis independently offers much more freedom in terms of length breadth and depth of the treatment. The researcher is the judge and he / she can make the product self – supporting. It can stand out in quality without pressure from external constraints.

In some theses, it may be necessary to establish some theory to describe the experimental techniques, than to report what was done on several different problems or different stages of the problem and then finally present a model or a new theory based on the researcher's work. For such as theses, the core chapter headings could be theory, materials and methods, problem–1, problem–2 etc., and proposed theory / model. For others it might be appropriates to discuss different techniques in different chapters.

Materials and methods vary enormously from thesis to thesis and may be absent in theoretical theses. It should be possible for another scholar to replicate exactly what the present researcher has done by following the description available in the thesis. Another scholar may do a similar study or experiment either in the same university or in a new set up in a foreign country.

When a researcher reports a theoretical work that is not original, he should include sufficient material to allow the reader to understand the arguments used and their physical bases. Sometimes, the researcher may present the theory from the beginning but should not reproduce pages of material that the reader could find in standard text books. One should not include theory that the researcher is not going to relate to his work. The researcher should concentrate more on the physical arguments as on equations.

While reporting theoretical work by the researcher, he can include more detail, but however, lengthy derivation can be moved to appendices. Order and style of presentation is very important.

In sum, the core chapters should contain detailed description of the theories and hypotheses involved, different problems and the methodology adopted in resolving them, the materials and methods used, the experimental investigations carried out, the diverse techniques resorted to during the long exercise of the research study, significant revelations during the course of the work and arguments that led the researcher to the conclusions arrived. Arguments may be conveniently presented as a series of numbered or bulleted points rather than as one chunk in a crowded paragraph.

Findings and discussions

The findings and discussions are often combined in theses as otherwise the findings have to be repeated from core chapters for the purpose of discussions. While the major findings have already been presented earlier, this section provides a more complete treatment of all the findings (results) and their significance. The discussion should center on the findings only and not on general discussion of the subject area. The nature of the discussion depend upon the specific research study. But, however some general comments would be in order.

Findings are usually a combination of verbal statements, numerical values, tables and graphs. Ih some cases the researcher may give the same result both in tabular and graphical form. The graph is useful for quickly showing overall behavior and trends, whereas the table is better for giving accurate numerical values. If there is a theoretical prediction for a result, which was measured, the

researcher should compare theory and experiment and discuss their agreement or disagreement.

The major aspects to be considered by the researcher in this section is what the findings mean? How they fit into existing body of knowledge? Are they consistent with current theories?, Do they give new insights?, and Do they suggest new theories or mechanisms? The researcher should objectively examine as how other people in the field might see. He should also see whether findings have any implications that do not relate to the questions that the researcher originally set out to answer.

Discussion on error analysis should also be incorporated. The researcher use error bar on the data unless the errors are very small. For single measurements, the bars should be on the basis of best estimate by the researcher in each coordinate. Fur multiple measurements, these should include standard error in the data.

Apart from comparing theory and experiment, the researcher must compare and relate his findings to those of other researchers in the field, if such results are available. If there are significant discrepancies, he should explain adducing reasons from his study. The practical importance of the findings of a research study in terms of converting it into a product, performance improvement, cost reduction or increase in productivity should also be discussed.

Conclusions and suggestions for further work

In this section, the researcher draw together in his mind all the findings and explain what conclusions he has reached based on the findings. These conclusions must be justified in terms of logical arguments, past experience, and his own judgements. It is imperative to make clear which statements are made based on facts and which are opinion. In the abstract, the researcher included conclusions in a very brief form, because it must also include other material. A summary of conclusions is usually longer and the researcher have more space to be more explicit and more careful with qualifications in this section. It is better to put the conclusions in point form.

It is often the case with scientific investigations that more questions than answers are produced. Of course, it is true that any topic can be explored ad

infinitum and human capacity is an admirable trait. It is conventional that a researcher suggest any interesting scope for further work by future researchers. He should also indicate the practical implication of his work and limitations faced by him so that future research workers can try to address some of them. This chapter should usually be reasonably short–a few pages. As with introduction, it is a good idea to ask some one who is not a specialist to rad this chapter and offer his / her comments before finalising the section.

List of References

A researcher while carryout a research go through many books, published articles, unpublished theses, patents, web pages and he / she refer them in his / her thesis. These references are usually placed at end of the text in a particular order as may be prescribed by the university. There is no single university accepted format for references, but many journals follow the style given below.

a) Book referred (example)

i. E.O. Doebelin, "Measurement Systems – Applications and Design", 5th ed., McGraw – Hill, New York, 2004.

ii. W. Strunk and E.B. White", The Elements of Style", 3rd ed., Maemillon, New York, 1978

b) Journal articles (example)

i. B. Wang, T. Kajita, T. Sun and G.C. Temes", High Accuracy Circuits for on chip Capacitive Rane Testing and Sensor Readout", IEEE Trans. Instrum & Meas., Feb. 1998, vol. 47, no.1, pp 16 – 20

ii. R.M. Horn and Robert O. Ritchie, Metal. Trans. A, 1978 vol.94, pp 1039 – 1053

c) Patent

i. J.K. Gustafsson", Analog – digital converter for a resistance bridge", Patent U.S. 3960010, June 1, 1976.

d) Paper published in a conference proceedings

i. V.J. Kumar, N.M. Mohan and V.G.K. Murti", Digital converter for Push-pull type Resitive Transducers", Proc. IEEE IMTC 2005, Ottawa, Canada, May 17 – 19, 2003, pp 422 – 425

e) Websites

A website may disappear or it may have been updated or changed completely. So reference to the website are usually less satisfactory. Nevertheless, there are some very useful authoritative sources. So if the rules of the university permits, a researcher may include it in his references cited.

i. Data Sheet, PIC 16F87XA,"28/40/44 Pin Enhanced Flash Microcontrollers," Microchip Technology inc., 2003, http://ww1.microchip.com /downloads/en/ Device Doc / 395826. pdf

ii. Mc Laughlin, Michael P, "A Tutorial on Mathematical Modelling". 1999 http://www,causascientia.org / math_stat / Tutorial. Pdf

f) Handbooks

i. "ASM Handbook", vol 20," Materials Selection and Design", ASM International, Materials Park, OH, 1997.

ii. D.A. Woodford, Design for High Temperature Applications, "ASM Handbook", vol 20, pp 573 – 588, ASM International, Materials Park OH, 1997

g) Private Communication

i. J.J. Doe, XYZ Company, Atlanta, PA, unpublished research, 1981

Appendices

Appendices are useful for documenting information of lesser importance or for removing from the main text those items, perhaps due to complexity would needlessly disrupt the smooth progress of the reader. Detailed mathematical derivations whose end result is the main interest to the researcher may for example be included in the Appendix. Other items which are typically included in appendices are : important and original computer programmes, data files that are too large to be represented simply in the result chapters, pictures or diagrams of results which are not important enough to keep in the main text. Appendices should be numbered serially using Arab numerals. eq: Appendix. 1, Appendix 2, etc.,

ORAL PRESENTATION AND DEFENSE

After the thesis is evaluated by the Examiners nominated by the university, the researcher has to appear before a viva – voce examination board constituted by the University to present his / her research work and defend it. This is an important stage in the Ph.D research project. Mostly it is a open viva, where after examination by the examination board, the audience may also raise queries and the researches is expected to answer their queries satisfactorily. The research should not consider it as a war where a group of experts are ranged against him. He should never take it as a defence, but just an opportunity for presentation of ideas which he extensively nurtured during the last few years.

As a person carried out the research work, he is sure of all the diverse aspects of the topic, perhaps for better than the members of the examination board before him. The objective of the scholar is to convince his audience that he had done his job well, his methodology is sound and the findings are useful. In experimental topics, all the findings of the scholar should be based on repeated experiments, which should be replicatable by anyone else.

It is better that the researcher to know in advance the duration he would get for making the presentation so that he can plan and make use of the time allotted most effectively. If the researcher is asked to finish the presentation in the next two minutes, when he is only midway he will end up badly bruised. This has to be avoided at all costs. Therefore a proper preparation is necessary.

If the researcher is allotted 20 to 30 minutes for oral presentation, he should decide as how many slides could be presented using power point presentation. Accordingly the researcher should prepare the slides based on his research work taking only the important aspects. His focus should be mainly on the problem taken up for study and how he approached the solution for the problem considered and major findings and conclusions. However, he should have enough back up slides, to answer any specific queries after the main presentation. After preparing the slides, it is better the researcher has a rehearsal not only by himself, but also with his supervisor so that he can take note of the views of his supervisor.

The researcher must discuss with his friends and colleagues on various aspects relating to his thesis and gather the possible questions that they contemplate.

Viva – voce examination is an exercise on oral communication for the researcher. The oral communication has several special characteristics: quick feedback by question and dialogue; impact of personal enthusiasm; impacat of visual aids; and the important influence of tone, emphasis, and gesture. A skilled speaker in close contact with an audience can communicate for more effectively than the cold, distant easily evaded written word. On the other hand, the organization and logic of presentation must be of a higher order for the oral than written communication.

Many opportunities for noise exist in oral communication. By proper preparation and delivery by the researcher coupled with good quality of the visual aids, the efficiency of the oral communication can be improved. The comments from others should be taken as valuable. So also, any new idea coming from the member of the examination panel should be welcomed.

The researcher should listen and comprehend the points made and / or queries raised before attempting to answer or clarify the points. The clarifications by the researcher should be clear and convincing. Some months would have elapsed between preparation and submission of thesis and holding of viva – voce examination. Therefore, before the viva – voce examination, the researcher once again thoroughly go through his report so that he / she can answer any queries or clarify points raised. The researcher should never go into arguments.

The confidence of the researcher gets boosted if he approach the viva – voce examination with a feeling that the expert panel is there to help him. After all the long effort of the researcher, it is reaching its concluding stage of success. Infact, the researcher is not alone, his supervisor also has s role in the entire process of research and thesis preparation.

INTELLECTUAL PROPERTY RIGHTS

The Intellectual Property

The term intellectual property refers to the protection of ideas with patents, copyrights, trade marks and trade secrets. These entities fall within the

broad area of property law and as such, can be sold or leased just like other forms of property. Also, just as property can be stolen or trespassed upon, so intellectual property can be infringed.

Copy right

Copyright is an intangible property. The subject matter of copy right, the thing protected is called a “work”. Copyright subsists through out India in a) original literary, dramatic, musical and artistic works b) cinematograph film and c) sound recording. Section 14 of the Indian Copyright Act of 1957, subsequently amended defines that a “Copyright means exclusive right to do or authorize the doing of acts in respect of a work or any substantial part there of”. The original Act of 1957 was amended in 1983, 84, 92, 94 and 1999. No copyright exist in any work except as provided in the Act. In this section, the discussion is restricted to only literary and computer related aspects only.

With regard to literary, dramatic or musical the owner has exclusive right to reproduce work in any material form including the storing of it in any medium by electronics means, to issue copies to the public, to perform or communicate the work to the public, and translate or make any adaptation of the work. The literary work includes tables and compilation and therefore Dissertation is prime facie a literary work. As foras computer programmes concerned, the owner has exclusive right to sell or give on hire or offer for sale or hire any copy of the progaramme. The 1999 amendment has added the right to sell or give on commercial rental or offer for sale or for commercial rental any copy of the programme.

The Act confers upon the author the economic rights, because exploitation of the work by himself or by licensing it for royalty bring economic benefits to the author. Thus, the objective is to encourage authors to do original works by giving them exclusive right for a limited period. It is a negative right to prevent others from copying their work.

In India, the copyright is awarded for the life of the author plus 50 years from the death of the author. For being the owner of copy right registration is not compulsory. An author of the work becomes owner thereof, the moment the work is created. Registration is not necessary either for the subsistence or for the

enforcement of copy right. As per section 48 of the Act, the Register of copy right is the prime facie evidence of particulars entered therein. therefore, it is advisable to register the copyright in the work.

Ownership and assignments

The ownership of copyright in a work is independent of the ownership to the physical material in which the work is fixed. For example, when you buy a book, you become the owner of the book, but the author of the book is the owner of copy right unless he / she assigned the copy right to a publishing company. While the author of any work is the first owner of copy right, when literary work is created by the author during the course of employment under a contract for service, the employer is the owner of copy right. A teacher who writes a book during his employment is the owner, since he is employed to teach and not to write.

A copyright can be assigned, but however it should be in writing and signed by the assignor or by his duly authorized agent (Sec. 19). Copyright is a multiple right, consisting of bundle of different rights in the some work, which can be assigned or licensed either as whole to one party or separately to different parties.

Rights and infringements

The Act recognizer several moral rights of the author, which exists, even after assignment of the work. These rights include claim authorship of the work: and to restrain and claim damages in respect of any distortion, mutilation, modification or other act in relation to the work which is done before the expiry of the term of copyright.

In the case of published work, the copy right will subsist in India if the work is published in India or outside India, the author at the time of publication (if alive at that date) or at the time of his death is a citizen of India. In the case of unpublished work, the copy right subsists in India, if the author at the time of making of work was a citizen or domicile of India.

If any of the acts specified in section 14, relating to the work is carried out by a person other than the owner or without license from the owner or

competent authority under the law, it constitutes infringement of copy right as per Sec 51 of the Act. The type of infringement will depend upon the nature of the work. However certain acts will not be construed as infringements as per the law. Permitted acts include fair dealing, educational purpose, literary and archive use, use associated with public administration, certain acts carried out by lawful users of computer programmes. The lawful users can take back-up copies, decompile progarmme in limited circumstances, copy and adopt programmes in ways consistent with lawful use.

Trade related intellectual property rights (TRIPS)

In order to harmonize with international practices, and in compliance with India's obligations under TRIPS, major legislative changes were brought in IPR. The copyright act was also amended. The member countries are required to comply with articles 1 to 21 of the Berne Convention. Under the literary work included computer programme tables and compilations including computer databases. Computer programme means a set of instructions expressed in words, codes, schemes or in any form including a machine readable medium, capable of causing a computer to perform a particular task or achieve a particular result.

The computer programme being a literary work, it should be original for being copyrighted. For this the author should spend sufficient labour, skill, judgement and capital to impart the work some quality or character which the raw material did not possess and which distinguishes the work from the raw material. After the amendment to copyright act in 1999, India has compiled with all the requirements of the Article in TRIPS.

India has one of the most modern Copyright protection laws in the world. The 1999 amendment (49 of 1999) not only made it fully compatible with the provisions of TRIPS agreement, but also additional such as satellite broadcasting, computer software and digital technology were brought under Indian Copyright protection. The other important developments was the issuance of International Copyright order 1999 extending the provisions of Copyright Act to the nationals of all world organizations (WTO). The Copyright Act prescribes mandatory punishments for piracy of copyrighted matter commensurate with the

gravity of the offence in order to dater infringement in compliance with TRIPS agreement.

Patent Law

As per Indian Patent Act, 'invention' means a new product or process involving an inventive step and capable of Industrial application and patent implies a patent granted for any invention under this Act. The basic objective of confering an Intellectual Property Right (IPR) by way granting patent right to a person is to give social recognition to him. This social recognition can further bring economic benefits to its holders. The award is in the form of monopolistic rights for his labour and efforts. A patent right granted by Indian Government gives the patentee the right to prevent others from making, using or selling the patented invention. The Indian Patent Law came into existence by way Patent Act 1970 (Act 39 of 1970). Several amendments have been made till date and the notable one was in 1999 to comply with TRIPS agreement.

Items which are not inventions.

The Indian Patent Act clearly defines items which cannot be considered as inventions. They include,

- any invention or commercial exploitation or intended use which would be contrary to law or morality or injurious to health.
- mere discovery of scientific principle or the formulation of an abstract theory
- discovery of any new form of known substance which does not result in the enhancement of the known efficacy of that substance or the mere discovery of any new property or new use for a known substance or the mere use of known process, machine or apparatus unless such known process result in new product or employs atleast one reactant.
- a substance obtained by mere admixture resulting only in the aggregation of the properties of the component thereof or a process for producing such substance.
- mere arrangement or duplication of known devices each functioning independently of one another in a known way.
- method of agriculture or horticulture

- mathematical or business method or a computer programme per se or algorithm.
- a literary, dramatic, music or artistic work or any other aesthetic creation whichever including cinemotographic works and T.V. production.
- topography of integrated circuits
- any process for the medicinal, surgical, curative prophylactic or other treatment of human beings or any process for the similar treatment of animals or plants.
- any invention relating to atomic energy falling within the sub-section (1) of Section 20 of the Atomic Energy Act 1962 (Act 33 of 1962)

Granting of patent rights

Aray person who is a true and first inventor, his assignes or the legal representative of a deceased inventor can make an application and apply to the Controller General of Patents, Designs and Trade Marks enclosing specifications, detailed drawing etc. The controller General refer it to appropriate examiners for their report. Once the controller after examination accepts the specification gives a notice to the applicant and also advertises in the official Gazetteer the fact that specification accepted and the drawings and specifications is open for the public. Four months time given to file any comments by opponents and opportunity given for hearing. Thereafter if everything is in order the Controller General grants the potent right to the applicant.

An Indian resident cannot apply for patent for an invention outside the country without prior permission He / she should have applied first to Indian authorities atleast six weeks before making application to outside.

The date of patent is the date of application filed by the applicant. While granting patent right, the authorities can impose conditions with regard to import and use by Government, limited use of patented item by others for experiment or research and instructions to pupil etc. The term of patent is fixed as 20 years from the date of filing application. But, however, the patentee has to pay renewal fee as prescribed to keep his patent right current.

Rights of Patentee

The Patent Act confers exclusive rights to the patentee to prevent third parties from making, using, offering for sale, selling or importing for those purposes that product in India. This is applicable for both product and process. The grantee or the proprietor has power to assign, grant license under, or otherwise deal with the patent and to give effectual receipts for any consideration for any such assignment, license and dealing.

The authority can compulsorily give license any time after three years of expiry from the date of sealing (grant) of a patent to any other person applied for grant of compulsory license on the ground that reasonable requests of public with respect to the patent have not been satisfied, patented invention not available to the public at a reasonably affordable price and the patented invention not worked in India.

General principles applicable to working of inventions

The guiding principles to check whether the patented invention work include.

- whether the invention for which patent granted worked in India on a commercial scale without delay
- patents are not granted merely to enable patentees to enjoy monopoly for the importation of a patented article
- whether the protection and enforcement of patent right contribute to the promotion of technological innovation and to the transfer and dissemination of technology.
- that patents granted do not impede protection of public health and nutrition.
- that they do not in anyway prohibit central Government in taking measures to protect public health.

The Act also provides for revocation of patent rights granted on the ground that the patented invention has not worked in India and that it has not satisfied the reasonable requirements of the public and that the patented invention is not available to the public at a reasonable price.

The Indian Patent Act is in accordance with the provisions of TRIPS agreement.

ETHICS IN RESEARACH

Research Ethics

Ethics can be defined as the system or code of morals of a particular person, religion, group or profession. Ethics is relative. It differs from one individual to another. Ethical behavior is an individual matter and a person's ethics and morals are built into his personality right from his childhood. However, the key forces that form the ethical behaviour of an individual include his / her family, peer influence, experience logged by the individual, his / her morals and values practiced and situations. Here, we look into ethical behaviour as individuals involved in scientific and social research works.

Research ethics involves the application of fundamental ethical principles to a variety of topics involving scientific research. These include the design and implementation of research involving physical experimentation, human and animal experimentation, indepth theoretical and other studies involved in various disciplines. It also include various aspects of academic scandal including scientific misconduct such as fraud, fabrication of data and plagiarism, whistle blowing and regulation of research etc., In medical field the research ethics is most developed. Many professional societies have also evolved ethical codes for the professionals to adopt.

The scientific research is built on a foundation of turst. Therefore scientist believe that the results reported by others are valid. The society also trusts that the results of research reflect a honest attempt by scientists to describe the world accurately and without bias. But this trust will endure only if the scientific community devotes itself to exemplifying and transmitting values associated with ethical conduct.

Many ethical issues have to be taken into serious consideration for research and they depend upon the nature of research work. For instance in the case of sociological research. not only permission of those involved in the study need to be obtained but also their rights to be protected and privacy and sensitivity to be maintained. Research ethics in medical context is dominated by

principalism. In research involving physical experimentation sincerity in reporting, safety of the individuals involved, environmental considerations etc are important.

As science becomes increasingly intertwined with major social, philosophical, economic and political issue, scientists become more accountable to the larger society of which they are a part. Therefore individual scientists and their institutions should periodically reassess the values and professional practices that guide their research as well as their efforts to perform their work with high integrity.

Integrity in Research

Integrity in research is essential for maintaining scientific excellence and for keeping the public trust. Integrity characterizes both researchers and the institutions in which they work. A committee set up by the National Academy of Sciences in USA observed, "For a scientist integrity embodies above all the individual's commitment to intellectual honesty and personal responsibility. It is an aspect of moral character and experience. For an institution, it is a commitment to creating an environment that promotes reasonable conduct by embracing standards of excellence, trustworthiness, and lawfulness……..Thus, integrity is an important trait both for individual researchers and for the institution in which they work.

For an individual researcher, integrity embodies commitment to intellectual honesty and personal responsibility for one's actions and to a range of practices that characterize the responsible conduct of research, including

- intellectual honesty in proposing, performing, and reporting research.
- accuracy in representing contributions to research proposals and reports.
- fairness in peer review
- collegiality in scientific interactions, including communication and sharing of resources.
- transferancy in conflict of interest or potential conflict of interest
- protection and safety of human subjects in the conduct of research.
- adherence to the mutual responsibilities between investigatiors.

Intellectual honesty requires that researchers present proposals and data honestly and communicate their best understanding of their work in writing and verbally. They should advocate for their research conclusions in the face of collegial skepalicism and must acknowledge errors. The researchers are not expected to report the work of others as if it were their own, which is plagiarism. Collegiality in scientific intarctions requires that investigators report research findings to the scientific community in a full, open and timely fashion. At the same time, it should be recognized that the scientific community is highly competitive. The researcher who first reports new and important findings gets credit with the discovery.

The faculty advisor (supervisor) is the mentor for an young researcher. His / her responsibilities include a commitment to continuous education and guidance of trainees, appropriate delegation of responsibility, regular review and constructive appraisal, fair attribution of accomplishment and authorship, and career guidance, as well as help in creating opportunities for employment and funding. For the trainee, the essential elements include respect for the faculty advisor, loyalty, strong commitment to science, dedication to the project, careful performance of experiments, precise and complete control of record keeping, accurate reporting of results and a commitment to oral and written presentations and publications.

Institutions seeking to create an environment that promotes responsible conduct by individual researchers that foster integrity must establish and continuously monitor structures, processes, policies and procedures that

- provide leadership in support of responsible conduct of research
- encourage respect for everyone involved in the research work
- promote productive interactions between trainees and faculty advisors
- advocate adherence to rules regarding all aspects of the conduct of research
- manage individual and institutional conflict of interests
- timely action against scientific misconduct.
- educating the researchers pertaining to integrity in the conduct of research

Promoting integrity through education is an important task of the institutions. It presents a special challenge because it requires a synthesis of ethics and science. When scientists and ethicists collaborate in the design and implementation of learning experiences, the students come to appreciate the complexity of problems that arise in the practice of science. Research advisors play a central role in the education of their trainees in the responsible conduct of research, not only by what they teach, but also by their own conduct. The committee set by the National Academy of Sciences, USA, which went into the question of research integrity emphasized the education of students and post doctoral follows, as they felt this is where the future lies.

It is the burden of research of research institutions to promote and monitor the responsible conduct of research. They must consistently provide the researchers with the resources they need to conduct research. These resources include leadership, training and education, policies and procedures as well as tools and support systems. Anyone needing assistance should have ready access to knowledgeable leaders.

Scientific misconduct

Another important aspect which needs to be considered is misconducts in scientific research. The scientific misconduct can be defined as “the intentional distortion of the research process by fabrication of data, text, hypothesis or methods from another researcher’s manuscript form or publication; or distortion of research process in other ways”. The consequences of scientific misconduct can be severe at a personal level for both perpetrators and any individual who exposes it.

The main motivators for researchers to commit misconduct are,

- Career pressure: The reputation of researcher lies on publication of high Profile scientific papers and get research grants. This motivates some persons to fabricate results.
- Laziness: Researchers introduce a fact that they believe is true without going to the trouble and difficulty of actually performing the experiments required.

- Lure of money: When grant is available especially in new technologies where there is no existing standard against which is compare
- Ideological considerations

Scientific misconduct is the violation of the standard codes of scholarly conduct and ethical behaviour in professional scientific research. The various forms if scientific misconduct is enumerated below.

- fabrication: The publication of deliberately false or misleading research. This is subdivided into obfuscation (omission of critical data or results) fabrication (actual making up of research data) falsification (manipulation of research data and process) bare assertion (making entirely unsubstantiated claims)
- plagiarism: The act of taking credit at attempting to take credit for the work of another. A subset is citation plagiarism ie. willful or negligent failure to appropriately credit other or prior discoverers. It is the most common type of misconduct.
- Self plagiarism: or multiple publication of the same content with different titles and / or in different journals.
- violation of ethical standards regarding human and animal experiments.
    - ghost writing: The phenomenon where some one other than the named author(s) make a major contribution.
    - conferring authorship on those who have not made substantial contributions to the research work. This is done by senior researchers who muscle their way on to the papers of inexperienced junior researchers as well as others that stake authorship in an effort to guarantee publication.
    - some academics consider suppression ie the failure to publish significant findings due to the results being adverse to the interests of the research or his / her sponsor a form of misconduct.

The authors and coauthors of scientific publications have a variety of responsibilities. Contravention of the rules of scientific authorship may lead to a charge of scientific misconduct. All authors are expected to keep all study data for later examination even after publication. The failure to keep data may be regarded as misconduct. All authors including co-authors are expected to make

reasonable attempts to check findings submitted to academic journals for publication. Simultaneous submission of scientific findings to more than one journal or duplicate publication of findings is usually regarded as misconduct.

In many countries there is no regulator to oversee the investigation of allegations of research misconduct. Universities therefore have no incentive and every disincentive to investigate allegations or act on the findings of such investigations if they vindicate the allegation.

The consequences of scientific fraud vary based on the severity of the fraud, the level of notice it receives and how long it goes undetected. For cases of fabricated evidence, the consequences can be wide ranging with others working to confirm or refute the false finding or with research agendas being distorted to address the fraudulent evidence.

The consequences of individuals indulged in misconduct has a reflection on the institutions that host or employ them and also on the participants in any peer review process that has allowed the publication of questionable research. A rang of actors involved in this. The persons who expose such cases can find themselves open to retaliation by a number of different means. These negative consequences for exposes of misconduct have driven the development of whistle blowers charter – designed to protect those who raise concerns in USA. A whistle blower is almost always alone in his fight – his career becomes completely dependent on the decision of alleged misconduct. Even organization feels shy of employing him on the pretext he / she is a trouble maker.

Now a days there are several tools to aid in the detection of plagiarism and multiple publication, thanks to development of Internet. A tool developed in 2006 by the University of Texas, called “Déjà vu” use the text mining algorithm eTBLAST and identified many instances of duplicate publication in biomedical area. There are tools to detect falsified data include error analysis. Any measurement usually have a small amount of error. Suppose in a particular case, the amount of error normally be in such measurements does not appear, one can conclude the data may have been forged. However, this alone is not sufficient to confirm. Further investigations have to be made.

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